DIEBOLD NIXDORF

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Project:

FISCAL FAMILY CONTROLLER (FFC) WITH TH230+ ROMANIA SPECIFICATION

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x	**Specification (Implementation)**
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1.16	225	*See chapter 1.3*	2019-03-28	Stefan Hoffmann

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Table of Contents

1 GENERAL 11

1.1 Foreseeable changes 11

1.2 References 11

1.3 Changes to previous FW versions 11

1.4 Known differences between MF-THF and FFC_TH230+ 16

1.4.1 Old commands syntax convention 16

1.4.2 New fiscal commands 16

1.4.3 Obsolete fiscal commands 20

1.4.4 Expired fiscal commands 21

1.5 Abbreviations 22

2 OVERVIEW 23

2.1 Purpose of this document 23

2.2 Introduction 23

2.3 Country ID's and codepages 24

3 MF MEMORY (MFMEM) 24

3.1 General 24

3.2 Hardware data (HWDATA) 25

3.3 MFMEM status 25

3.4 Block handling (logical and physical blocks) 26

3.5 MFMEM physical block status 28

4 ELECTRONIC JOURNAL (EJ) 29

4.1 General 29

4.2 Electronic journal medium 29

4.3 Electronic journal organization 30

4.4 Electronic journal status 30

4.5 Unique medium identification (UMID) 32

5 FISCAL FUNDAMENTALS 32

5.1 Fiscal Status 32

5.1.1 Request the fiscal status 32

5.1.2 MFC status byte 1 32

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5.1.3 MFC status byte 2 34
5.2 ASB Handling (MF ASB) 36
5.3 Fiscal Firmware (Flow chart) 38
5.3.1 Power On procedure 38
5.3.2 Idle Loop 40
5.4 Receipt Types 41
5.4.1 General 41
5.4.2 Retail sales receipt (legal receipt) 42
5.4.3 Generic receipt (illegal receipt) 44
5.4.4 Cash in/out receipts (illegal receipt) 46
5.4.5 Retail sales receipt with tax ID 46
5.4.6 Payment correction receipt 48
5.5 Counters and totalizers 49
5.5.1 Totalizers 49
5.5.2 Counters 51
5.5.3 Size 53
5.6 Payment list handling 54
5.7 Logical block types 55
5.7.1 Overview 55
5.7.2 SNO block 56
5.7.3 EOD block 57
5.7.4 TKD block 58
5.7.5 EJ change block 58
5.7.6 TH block 58
5.7.7 VAT block 59
5.7.8 CRY block 59
5.7.9 FW block 60
5.7.10 Block restriction 61
5.8 EJ medium handling 63
5.8.1 EJ medium initialization (Flow chart) 63
5.8.2 Electronic journal operation 64
5.9 EJ data security 64
5.9.1 EJ file logical structure 64
5.9.2 EJ verify utility 65

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6 PRINTER COMMANDS 67

7 FISCAL COMMANDS 74

7.1 General rules 74
7.1.1 Parameters 75
7.2 Notation 75
7.2.1 Restricted Text 76
7.3 INIT commands 77
7.3.1 INIT_HWDATA 77
7.3.2 INIT_PRE_FORMAT 77
7.3.3 INIT_FULL_FORMAT 78
7.3.4 INIT_WRITE_SNO 78
7.4 SET commands 80
7.4.1 SET_DATE_TIME 80
7.4.2 SET_VATRATES 81
7.4.3 SET_HEADER 83
7.4.4 SET_CURRENCY_COUNTRY 86
7.4.5 SET_PAYMENT_LIST 89
7.4.6 SET_PAYMENT_LIST_VOID 90
7.4.7 SET_WORKING_MODE 91
7.5 DAY commands 93
7.5.1 DAY_BEGIN 93
7.5.2 DAY_END 93
7.5.3 DAY_XREPORT 95
7.5.4 Z/X Report 96
7.6 RECEIPT commands 100
7.6.1 RECEIPT_BEGIN 100
7.6.2 RECEIPT_END 101
7.6.3 RECEIPT_END_MFLINE 105
7.7 ARTICLE commands 106
7.7.1 ARTICLE_SELL 106
7.7.2 ARTICLE_VOID 108
7.7.3 ARTICLE_DISCOUNT 109
7.7.4 ARTICLE_DISCOUNT_VOID 110
7.7.5 ARTICLE_SURCHARGE 111

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7.7.6 ARTICLE_SURCHARGE_VOID 113
7.7.7 ARTICLE_RETURN 114
7.7.8 ARTICLE_RETURN_VOID 115
7.7.9 ARTICLE_FUEL_SELL 116
7.7.10 ARTICLE_FUEL_VOID 120
7.7.11 ARTICLE_TAX 120
7.7.12 ARTICLE_TAX_VOID 121
7.8 SUBTOTAL command 123
7.8.1 SUBTOTAL_LINE 123
7.8.2 SUBTOTAL_DISCOUNT 124
7.8.3 SUBTOTAL_DISCOUNT_VOID 127
7.8.4 SUBTOTAL_SURCHARGE 128
7.8.5 SUBTOTAL_SURCHARGE_VOID 129
7.9 TOTAL command 131
7.9.1 TOTAL_LINE 131
7.10 PAYMENT commands 134
7.10.1 PAYMENT_IN 134
7.10.2 PAYMENT_IN_VOID 135
7.10.3 PAYMENT_OUT 136
7.11 FREEPRINT commands 138
7.11.1 FREEPRINT_RESTRICTED 138
7.11.2 FREEPRINT_NORMAL 139
7.12 SPECIAL commands 141
7.12.1 SPECIAL_CASH_IN 141
7.12.2 SPECIAL_CASH_OUT 142
7.12.3 SPECIAL_CUT 143
7.12.4 SPECIAL_PRINT_HEADER_CUT 143
7.12.5 SPECIAL_ALL_VOID 144
7.12.6 SPECIAL_CMD_RESTART 145
7.13 REPORT commands 146
7.13.1 General Note 146
7.13.2 REPORT_BLOCK2BLOCK 146
7.13.3 REPORT_DATE2DATE 151
7.13.4 REPORT_EOD2EOD 153
7.13.5 REPORT_TOTAL_DATE2DATE 154

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7.14 TRAINING commands 157
7.14.1 General Note 157
7.14.2 TRAINING_BEGIN 157
7.14.3 TRAINING_END 158
7.15 GET commands 160
7.15.1 General Note 160
7.15.2 GET_RECEIPT_TOTALIZER 160
7.15.3 GET_RECEIPT_BRUTTO_SALES 160
7.15.4 GET_RECEIPT_VAT_AMOUNT 161
7.15.5 GET_RECEIPT_PAYMENT 161
7.15.6 GET_RECEIPT_STATISTIC (in preparation) 162
7.15.7 GET_DAY_TOTALIZER 162
7.15.8 GET_DAY_BRUTTO_SALES 162
7.15.9 GET_DAY_VAT_AMOUNT 163
7.15.10 GET_DAY_PAYMENT 163
7.15.11 GET_DAY_STATISTIC 163
7.15.12 GET_DAY_DRAWER_CONTENT 164
7.15.13 GET_DAY_RECEIPT_TYPE_CNT 164
7.15.14 GET_DAY_RECEIPT_GROUP_CNT 165
7.15.15 GET_DAY_EVENT_CNT 165
7.15.16 GET_GRAND_BRUTTO_SALES 165
7.15.17 GET_GRAND_VAT_AMOUNT 166
7.15.18 GET_GRAND_FISCAL_CNT 166
7.15.19 GET_MFMEM_CNT 166
7.15.20 GET_MFMEM_LAST_BL_DATE 167
7.15.21 GET_MFMEM_LAST_BL_NUM 167
7.15.22 GET_MFMEM_LAST_LOGICAL_BL_NUM 168
7.15.23 GET_MFMEM_BL_NUM 168
7.15.24 GET_MFMEM_BL_HEADER 169
7.15.25 GET_DATE_TIME 170
7.15.26 GET_VAT_RATES 170
7.15.27 GET_CURRENCY_COUNTRY 171
7.15.28 GET_PAYMENT_LIST 171
7.15.29 GET_ERROR_LIST 171
7.15.30 GET_FIRMWARE_VERSION 172

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7.15.31 GET_TKD_INFO 172
7.15.32 GET_HEADER 173
7.15.33 GET_SNO_NUMBER 174
7.15.34 GET_DATE_FIRST_RECEIPT 174
7.15.35 GET_DEVICE_CONNECTED 175
7.15.36 GET_WORKING_MODE 175
7.15.37 GET_FA_STATUS 175
7.15.38 GET_FA_DAILY_STATUS 176
7.16 EJ_ commands 178
7.16.1 EJ_INIT 178
7.16.2 EJ_REPRINT_T2T_BY_SESSION 179
7.16.3 EJ_REPRINT_T2T_BY_DATE_TIME 181
7.17 EJ_GET commands 184
7.17.1 EJ_GET_STATUS 184
7.17.2 EJ_GET_SIZE 184
7.17.3 EJ_GET_FREE_SPACE 184
7.17.4 EJ_GET_INIT_INFO 185
7.17.5 EJ_GET_MEDIUM_INFO 185
7.17.6 EJ_GET_LAST_MEDIUM_INFO 186
7.17.7 EJ_GET_EOD_ENTRY 186
7.17.8 EJ_GET_NUMBER_OF_EOD 187
7.17.9 EJ_GET_EOD_RANGE 187
7.17.10 EJ_GET_DATE_RANGE 187
7.17.11 EJ_GET_DATA 188
7.17.12 EJ_GET_EOD_NUMBER_BY_DATE 189
7.17.13 EJ_GET_INFO_PER_SESSION 189
7.18 TEST commands - general 191
7.18.1 TEST_READ_MFMEM_STATUS 191
7.18.2 TEST_MFMEM_FORMAT 191
7.18.3 TEST_MFMEM 191
7.18.4 TEST_READ_RTC 192
7.18.5 TEST_READ_MFMEM_BL 192
7.18.6 TEST_READ_MFMEM 192
7.18.7 TEST_PRINT_SELFTEST 193
7.18.8 TEST_PRODUCTION 193

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7.19 TAX TERMINAL commands 194
7.19.1 General Note 194
7.19.2 TAX_TERMINAL_PAIR_REQUEST 194
7.19.3 TAX_TERMINAL_FIRMWARE_UPDATE 195
7.19.4 TAX_TERMINAL_TEST 196
7.19.5 TAX_TERMINAL_IMPORT_XML_PROFILE 196
7.19.6 TAX_TERMINAL_EXPORT_XML_BY_SESSION 196
7.20 MFC commands 197
7.20.1 MFC_STATUS 197
7.20.2 MFC_SYNC 197
7.20.3 MFC_TEST 198
7.20.4 MFC_INFO 198
7.20.5 MFC_OPEN_D 199
7.20.6 MFC_READ_LAST_INTERRUPTED_CMD 200
7.20.7 MFC_READ_HOST_DATA 200

8 ERROR HANDLING 201
8.1 Error messages 201
8.2 FM error codes 213
8.3 EJ error codes 214
8.4 Supplementary checks error codes 214

9 INITIALIZATION / FISCALIZATION 216

10 SERVICE RESET (TKD RESET) 216

11 SERVICE MENU 218
11.1 Menu handling 218
11.1.1 Selecting the service menu 218
11.1.2 Controlling the Service Menu 219
11.2 LED Service menu 220
11.2.1 Display Service menu 220
11.2.2 Language 220
11.2.3 Print Selftest 220
11.2.4 XML files to USB 220
11.2.5 TT Test 220
11.2.6 Exit 221

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12 APPENDIX 221
12.1 Printer and MF LED's 221
12.2 Tax Terminal LED's 223
12.3 Updating the fiscal firmware 223
12.4 Fiscal hardware 224
12.4.1 Fiscal Family Controller (FFC) 224
12.4.2 Tax Terminal 224
12.5 RS232 interface 225
12.6 Codepage 852 (Multilingual I + Euro) 226

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1 GENERAL

1.1 Foreseeable changes

None

1.2 References

[R1] Romanian Fiscal Law
[R2] TH230+ Programmers Guide
[R3] TH230 Firmware Upgrade Concept
[R4] SD Specifications Part 1 Physical Layer Simplified Specification Version 2.00 September 25, 2006
[R5] RFC6234 - Secure Hash Algorithm 2 (SHA2) / 256 Bits – Secure Hash Standard
[R6] Hardware Specification for Fiscal Family Controller TH230+; 0170243901, A, 1
[R7] WN Enhanced Serial Protocol; 0170102235
[R8] TH230-MF Service Guide

1.3 Changes to previous FW versions

Version	*Date*	*Changes*
01.19	**2019-11-13**	
		- bug fix EJ_PRINT printout of tracks in the context SPECIAL_ALL_VOIDS
01.18	**2019-08-26**	
		- bug fix SET_CUT_MODE
01.17	**2019-08-13**	
		- bug fix TKD reset after canceled fiscal ticket - SET_DATE_TIME fixed no escape from blockade after EM_RTC_BATTERY - New command chapter 7.4.8 SET_CUT_MODE (partial cut selectable)
01.16	**2019-03-28**	
		- Unconformity Report 2019-03: In all MF reports from fiscal memory the block of EJ initialization must be printed

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		- Unconformity Report 2019-03: XML <coteZ> section must contain all enabled VAT categories, even if there are none sales
01.15	**2019-02-28**	
		- fixed Report X footer in case of printout interruption by paper-end or power-off (ICI unconformity report) - fixed typo in SET_HEADER info ticket - implemented transfer of "FFC firmware version" and "Last firmware update time" to Tax Terminal
01.14	**2019-01-10**	
		- fixed missing VAT records (<cote> section) in XML receipt data in case of "cmd interrupted" by power-off
01.13	**2019-01-08**	
		- realtime command DLE ENQ checks the validity (1 or 2) of the parameter n (barcode 128 unexpected cut bug) - bugfix in TKD Reset and EJ SHA printout/ entry in session table
01.12	**2018-09-07**	
		- RECEIPT_BEGIN: Fixed an error which may lead to "ticket header only" lines in EJ and gaps in fiscal ticket counter sequence if the printer cover is open while printing
01.11	**2018-07-25**	
		- Fixed possible dead-lock after interruption of command DAY_END (error EM_EJ_SESSION_NOT_OPEN at SPECIAL_CMD_RESTART) - REPORT_DATE2DATE: Fixed possible return of error EM_MFMEM_BL_NOT_EXIST if the range doesnot contain any tickets (empty report expected)
01.10	**2018-07-06**	
		- Unconformity Report 2018-06-27, #1/4: Additionally check Tax Terminal SD card at startup. Additionally check EJ, Tax Terminal crypto chip and SD card at EJ_INIT, DAY_BEGIN, RECEIPT_BEGIN, all REPORT/EJ_REPRINT commands, all SET commands and XML export. Reject these commands if check fails. - Unconformity Report 2018-06-27, #3: Removed print line "*** BON NEFISCAL *** BON NEFISCAL ***" from all EJ_REPRINT reports - Unconformity Report 2018-06-27, #6: Added “TOTAL TVA” line (sum of all VAT values except E) on layout of fiscal ticket

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		- Unconformity Report 2018-06-27, #7: Set EJ near full condition to 2.200 print lines before EJ end. Fixed Z report line of free lines in EJ. - Unconformity Report 2018-06-27, #9: Fixed wrong "totTva" calculation in XML file - Unconformity Report 2018-06-27, #10: Fixed timestamp on daily XML file - Unconformity Report 2018-06-27, #11: Only the first SPECIAL_CASH_IN command with type “init float” of a fiscal day is handled as “init float”. All further commands with the same type are handled as type “extra float”. - Unconformity Report 2018-06-27, #13: Fixed "zero value" VAT records in XML file - Fixed wrong printout of "CASH IN SERTAR" and "TOTAL CASH IN SERTAR" values in reports - Removed fiscal logo in printouts of all REPORT commands - Impl. power-off timestamp transfer to Tax Terminal (for ANAF event reports) ***After firmware update a Tax Terminal re-pair is necessary!***
01.09	**2018-06-21**	
		- Changed handling in EJ near end situation. An open receipt is closed and a Z-Report is printout automatically.
01.08	**2018-06-11**	
		- Bugfixing in SET_HEADER
01.07	**2018-06-08**	
		- TEST_READ_MFMEM_BL: Changed min. value of <*BlockNo*> from 0 to 1 - Bugfix: Removed incorrect '@' from EJ (occurred before ticket header from 2nd ticket) - Bugfixing in SET_HEADER. Detection of printer commands improved.
01.06	**2018-05-31**	
		- Added new commands TAX_TERMINAL_IMPORT_XML_PROFILE and TAX_TERMINAL_EXPORT_XML_BY_SESSION - MFC status toggling fixed in MFC_EJ_FULL und cover open situation - Change the line space to 1/6 inch for the new big font in TH230+. **Documentation** - Improved description of command MFC_TEST (empty answer)

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01.05	**2018-05-23**	
		- TEST_READ_MFMEM parameter <GetIndex> has been extended from five to six digits. - TEST_READ_MFMEM_BL the parameter <*BlockNo*> addressed physical blocks - If a legal ticket is aborted with SpecialAllVoid, then this ticket is complete removed from the EJ. - Working mode correctly saved and reinitiazied after TKD reset. - REPORT commands: ignore "EJ full" status **Documentation** - Bugfix in description of the checksum over a physical block in MFMEM - Fixed indexes in EOD block (5.7.3) and added “Working Mode” field
01.04	**2018-05-09**	
		- Implemented new request from ANAF: Reports from MF and EJ to be downloaded on request to USB stick -> Requires CoMo firmware >= 1.1.30869 ! **Documentation** - Added new <Station> parameter option “32 - External Storage” for all EJ_REPRINT and REPORT commands
01.03	**2018-04-26**	
		Bugfixed and changes - FREEPRINT: ignore "Station" value and always print on paper and - depending on ticket type - into EJ as well - SET_DATE_TIME: sync with CoMo immediately - On printing barcode the text '[BARCODE]' is written to EJ now - Fixed implementation of Unconformity Report point #5 (lines "B O N F I S C A L" and "BON NEFISCAL" only for EJ) - The Error blink codes for Blockade are set when MFMEM_FULL is reached. **Documentation** - FREEPRINT_NORMAL: Added note about text [BARCODE] in EJ
01.02	**2018-04-25**	
		Bugfixed and changes - Unconformity Report #5: Inside of EJ file the fiscal ticket must finish with the lines: B O N F I S C A L BON NEFISCAL

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		- Unconformity Report #17: When ICI remove SD Card from CoMo the printer they can still issue fiscal tickets on the printer - Write FREEPRINT on EJ if the receipt is also on EJ - Allow TT_FIRMWARE_UPDATE in non-fiscal mode (RBC production requirement) - Fixed missing PAYMENT lines in EJ - EJ: filter out ESCRn (end of X/Z report if MFMEM near full) - Correct the Z-Report footer printout in MFMEM-NEAR-FULL and MFMEM-FULL situations - Change the value of status switch to MFMEM-NEAR-FULL = EOD# 7785 and MFMEM-FULL = EOD# 7845 - Fixed "arbitrate" error after SPECIAL_ALL_VOID **Documentation** - FREEPRINT_NORMAL: Added note about text [BARCODE] in EJ
01.01	**2018-04-24**	
		Bugfixed and changes - Unconformity Report #4: Inside of EJ file all the lines " * BON DUPLICAT DIN JURNAL * BON NEFISCAL * " must be deleted - Unconformity Report #6: Inside of EJ there must be no control or formatting characters except CRLF - Unconformity Report Add-on #1: FREEPRINT commands: fixed handling of "Station" parameter (only "2" allowed and always write to EJ) - Unconformity Report Add-on #2: EJ_REPRINT command with Station "16" must not print the message digest on paper - Unconformity Report Add-on #3: SPECIAL_CASH_IN with parameter Initial Float must accept also the value zero - TT_FIRMWARE_UPDATE: do not check CoMo pairing and "firmware available" flag (production requirement) - Unconformity Report #1: The export of MF and EJ report in the file with extension MF (using MFC_READ_HOST_DATA and JavaPOS/OPOS driver) must be a text file (no XML) - Unconformity Report #3: Structure of EJ file must have only fiscal tickets and Z reports - Unconformity Report #7: They do not agree OFFSET track table lines at the end of EJ file - Unconformity Report #8: On EJ file after Z report must have Message Digest - Unconformity Report #2: Calculation for MFMEM_NEAR_FULL corrected
01.00	**2018-03-22**	**Initial Release**

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		- Changed version from 00.49 to 01.00

1.4 Known differences between MF-THF and FFC_TH230+

1.4.1 Old commands syntax convention

Please note that software applications which originally have been developed in order to support MF-THF will have to be adapted to the new FFC-TH230+ interface as certain old commands which have been changed and/or will not be supported any longer. Please check the following delta list in detail!

1.4.2 New fiscal commands

Command	FFC-TH230+ Syntax	Comment
INIT_HWDATA	**ESC MFB** s a **ESC MFB1** f n <HardwareData> **ESC MFE**	The old command (**ESC MFB** K A <HWData> **ESC MFE**) is still supported but not recommended for new implementations. New hardware data is defined for the new Fiscal Family Controller (FFC) and Fiscal Memory type (PFM). See chapter 3.2 Hardware data (HWDATA).
INIT_PRE_FORMAT	**ESC MFB** s f **ESC MFE**	Partial formatting used in production line only; replaced: **INIT_FORMAT_MFMEM** ESC MFB K B 3 ; 0 ESC MFE
INIT_FULL_FORMAT	**ESC MFB** s b **ESC MFE**	The fiscal memory (PFM) will be full formatted by the end-user. Replaced: **INIT_FORMAT_MFMEM** ESC MFB K B 1 ; 0 ESC MFE
INIT_WRITE_SNO (INIT_WRITE_SNO_BL)	**ESC MFB** s c **ESC MFB1** d r <FiscalID> **ESC MFB1** d s <CIF> **ESC MFB1** d u <FiscalSerialNumber> **ESC MFE**	The old command (**ESC MFB** K C <SystemID> ; <CountrySpecificInfo> **ESC MFE**) **is not supported.** CIF information precondition for fiscalisation.
SET_DATE_TIME (replaced SET_TIME)	**ESC MFB** h e **ESC MFB1** e k <DateTime> **ESC MFE** <DateTime>: "DD.MM.YY hh:mm:ss"	The old command **ESC MFB** I <TimeDate> **ESC MFE** with <TimeDate>: ss,mm,hh;DD,MM,YY is still supported but not recommended for new implementations.
SET_VATRATES	**ESC MFB** h a **ESC MFB1** c b *<VATRateA>* **ESC MFB1** c c *<VATRateB>* **ESC MFB1** c d *<VATRateC>* **ESC MFB1** c e *<VATRateD>* **ESC MFB1** c f *<VATRateE>* **ESC MFB1** c g *<VATRateF>* **ESC MFB1** c h *<VATRateG>* **ESC MFB1** c i *<VATRateH>* **ESC MFE**	The old command supported only 4 VAT rates. With "99,99%" it is possible to switch off the VAT category G and H. Category E and F must be set to 0000.

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SET_HEADER	**ESC MFB** h b **ESC MFB1** b l <*ReceiptHeaderLine1*> **ESC MFB1** b m <*ReceiptHeaderLine2*> **[ESC MFB2** b o < *ReceiptHeaderLine3*>] **[ESC MFB2** b p < *ReceiptHeaderLine4*>] **ESC MFB1** b q < *ReceiptHeaderLine5*> **ESC MFB2** d o <CIF ID> **ESC MFE**	New command structure and parameter. **Old command not supported!**
SET_CURRENCY_COUNTRY	**ESC MFB** h d **ESC MFB1** i m <*CurrencyLabel*> **ESC MFE**	New command replaces SET_CURRENCY (“h c”).
SET_PAYMENT_LIST	**ESC MFB** h s **ESC MFB1** g a <PaymentType> **ESC MFB1** g e <PaymentName> **ESC MFB1** g f <PaymentCategory> **ESC MFE**	
SET_PAYMENT_LIST_VOID	**ESC MFB** h t **ESC MFB1** g a <PaymentType> **ESC MFE**	This command deletes one payment list entry.
RECEIPT_END	**ESC MFB b b** **[ESC MFB2** b h <TaxID>**] ESC MFE**	New optional parameter for customer CIF number.
RECEIPT_END_MFLINE	**ESC MFB b d** **[ESC MFB2** b h <TaxID>**] ESC MFE**	New optional parameter for customer CIF number.
ARTICLE_SELL	**ESC MFB f a** **ESC MFB1 b a** <ArticleName> **ESC MFB1 a a** <SinglePrice> **ESC MFB1 d d** <Quantity> **ESC MFB1 c a** <VATCategory> **ESC MFB1 b j** <QuantityUnit> **ESC MFB1 d j** <Article Category> **[ESC MFB2 b e** <PrintLinePost>] **ESC MFE**	Two code variant available (here code 2), Code 1 is compatible to the old version.
PAYMENT_IN	**ESC MFB o a** **ESC MFB1** g a <PaymentType> **ESC MFB1** g b <PaymentValue> **ESC MFE**	New command group.
PAYMENT_IN_VOID	**ESC MFB o c** **ESC MFB1** g a <PaymentType> **ESC MFB1** g b <PaymentValue> **ESC MFE**	New command group.
PAYMENT_OUT	**ESC MFB o b**	New command group.

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	ESC MFB1 g a <PaymentType> **ESC MFB1** g b <PaymentValue> **ESC MFE**	
SPECIAL_CASH_IN	**ESC MFB i** a **ESC MFB1** a g <CashValue> **ESC MFB1** f p <CashDrawBalanceType> **ESC MFE**	New command group.
SPECIAL_CASH_OUT	**ESC MFB i** b **ESC MFB1** a g <CashValue> **ESC MFB1** f p <CashDrawBalanceType> **ESC MFE**	New command group.
SPECIAL_CUT	**ESC MFB i** l **[ESC MFB2** f f <CutType> **ESC MFE**	New command replaces SPECIAL_PRINT_HEADER_CUT
GET_RECEIPT_PAYMENT	**ESC MFB** k z **ESC MFE**	Requests the payment amounts per receipt.
GET_RECEIPT_STATISTIC (in preparation)	**ESC MFB** w t **ESC MFE**	Requests the statistic amounts e.g. CIF amounts per receipt.
GET_DAY_PAYMENT	**ESC MFB** k y **ESC MFE**	Requests the payment amounts per day.
GET_DAY_STATISTIC	**ESC MFB** w u **ESC MFE**	Requests the statistic amounts e.g. CIF amounts per day.
GET_DAY_DRAWER_CONTENT	**ESC MFB** r m **ESC MFE**	Requests the amount of cash drawer balance types per day.
GET_DAY_RECEIPT_TYPE_CNT	**ESC MFB** w j **ESC MFE**	Requests all receipt type counters. Replaces the old command GET_DAY_FISCAL_CNT
GET_DAY_RECEIPT_GROUP_CNT	**ESC MFB** w k **ESC MFE**	Requests all receipt group counters.
GET_DAY_EVENT_CNT	**ESC MFB** w l **ESC MFE**	Requests all fiscal event counters.
GET_MFMEM_LAST_BL_DATE	**ESC MFB** r z **ESC MFE** (replaced ESC MFB r a ESC MFE)	This command requests the date and time for each block type last written to the fiscal memory. **New date and time format.**
GET_MFMEM_LAST_BL_NUM	**ESC MFB** k m **ESC MFE**	
GET_MFMEM_LAST_ LOGICAL_BL_NUM	**ESC MFB** r d **ESC MFE**	This command replaces GET_MFMEM_LAST_BL_ADDRESS

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GET_MFMEM_BL_NUM	**ESC MFB w v** **ESC MFB1 d a <BlockTypeNo>** **ESC MFB1 f d <BlockType >** **ESC MFE**	
GET_MFMEM_BL_HEADER	**ESC MFB w w** **ESC MFB1 d a <BlockNo>** **ESC MFE**	
GET_DATE_TIME	**ESC MFB** w b **ESC MFE ESC + 09h** new answer format: <DateTime>: "DD.MM.YY hh:mm:ss"	The old command **ESC MFB** k k **ESC MFE** with old answer format (<Time>;<Date>: ss,mm,hh;DD/MM/YY) is still supported but not recommended for new implementations.
GET_PAYMENT_LIST	**ESC MFB** r w **ESC MFB1** g a <PaymentType> **ESC MFE**	This command requests the content of one entry in the payment list.
GET_ERROR_LIST	**ESC MFB** k x **ESC MFE**	The separator signs have been changed: New format: ("DD.MM.YYYY hh:mm:ss") Old format: ("DD/MM/YY hh-mm-ss")
GET_FIRMWARE_VERSION	**ESC MFB** w c **ESC MFE** New answer format: ESC s DLH DLL 09H ESC MFB <*FirmwareVersion*> ESC MFE (for the former information *MFCStatusByte1* use **MFC_STATUS** command and for LastError **MFC_SYNC** or **GET_ERROR_LIST**)	This command replaces GET_VERSION. The old command **ESC MFB** r f **ESC MFE** with the old answer format: MFC StatusByte1/last error/firmware version is still supported but not recommended for new implementations.
GET_TKD_INFO	**ESC MFB** r g **ESC MFE**	Requests all fiscal event counters.
GET_HEADER	**ESC MFB** r y **ESC MFE**	This command requests the current receipt header.
GET_SNO_NUMBER	**ESC MFB** w a **ESC MFE**	This command requests the serial number defined by INIT_WRITE_SNO.
GET_DATE_FIRST_RECEIPT	**ESC MFB** r b **ESC MFE**	This command requests the date and time of the first RECEIPT_BEGIN command after DAY_OPEN from the MF controller.
TEST_READ_MFMEM_STATUS	**ESC MFB** l b **ESC MFE**	Sends back the MFMEM status byte.
MFC_STATUS	**ESC + 07h**	Sends back the MFC status. This command is intended to replace the command MFC_INFO (see 1.4.3 Obsolete fiscal commands)
new command group EJ and EJ_GET		

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1.4.3 Obsolete fiscal commands

These commands are not recommended for new implementations. They will not be supported in the future.

Command	Syntax	Comment
MFC_INFO	**ESC + 0Ah**	It is recommended to use the commands: **MFC_STATUS, GET_DATE_TIME, GET_DAY_RECEIPT_GROUP_CNT, GET_MFMEM_LAST_BL_DATE, GET_SNO_NUMBER**
TEST_READ_RTC	**ESC MFB** l g **ESC MFE**	It is recommended to use the command **GET_DATE_TIME**
INIT_HWDATA	**ESC MFB** K A <HWData> **ESC MFE**	It is recommended to use the new syntax: **ESC MFB** s a **ESC MFB1** f n <HardwareData> **ESC MFE**
INIT_FORMAT_MFMEM	**ESC MFB** K B 3 ; 0 **ESC MFE**	It is recommended to use the command **INIT_PRE_FORMAT** and **INIT_FULL_FORMAT**
INIT_WRITE_SNO_BL	**ESC MFB** K C <SystemID> ; <CountrySpecificInfo> **ESC MFE**	It is recommended to use the new syntax for **INIT_WRITE_SNO**: **ESC MFB** d x … **ESC MFE**
SET_HEADER	**ESC MFB** h b ESC MFB1 bb <ReceiptHeader> **ESC MFE**	It is recommended to use the new: **ESC MFB** h b **ESC MFB1** b l <ReceiptHeaderLine1>… **ESC MFE**
SET_TIME	**ESC MFB** l <TimeDate> **ESC MFE**	It is recommended to use the new syntax for **SET_DATE_TIME**: **ESC MFB** h e **ESC MFB1** e k <DateTime> **ESC MFE**
GET_DATE	**ESC MFB** k k **ESC MFE**	It is recommended to use the new syntax for **GET_DATE_TIME**: **ESC MFB** w b **ESC MFE**
GET_VERSION	**ESC MFB** r f **ESC MFE**	It is recommended to use the commands: **GET_FIRMWARE_VERSION, MFC_STATUS, MFC_SYNC** or **GET_ERRORLIST**
TEST_MFMEM_FORMAT	**ESC MFB** L C **ESC MFE**	It is recommended to use the new syntax: **ESC MFB** l c **ESC MFE**
TEST_READ_RTC	**ESC MFB** L G **ESC MFE**	It is recommended to use the new syntax: **ESC MFB** l g **ESC MFE**
TEST_READ_MFMEM_BL	**ESC MFB** L H <BlockNo> **ESC MFE**	It is recommended to use the new syntax: **ESC MFB** l h **ESC MFB1** d a <*FromValue*> **ESC MFE**
TEST_READ_MFMEM	**ESC MFB** L J <MFMEMPage> **ESC MFE**	It is recommended to use the new syntax:

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		ESC MFB l j **ESC MFB1** d c <GetIndex> **ESC MFE**
TEST_PRINT_SELFTEST	**ESC MFB** L O **ESC MFE**	It is recommended to use the new syntax: **ESC MFB** l o **ESC MFE**
MFC_OPEN_D	**ESC + 0DH** <DISPType> <DISPInterfaceType> <DISPInterfaceNo> <DISPConnection> <PRTInterfaceType> <PRTInterfaceNo>	There is no equivalent Feature available, the attached display is opened automatically by the fiscal controller

1.4.4 Expired fiscal commands

Command	Syntax	Comment
GET_BUTTON	**ESC MFB** k n **ESC MFB1** d c <GetIndex> **ESC MFE ESC + 09h**	No buttons exist.
TEST_SELFTEST	**ESC MFB** L A **ESC MFE ESC + 09H**	Use GET_VERSION.
TEST_GET_ VERS	**ESC MFB** L F **ESC MFE ESC + 09H**	Use GET_VERSION.
TEST_GET_BASIC_VERS	**ESC MFB** L M **ESC MFE ESC + 09H**	
TEST_READ_EPROM	**ESC MFB** L I **ESC MFE ESC + 09H**	
TEST_DISP_INTERFACE	**ESC MFB** L L **ESC MFE ESC + 09H**	
TEST_GET_TKD_INFO	**ESC MFB** L N **ESC MFE ESC + 09H**	Use GET_TKD_INFO.
TRAINING_BEGIN	**ESC MFB** l a **ESC MFE**	use TRAINING_BEGIN
TRAINING_END	**ESC MFB** l b **ESC MFE**	use TRAINING_END
NONFISCAL_BEGIN	**ESC MFB** n a **ESC MFE**	Nonfiscal mode not allowed
NONFISCAL_END	**ESC MFB** n b **ESC MFE**	Nonfiscal mode not allowed

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1.5 Abbreviations

ASB	Automatic Status Back
ASCII	American Standard Code for Information Interchange
BCD	Binary-Coded Decimal
BL	Block
BMP	Bitmap Picture
CHK	Check (file extension)
CIF	Code Identification Fiscal – Customer Tax Number
CIF_ID	Customer Tax ID
CMD	Command
CMOS	Complementary Metal-Oxide-Semiconductor - Memory on the motherboard
CNT	Counter
CPU	Central Processing Unit
CRC16	Cyclic redundancy check - a check for data correctness
CRY	Currency
DH	Digest Header (EJ information)
DISP	Display
EB	Encryption Block
EJ	Electronic Journal
EJC	Electronic Journal Compressed (file extension)
EJD	Electronic Journal Decompressed (file extension)
EOD	End Of Day
EPROM	Erasable Programmable Read Only Memory
ESC	Escape
ESD	Electro Static Discharge
ESFP	
FAT16/FAT32	File Allocation Table (file system)
FFC	Fiscal Family Controller (printer type designation)
FM	Fiscal Modul
FW	Firmware
HRI	Human Readable Interpretation of barcode (Plain Text)
HWDATA	Hardware Data
KE	Public Key Exponent (encryption)
KM	Public Key Module (encryption)
LED	Light-Emitting Diode
LEI	Romanian currency code
LF	Line Feed
MD	Message Digest (EJ information)
MF	Misuratore Fiscale (fiscal measuring device)
MFC	MF Controller
MFMEM	MF Memory
MF-THF	Module Fiscal Thermo Printer (Fiscal)
MMC	MultiMediaCard
NV-RAM	Non-Volatile RAM
OPOS	OLE for POS (Object Linking and Embedding for POS)
PC_	Payment Category
PDF	A type of 2-dimentional barcode

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PFM	Protected Fiscal memory
POS	Point of Sale
PRT	Printer
PS	Padded String
PS_	Payment Status
PT_	Payment Type
RAM	Random Access Memory
ROM	Read Only Memory
RSA	Smart card public key cryptographic algorithms
RTC	Real Time Clock
SC	Smart Card - does the signature for the EOD
SD	Secure Digital Card (mass storage)
SG	Digital signature for a smart card record
SHA-2	Secure Hash Algorithm 2 (encryption 256 bit)
SNO	Serial Number
TH	Ticket Header
TKD	Technical Service
TT	Tax Terminal
UMID	Unique Medium Identifier
VAT	Value Added Tax
WN	Wincor-Nixdorf

2 OVERVIEW

2.1 Purpose of this document

This specification describes the functionality of the POS printer TH230+ in connection with the Fiscal Family Controller (FFC) for Romania.

This document can be seen as supplemental information to the programmer's guide for the development of a BEETLE application, an OPOS service object or a driver based on the printer unit. The fiscal firmware structure and the detailed description of the MF commands are included in this document.

2.2 Introduction

To ensure compliance with national tax laws, some countries have legal descriptions of how to administrate and store tax relevant sales data in retail environments.

The solution to these requirements is a hardware and software extension of the standard.

The communication path within the printer firmware (between communication interface and printer) will be intercepted by the fiscal firmware unit. The data flow to the printer is analysed and it is guaranteed that all inadmissible print commands are rejected; the tax relevant data are collected and finally - at the end of day - unerasably stored in the Fiscal Memory (PFM).

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The storage capacity of the MF memory (PFM) extends up to at least 5 years. Due to fiscal security aspects the MF memory is inaccessible and is embedded in resin during the assembly process.

2.3 Country ID's and codepages

Country	Letter ID	FW ID	Codepage
Italy	I	01	437
Greece	GR	02	813
Hungary	HUN	03	852
Turkey	TR	04	857
Poland	PL	05	852
Czech Republic	CZH	11	
Russia	RUS	13	
Romania	RO	14	852
Bulgaria	BUL	15	866

3 MF MEMORY (MFMEM)

3.1 General

The fiscal memory is a PFM (**P**rotected **F**iscal **M**emory) with 4 Mbytes capacity.

The hardware data block (HWDATA) is located at address 0x000000 of the MFMEM and consists of 256 bytes for hardware information, which will be programmed during the manufacturing process. The first four bytes of the HWDATA reflects the status of the fiscal memory and is called MFMEM Status.

The fiscal area holds information about all fiscal relevant operations. It is organized in 16.384 blocks with a size of 256 bytes. The first byte of each user block reflects the status of this block and is called MFMEM Block Status (see chapter 3.4 **Block handling (logical and physical blocks)** for more details). To store more data than fits into one block, the physical block(s) will be grouped to a logical block. More details can be found in chapter 3.4.

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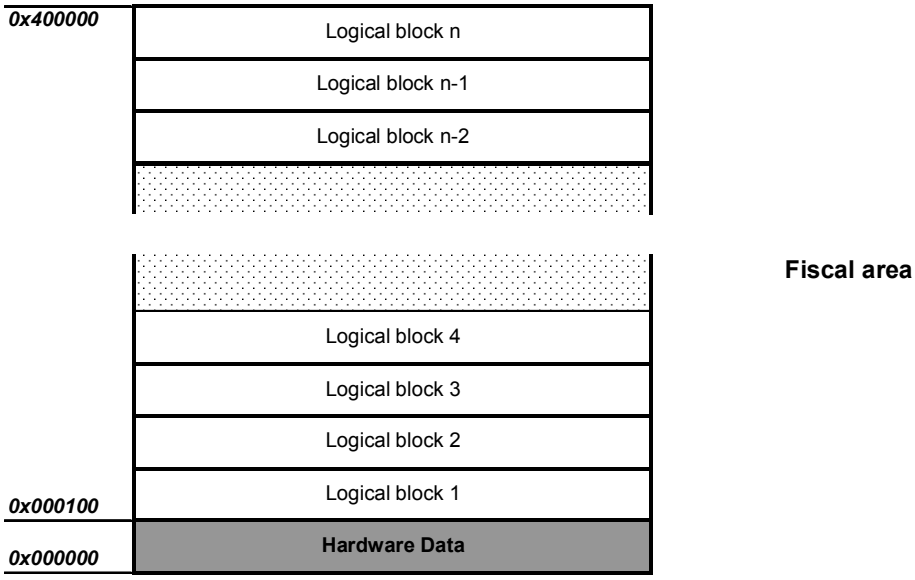

3.2 Hardware data (HWDATA)

The hardware data contains important system information which is necessary for normal operation. It also allows the correct interpretation of the fiscal memory data outside a fiscal printer.

The structure of the HWDATA is as follows:

Address	Size	Name	Description	Value
0x00	4	MFMEM status	Overall status of the fiscal memory	see 3.3 MFMEM status
0x04	2	Block length	Length of a physical block in bytes	0x0100 (256)
0x06	2	Number of blocks	Number of possible physical blocks	0x4000 (16384)
0x08	2	EPROM size	Size of EPROM in Kbyte	0x1000 (4096)
0x0A	8	Test pattern	Test pattern for verification	0x01 0x03 0x07 0x0F 0x1F 0x3F 0x7F 0xFF
0x12	1	Printer type	Type of printer (FFC/TH230+)	0x3D
0x13	1	Block structure	Identifies the kind of MFMEM organization and the structure of the master/slaves blocks	0x01
0xFF	236	Unused	Reserved	0x00

3.3 MFMEM status

The first byte in the MFMEM (physical address 0x000000) represents the MFMEM Status. This status gives detailed information about the actual state of the MFMEM.

7	6	5	4	3	2	1	0

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MFMEM_ DEFECT	MFMEM_ FULL	MFMEM_ NEAR_FULL	MFMEM_ VAT_BURNED	MFMEM_ SNO_BURNED	MFMEM_ FORM_END	MFMEM_ FORM_BEG	MFMEM_ HWDATA

Bit	Function	Value	
		0	1
0	MFMEM_HWDATA	Hardware data not present	Hardware data present
1	MFMEM_ FORM_BEG	Format process not started	Format process started
2	MFMEM_FORM_END	Format process not finished	Format process finished Memory is completely formatted
3	MFMEM_SNO_BURNED	No serial number block (SNO) present	Serial number block (SNO) present
4	MFMEM_VAT_BURNED	No VAT block present	VAT block present
5	MFMEM_NEAR_FULL	Memory for more than 60 EOD blocks available	Memory for less or equal than 60 EOD blocks available
6	MFMEM_FULL	Memory is not full	Memory is full (MFC is blocked)
7	MFMEM_DEFECT	Memory is valid	Memory is defect (MFC is blocked)

3.4 Block handling (logical and physical blocks)

To allow the storage of more data than fits in one physical block, the data will be split into multiple physical blocks. The resulting physical blocks will be grouped to a logical block.

The first physical block of the logical block is called the master block and all following physical blocks are slave blocks. It is possible for a logical block to consist of only one physical block. The grouping of the blocks is represented by the bits MFMEM_BL_MASTER and MFMEM_LAST_BL in the MFMEM block status.

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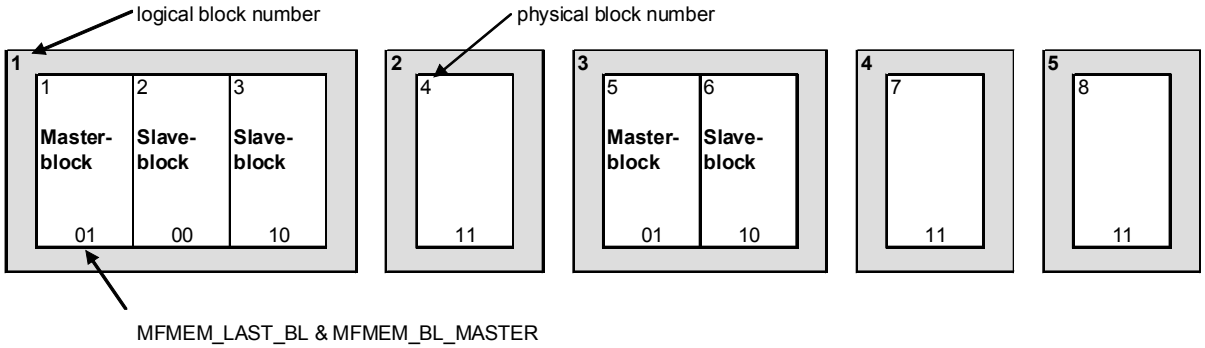
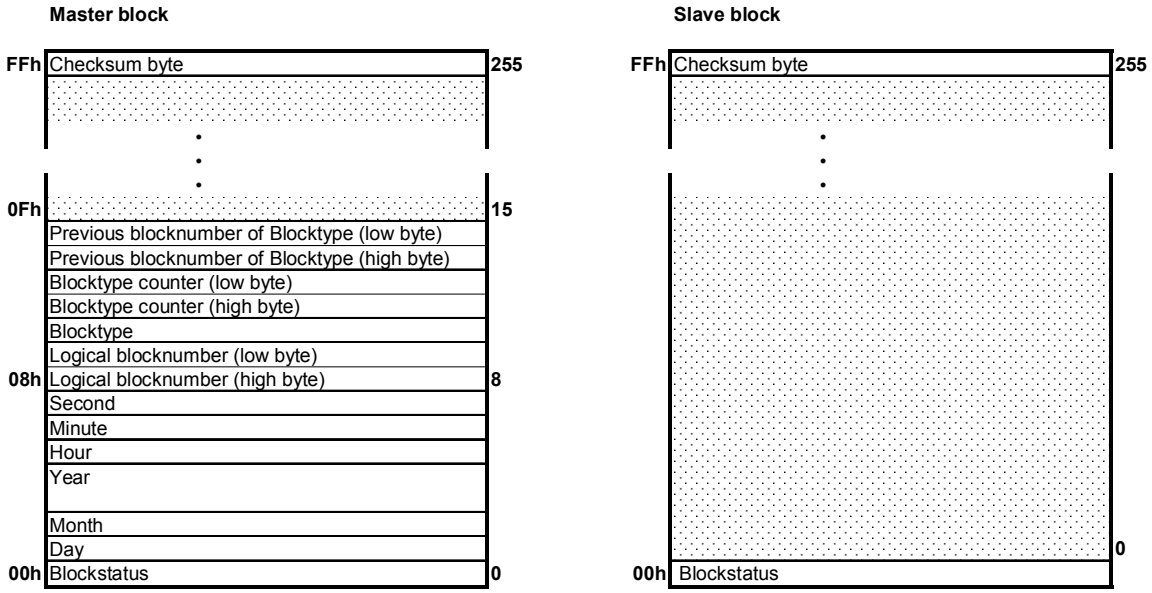


Every physical block has a defined header. This header is different for a master and a slave block. Therefore a slave block can carry more payload data then a master block. A master block can carry up to 242 bytes and a slave block up to 254 bytes of data.

The header of a master block consists of the physical block status, the creation date, the logical block counter, the blocktype and the blocktype counter. The header of a slave block consists just of the physical block header. The last byte of a physical block is always the block checksum (XOR of all data from 0x00 to 0xFE).

The definitions and descriptions of the content of the different blocktypes are described in chapter *5.7* Logical block types



Element	Description	Master	Slave
Checksum	Checksum over the complete physical block	X	X

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Blocktype counter (low/high)	Number of blocks of the kind specified by ***blocktype*** in the MFMEM	X	
Blocktype	Specifies the blocktype (e.g: EOD / TKD…)	X	
Logical blocknumber (low/high)	Number of the logical block	X	
Year / Month / Day	BCD coded creation date of logical block	X	
Blockstatus	Status of physical block (refer to chapter 3.5 MFMEM physical block status" for more details)	X	X

When reading a logical block from memory, the block checksum will be verified. If this verification fails the block will be read out 2 more times. If no block can be read out with the correct checksum verified, a hard blockade is set. In this case it is very possible that some bits in MF memory are skipped and that the printer can not be guaranteed to be working correctly anymore. If the second reading failed again with other result, a third reading attempt is started and in case of an error, a blockade is set since there could have been ESD problems. This blockade can be removed by executing a TKD (see chapter 10 SERVICE RESET (TKD RESET)) reset.

3.5 MFMEM physical block status

The first byte of every physical block is the block status. It gives detailed information about the status of this physical block.

7	6	5	4	3	2	1	0
MFMEM_ BL_DEFECT	Reserved	Reserved	MFMEM_ LAST_BL	MFMEM_ BL_MASTER	MFMEM_ BL_BURN_END	MFMEM_ BL_BURN_BEG	MFMEM_ BL_FORM

Bit	Function	Value	
		0	**1**
0	MFMEM_ BL_FORM	Block is not formatted	Block is formatted
1	MFMEM_BL_BURN_BEG	Block is empty	Block burning in progress
2	MFMEM_BL_BURN_END	Block burning not finished	Block burning finished

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3	MFMEM_BL_MASTER	Slave block	Master block
4	MFMEM_LAST_BL	One or more physical blocks are following	Last physical block
5	Reserved	-	-
6	Reserved	-	-
7	MFMEM_BL_DEFECT	Block is valid	Block is faulty

4 ELECTRONIC JOURNAL (EJ)

4.1 General

According to the Romanian fiscal law all fiscally relevant data has to be registered on a suitable non-paper support (electronic journal). The following requirements have to be fulfilled:

- Electronic journal data must not be altered or alterable.
- In the absence or malfunction of the electronic journal, the printer must stop working.
- Every electronic journal must be initialised with a unique ID.
- It must be possible to match the data with the device (printer) which stored this data.
- The printer application has to provide reading and printing methods to restore the electronic journal data: between two dates, by receipt number and between two receipts (printer + PC application)
- The printout of the entire contents of the electronic journal and the transmission of the electronic journal data to another computer by means of “standard methods” must be possible.
- Information about the remaining space for data storage on the electronic journal and indication of full and nearly full status must be available.
- Electronic journal errors or the removal of the electronic journal will result in the registration of the error and suspension of printer functionality.
- The electronic journal medium (card) must be placed inside of the controller under the seal protection, as per fiscal law.

4.2 Electronic journal medium

The physical electronic journal medium is a SecureDigital Memory Card (refer to [R4] SD Specifications Part 1 Physical Layer Simplified Specification Version 2.00 September 25, 2006
).

The SD memory Card is a very small, non-volatile storage medium with a capacity of up to 32 GByte.

The SD card must be at least 8 GB in capacity and formatted by Wincor Nixdorf, that way we can ensure that only verified card types are used.. The card will be reformatted during the initialization process of the fiscal firmware.

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4.3 Electronic journal organization

The SD card is formatted as a volume separated in two partitions during its initialization.

The first partition is formatted with FAT32 file system structure and the second partition is formatted with FAT16 file system structure.

The first partition contains the Electronic Journal files as defined by the fiscal law.

The second partition is a transaction file containing a record for every ticket registered in the current EJ.

The EJ file contains all data printed by the MFC during a given EJ session i.e. between a fiscal day open and a fiscal day close. The integrity of EJ data is ensured by the a SHA256 checksum of the EJ file.

Applying the SHA-2 hashing algorithm with 256 bit to the data printed on the receipt station and recorded to the electronic journal media, a message digest has to be calculated. The message digest will be recalculated any time new data is printed and recorded.

The content of the second partition takes a role in the RAM restoration of the TKD procedures.

4.4 Electronic journal status

The electronic journal status consists of two bytes; each with eight independent status bits. This status reflects the current physical and logical state of the electronic journal medium. The EJ status can be obtained with the command EJ_GET_STATUS (see chapter 7.17.1 EJ_GET_STATUS)

15	14	13	12	11	10	9	8
MEDIUM_ AUTHORIZED	MEDIUM_VALID	Reserved	Reserved	MEDIUM_PRESE NT	WRITE_PROTEC TED	Reserved	Reserved

7	6	5	4	3	2	1	0
Reserved	MEDIUM_END	MEDIUM_FULL	MEDIUM_NEAR_ FULL	DATA_WRITE	TRACK_OPEN	SESSION_OPEN	MEDIUM_ INITIALIZED

Bit	Function	Value	
		0	1
0	MEDIUM_INITIALIZED	The current EJ medium is ***not*** initialized for use as a	The current EJ medium is initialized for use as a

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		Wincor Nixdorf EJ medium.	Wincor Nixdorf EJ medium.
1	SESSION_OPEN	No session open	Session is open
2	TRACK_OPEN	No track open	Track is open
3	DATA_WRITE	A write cycle to the EJ medium is ***not*** in process	A write cycle to the EJ medium is in process
4	MEDIUM_NEAR_FULL	The remaining free capacity of the EJ medium has ***not*** reached the near full limit	The remaining free capacity of the EJ medium has exceeded the near full limit
5	MEDIUM_FULL	The EJ medium is ***not*** full	The EJ medium is full. No more write operations to the medium will be possible. It is possible to issue a DAY_END Z report
6	MEDIUM_END	The EJ medium is **not** full	The EJ medium is completely full. No more write operations to the medium will be possible
7	Reserved	-	-
8	Reserved	-	-
9	Reserved	-	-
10	WRITE_PROTECTED	The EJ medium is ***not*** physically write protected	The EJ medium is physically write protected
11	MEDIUM_PRESENT	There is ***no*** EJ medium present	An EJ medium is present
12	Reserved	-	-
13	Reserved	-	-
14	MEDIUM_VALID	The EJ medium is ***not*** a valid Wincor Nixdorf EJ medium	The EJ medium is a valid Wincor Nixdorf EJ medium
15	MEDIUM_AUTHOTRIZED	The EJ medium is ***not*** the authorized medium for normal fiscal operations (write access not allowed)	The EJ medium is the authorized medium for normal fiscal operations (write access allowed)

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4.5 Unique medium identification (UMID)

The unique medium identifier is used to clearly identify an electronic journal medium independently of its physical medium type. The UMID will be provided by the fiscal firmware and uses medium type specifics to build this ID.

The UMID is 20 characters long. The first character specifies the medium type. The following 19 characters can be any alphanumeric character. (e.g. `SXXXXXXXXX03201C4D64`)

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Medium type	Alphanumeric character																		

The following list shows all possible medium types

- S SD card
- C Compact flash
- I Internal flash memory

5 FISCAL FUNDAMENTALS

5.1 Fiscal Status

The fiscal status consists of two byte (MFC_STATUSBYTE_1 and MFC_STATUSBYTE_2), each with eight independent status bits. This status can be requested by the application.

5.1.1 Request the fiscal status

The status information presents the current status of the fiscal device at the time of the request, and can be read by the application with the command MFC_STATUS (Chapter 7.20.1) (buffered command, not available in real-time condition). If the Automatic Status Back (ASB) is enabled by the command <GS> a *n,* any change of the status will be sent automatically.

5.1.2 MFC status byte 1

Bit	Value	Description	Related Commands and Notes
0 – MFC_BLOCKADE	0	MFC system is in normal operation mode.	
	1	MFC system is blocked due to a severe error (service call needed).	
1 – MFC_TRAINING_MODE	0	Normal operation.	TRAINING_END
	1	Training mode active.	TRAINING_BEGIN
2 – MFC_DAY_OPEN	0	Fiscal day is closed.	DAY_END
	1	Fiscal day is opened.	DAY_BEGIN
Whx 3 – MFC_RECEIPT_OPEN	0	Fiscal receipt is closed.	RECEIPT_END RECEIPT_END_MF_LINE SPECIAL_ALL_VOID
	1	Fiscal receipt is opened.	RECEIPT_BEGIN

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Bit	Value	Description	Related Commands and Notes
4 – MFC_REPORT_OPEN	0	Fiscal report is closed.	The status bit will be cleared when the REPORT_ command is finished. SPECIAL_ALL_VOID
	1	Fiscal report is opened.	DAY_XREPORT REPORT_BLOCK2BLOCK REPORT_DATE2DATE REPORT_EOD_BLOCK2BLOCK REPORT_TOTAL_DATE2DATE
5 – MFC_DOCUMENT_OPEN	0	Fiscal document is closed.	
	1	Fiscal document is opened.	There is no command to set this status in Romania.
6 – MFC_RECEIPT_TOTAL	0	Fiscal receipt is opened.	RECEIPT_END SPECIAL_ALL_VOID
	1	Total sum is already valid and the total line has been printed.	TOTAL_LINE
7 – MFC_TKD_RESET	0	No service reset active.	Will be cleared automatically if the service reset is completed correctly.
	1	A service reset is currently active.	Set when a service reset is executed.

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5.1.3 MFC status byte 2

Bit	Value	Description	Related Commands and Notes
0 – MFC_FISCAL_MODE	0	The fiscal mode is inactive.	• INIT_HWDATA • INIT_FULL_FORMAT
	1	The fiscal mode is set.	• INIT_WRITE_SNO
1 – MFC_PRINTOUT_INTERRUPTED	0	Normal operation without power fail or printer error	SPECIAL_ALL_VOID
	1	Error during printing operation. A complete printout cannot be guaranteed.	The status bit MFC_PRINTOUT_INTERRUPTED will be set if a complete printout of a memory report or receipt is interrupted by a power fail or a printer error. In such events the system cannot guarantee the printed output. To go on with fiscal printing it is necessary to send the command SPECIAL_ALL_VOID, thus cancelling the operation.
2 – MFC_CMD_INTERRUPTED	0	Normal operation without interrupted command.	The status bit MFC_CMD_INTERRUPTED will be cleared if the interrupted command is executed completely.
	1	One command has been interrupted due to power fail or due to an error condition.	The status bit MFC_CMD_INTERRUPTED will be set if a fiscal printout is interrupted by a power fail or a printer error (carrier movement or home detection error). The only command to go on with fiscal printing is to send `SPECIAL_CMD_RESTART` (Chapter 7.12.6). Which command was interrupted could be read by the application with the command `MFC_READ_LAST_INTERRUPTED_CMD` (Chapter 7.20.6). The status bit MFC_CMD_INTERRUPTED will be cleared if the interrupted command is executed completely.
3 – EJ_NEAR_FULL	0	The EJ memory is not near full.	
	1	The EJ memory is near full. If less than one average sales receipt (max. 2.200 print lines) additional to EJ full condition.	This is an information bit only that the full condition will appear soon. No immediate action is needed.
4 – EJ_FULL	0	The EJ memory is not full.	

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Bit	Value	Description	Related Commands and Notes
	1	The EJ memory is full. In all cases it is guarantee that the last Z-Report is correct finished and stored in EJ (this is down by reserving of 500 print lines).	Only dedicated actions are possible: MFC_RECEIPT_OPEN = 1: TOTAL_LINE, RECEIPT_END or SPECIAL_ALL_VOID commands MFC_RECEIPT_TOTAL = 1: RECEIPT_END or SPECIAL_ALL_VOID commands MFC_DAY_OPEN = 1: DAY_END command MFC_REPORT_OPEN = 1: SPECIAL_ALL_VOID
5 – EJ_DATA_PENDING	0	No EJ data to be transferred.	Not used in Romania.
	1	EJ data has to be transferred to the host.	
6 – Unused	0		
	1		
7 – Unused	0		
	1		

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5.2 ASB Handling (MF ASB)

The **A**utomatic **S**tatus **B**ack function is a feature which is available in printer designs. If ASB is enabled, the printer issues information when an error or a change of the printer status occurred (paper status, printer status, printer errors). With ASB, the application does not have to poll the printer to get information current status information, for example about presence of paper.

The MF controller also generates its own ASB. They are issued if an MF error occurred or the MF status has changed.

The following is a description of the ASB handling in the MF controller.

Default:

Printer ASB off

MFC ASB off (however the MFC sends a single ASB with the error EM_POWER_FAIL after power-on)

Printer initialization with <ESC> "@" sets the default to:

Printer ASB off

MFC ASB off

Sending of <GS> "a" 00H sets the default to:

Printer ASB off

MFC ASB off

Sending of <GS> "a" n (n != 00H) sets the default to:

Printer ASB on

MFC ASB on

Sending the command **<GS> a nH** (n != 00H) leads to seven bytes of MF ASB information followed by four bytes printer ASB information. The content of the printer ASB is described in the TH230+ Programmers Guide

If the MF Controller is switched on with a power-on error (for example battery empty), it sends the first ASB immediately after power-on:

1. ASB: EM_POWER_FAIL

The second ASB is send, when the MF controller receives the command **<GS> "a" n** (n != 00H):

2. ASB: EM_RTC_BATTERY_EMPTY

Because of this, the error message “battery empty” is not lost, if MF printer and host system are switched on together.

<u>Note:</u>

If the MF Controller switches on the printer ASB, it sends **<GS> a FFH** to the printer. This enables all ASBs.

The complete **MF ASB** message is as follows:

1st byte:	ESC
2nd byte:	'm'
3rd byte:	MFC error

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4th byte:	MFC_STATUSBYTE_1
5th byte:	MFC_STATUSBYTE_2
6th byte:	CNT_FISCAL_TICKET low byte (ticket counter)
7th byte:	CNT_FISCAL_TICKET high byte (ticket counter)

More details about the printer ASB can be found in the TH230+ Programmers Guide

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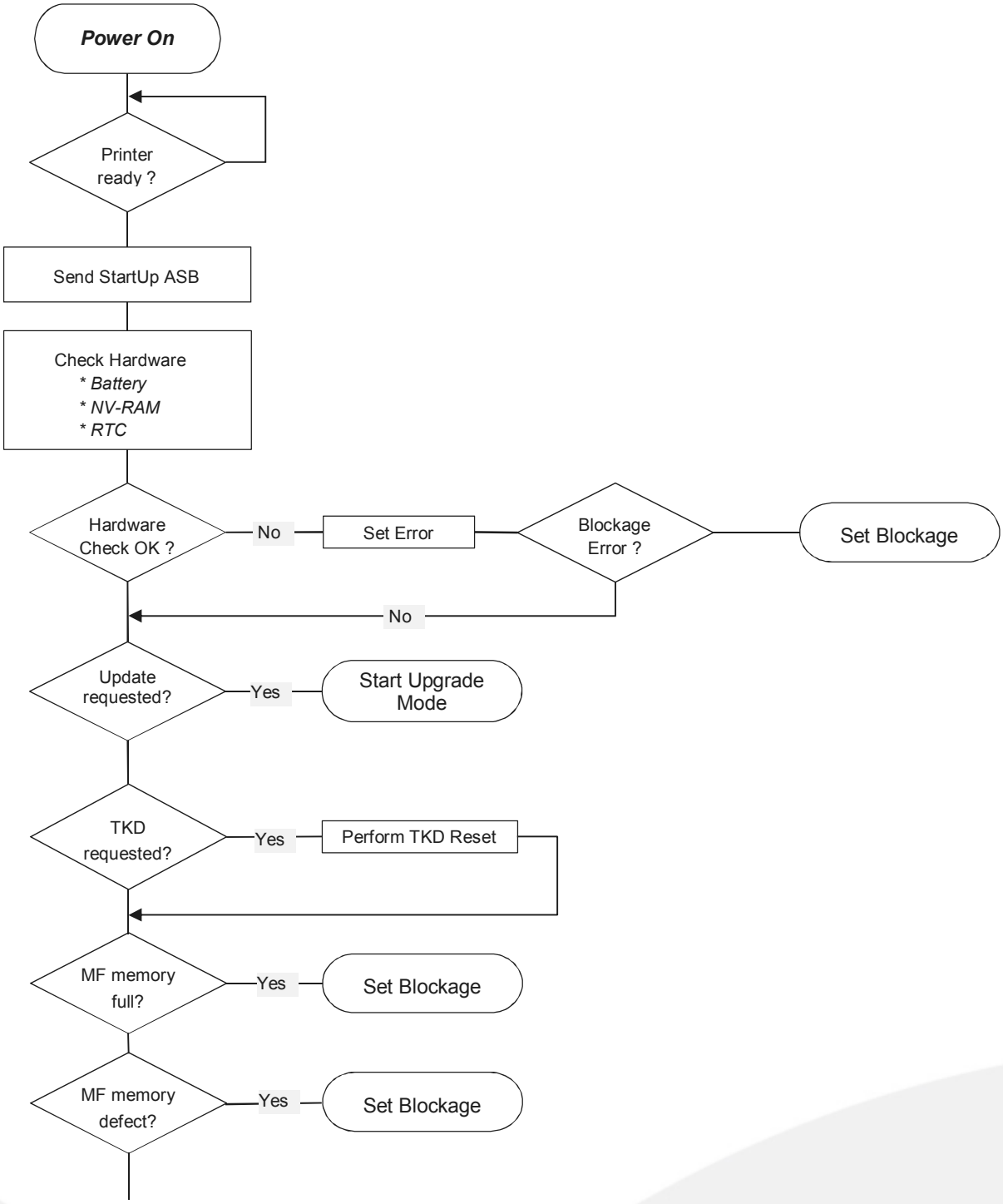
5.3 Fiscal Firmware (Flow chart)

5.3.1 Power On procedure



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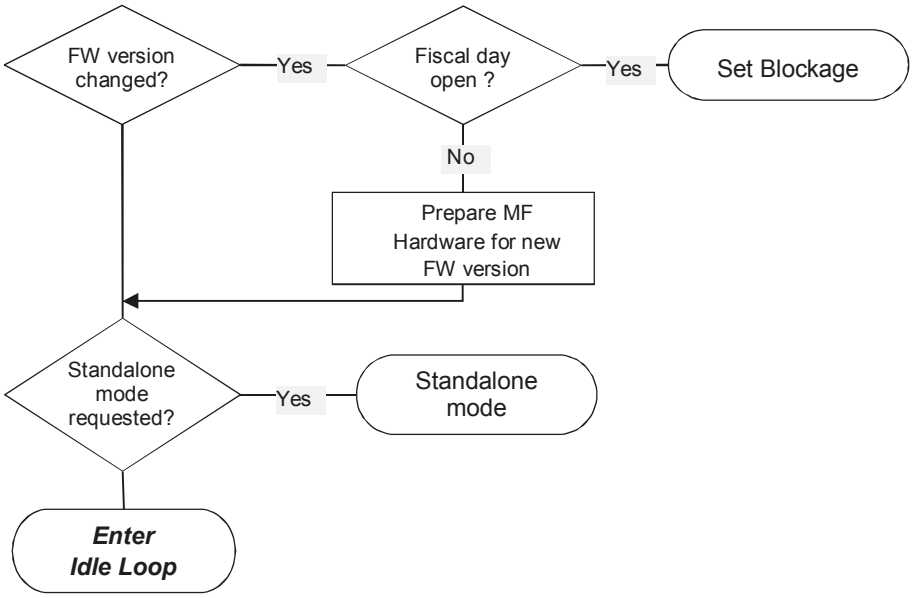

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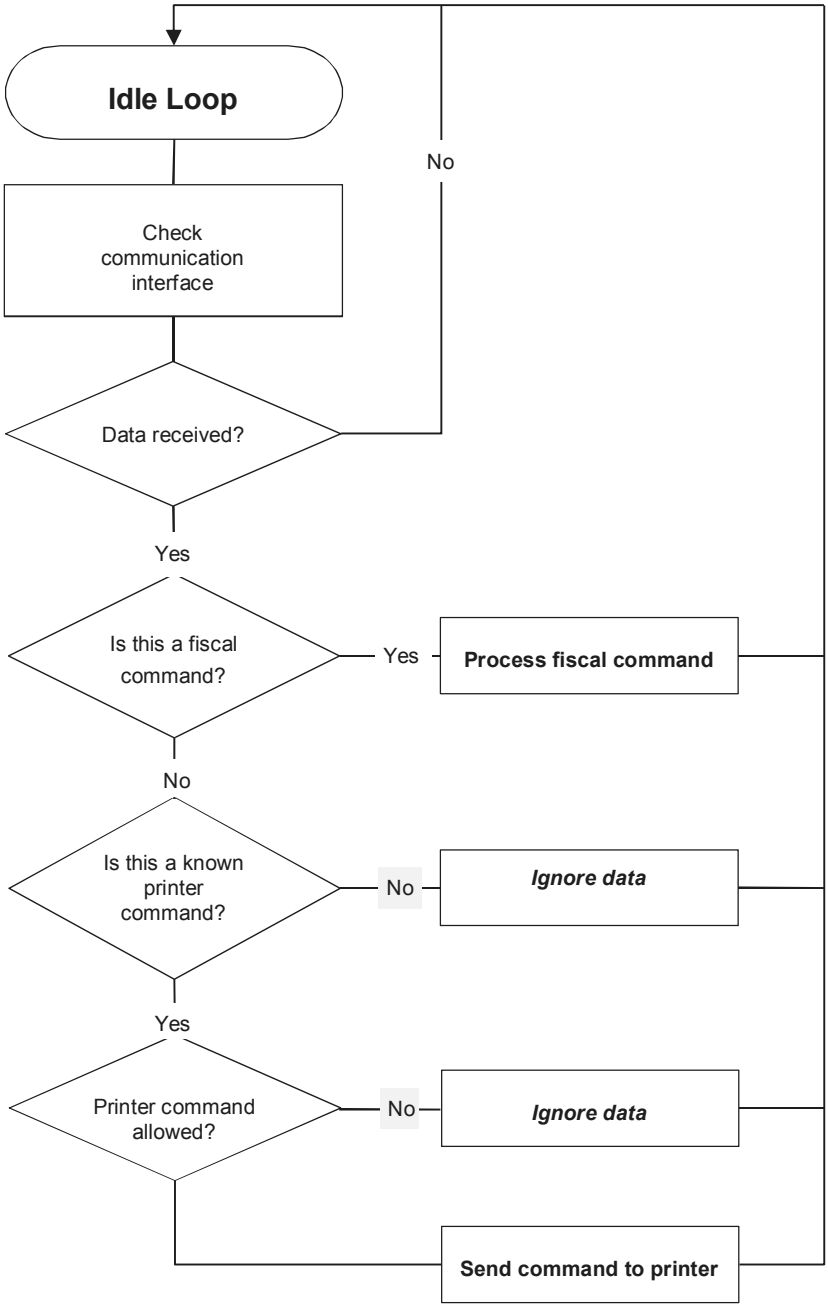
5.3.2 Idle Loop

After the successful power on procedure, the fiscal firmware starts waiting for data from the communication interface. The fiscal firmware resides in this state until new data has been received.



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5.4 Receipt Types

5.4.1 General

Receipt printing is processed according to the following rules:

Each receipt includes a header information part, a receipt body part and a footer part. A correct receipt must always be cut by the printer cutter.

There are three different receipt types: legal receipts, illegal receipts and administrative receipts. “Legal receipts” (RETAIL_SALES_RECEIPT, RETAIL_SALES_RECEIPT_WITH_TAX_ID) are official income documents and increment the CMOS totalizers. “Illegal Receipts” (GENERIC_RECEIPT, CASH_IN, CASH_OUT, PAYMENT_CORRECTION) have no fiscal footer but are also stored in EJ. “Informative tickets” print information (e.g. after SET_ commands).

After a legal receipt is started, it can be turned illegal (e. g. SPECIAL_ALL_VOID) instead of ending it legally. If an illegal receipt type is started, it can never be converted to a legal receipt.

Receipt types used in Romania:

	Description	Receipt Type	EJ registered
“1”	Retail Sales Receipt	Legal	Yes
“2”	Generic Receipt	Illegal	Yes
“3”	Cash in Receipt	Illegal	Yes
“4”	Cash out Receipt	Illegal	Yes
“5”	Retail Sales Receipt with tax ID	Legal	Yes
“6”	Payment Correction	Illegal	Yes

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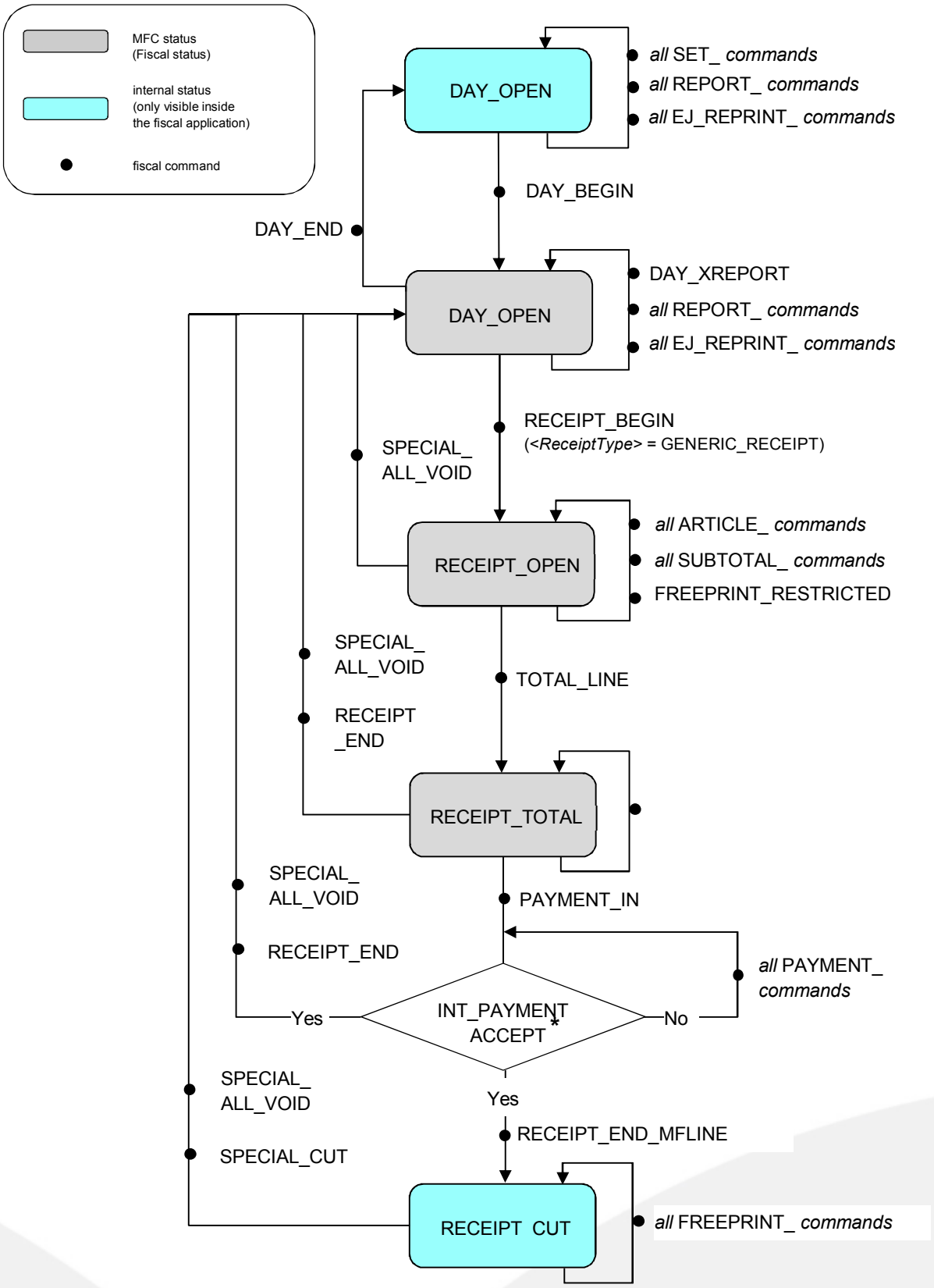
5.4.2 Retail sales receipt (legal receipt)



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```
* SUM(all PAYMENT_IN_ commands) – SUM(all PAYMENT_OUT_ commands) -
RECEIPT_TOTAL_SALES  ==  0 ?
```

For a “Retail Sales Receipt” (Legal Receipt) to be closed as fiscal, the following rules must be met: The parameter <*TotalValue*> in TOTAL_LINE command must be equal to the ticket totalizer in NVRAM of MFC.

All items specified in the “Data Storage” table have been successfully updated.

Printing is concluded with no errors. If any of these aspects are not met, the RECEIPT_END command fails with an error code. The only alternative is to issue a SPECIAL_ALL_VOID or SPECIAL_CMD_RESTART (Chapter 7.12.6)command.

Data Storage

To close a “Retail Sales Receipt” (Fiscal Receipt), the following CMOS data will be updated.

Field ID	Accumulates	Short Explanation
R1	BRUTTO_SALES	Array with day totalizers: Day.Tot[Rx]
R2	VOID	
R3	DISCOUNT	
R4	SURCHARGE	
R5	RETURN	
VA to VH	VAT_KA..D, G, H (KE is always zero, KF is reserved for Road Tax)	Array with VAT amounts: Day.VAT[ ]
C1	Legal receipt counter is incremented by 1.	Array with day counters: Day.Cnt[LEGAL_RECEIPT]
D1	Retail sales receipt counter is incremented by 1.	Array with day type counters: Day.CntType[RETAIL_SALES_RECEIPT]
BA...BH	BRUTTO_SALES according to VAT categories and Road Tax (BF)	Array with all accumulated brutto amounts: Day.BruttoSales[Bx]

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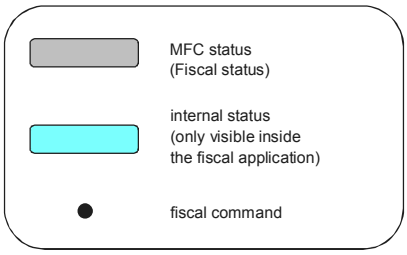
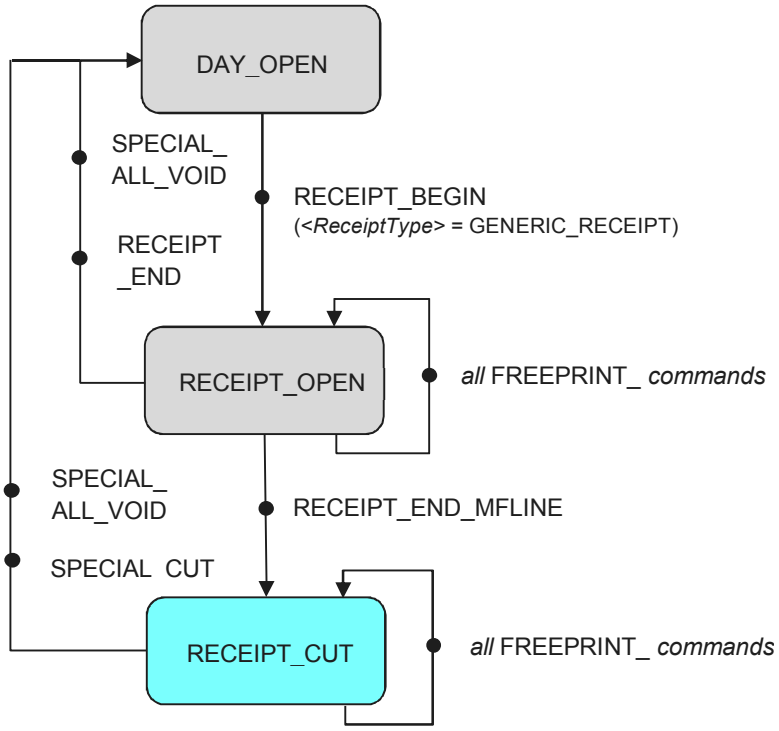
5.4.3 Generic receipt (illegal receipt)





This receipt type will be used by the application to print reports and other informational documents, which are not “fiscal documents”.

The application begins printing by issuing a RECEIPT_BEGIN command and providing *<ReceiptType>* = GENERIC_RECEIPT.

This receipt type is comprised only of FREEPRINT_NORMAL commands (see 7.11.2 FREEPRINT_NORMAL) which can be used to print all information. If fiscal commands are sent when an illegal receipt is open, it will be rejected with EM_CMD_WRONG_ORDER.

Data Storage

If a GENERIC_RECEIPT is closed (always as non-fiscal [illegal]), the following CMOS data will be updated:

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Field ID	Accumulates	Short Explanation
C2	Illegal receipt counter is incremented by 1.	Array with the day counters: stDay.auiCnt[ILLEGAL_RECEIPT]
D2	Generic receipt counter is incremented by 1.	Array with the day counters: stDay.auiCntType[GENERIC_RECEIPT]

Appearance

This example shows how FREEPRINT_NORMAL can be used to print information in a GENERIC_RECEIPT.

```
  ****** W I N C O R - N I X D O R F ******

            receipt header line 2

            receipt header line 3

            receipt header line 4

      Cod Fiscal C.I.F.: 12345678901234


****  BON NEFISCAL  ****  BON NEFISCAL  ****

ORA                                     SUMA

08:00 - 09:00                       1,239,87

09:01 - 10:00                       4,537,56

10:01 - 11:00                      12,347,35

11:01 - 12:00                      25,692,34

12:01 - 13:00                      20,983,03

14:01 - 15:00                      13,240,03

15:01 - 16:00                       9,242,21

                                -------------

TOTAL                             87.282,39

 TVA A  00,00%                          0,00

 TVA B  19,00%                          0,48

 TVA C  32,25%                          7,00

 TOTAL ALTE TAXE F                      1,47


EJTRZ: 00002

ID UNIC:  BU4710123420170626120000000010001

BNF.00001    DATA:15/09/2008    ORA:15-12-21

               RL   1234567890

****  BON NEFISCAL  ****  BON NEFISCAL  ****
```

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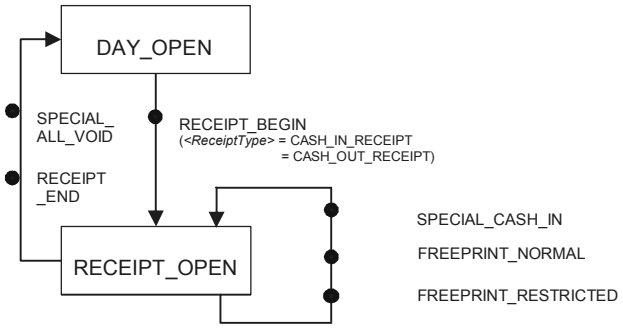
5.4.4 Cash in/out receipts (illegal receipt)



This receipt type will be used to manage the cash drawer content (e.g. Paid In/Out, Pick-Up, Initial and Extra Float)

This receipt type is comprised only of SPECIAL_CASH_IN, FREEPRINT_NORMAL and FREEPRINT_RESTRICTED commands.

Data Storage

Field ID	Accumulates	Short Explanation
D1	CASH_IN_DRAWER	Array with day totalizers: Day.CashDrawBalance[Rx]
D2	INITIAL_FLOAT	
D3	EXTRA_FLOAT	
D4	PICK_UP	
D5	PAID_IN	
D6	PAID_OUT	
C2	Illegal receipt counter is incremented by 1.	Array with the day counters: Day.auiCnt[ILLEGAL_RECEIPT]
D3/4	Cash In/Out counter is incremented by 1.	Array with the day counters: Day.CntType[CASH_IN/OUT]

5.4.5 Retail sales receipt with tax ID

With this ticket type the Retail Sales Receipt has an additional line with Customer tax ID as shown below:

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```
****** W I N C O R - N I X D O R F ******
          receipt header line 2
          receipt header line 3
          receipt header line 4
     Cod Fiscal C.I.F.: 12345678901234

            B O N  F I S C A L
                                    Lei
     1,000 MMM x         0,20
Service                            0,20 A
     1,000 MMM x         3,00
Motor Oil                          3,00 B
Regular Fuel                      28,70 D
   36,78800 L   x         0,78
TAXA DRUM Regular Fuel             1,47 F
   36,78800 L   x         0,04
                              -----------
TOTAL                             33,37

 TVA A  00,00%                      0,00
 TVA B  19,00%                      0,48
 TVA C  32,25%                      7,00
 TOTAL ALTE TAXE F                  1,47

C.I.F. Client: TAX_ID113344

EJTRZ: 00001
ID UNIC:  BU4710123420170626120000000010001
BF. 00001    DATA:15/09/2008    ORA:15-12-21
            RL   1234567890
            B O N  F I S C A L
```

The rest of the flow follows the normal retail sales receipt.

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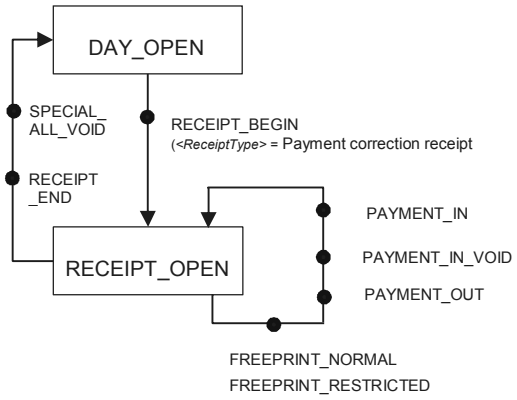
5.4.6 Payment correction receipt



This receipt type will be used to correct the transaction with different payment varieties.

This receipt type is comprised only of PAYMENT_IN, PAYMENT_IN_VOID and PAYMENT_OUT commands.

The PAYMENT_IN accepts all valid payment types like Cash, Credit etc. PAYMENT_OUT can only be carried out under the condition that resulting sum should not be less than zero and if the CASH type is preferred for out payment then it should not be lesser than what Cash Drawer holds.

Appearance

This example shows the layout of some fiscal lines printed on a valid PAYMENT_CORRECTION_RECEIPT.

Example:

```
  ****** W I N C O R - N I X D O R F ******

            receipt header line 2

            receipt header line 3

            receipt header line 4

     Cod Fiscal C.I.F.: RO12345678901234


****  BON NEFISCAL  ****  BON NEFISCAL  ****

                                       Lei

CASH                                 10,00

CARD                                 10,00 -


EJTRZ: 00001

BNF. 00001    DATA:15/09/2008    ORA:15-12-21

                 1234567890

 ****  BON NEFISCAL  ****  BON NEFISCAL  ****
```

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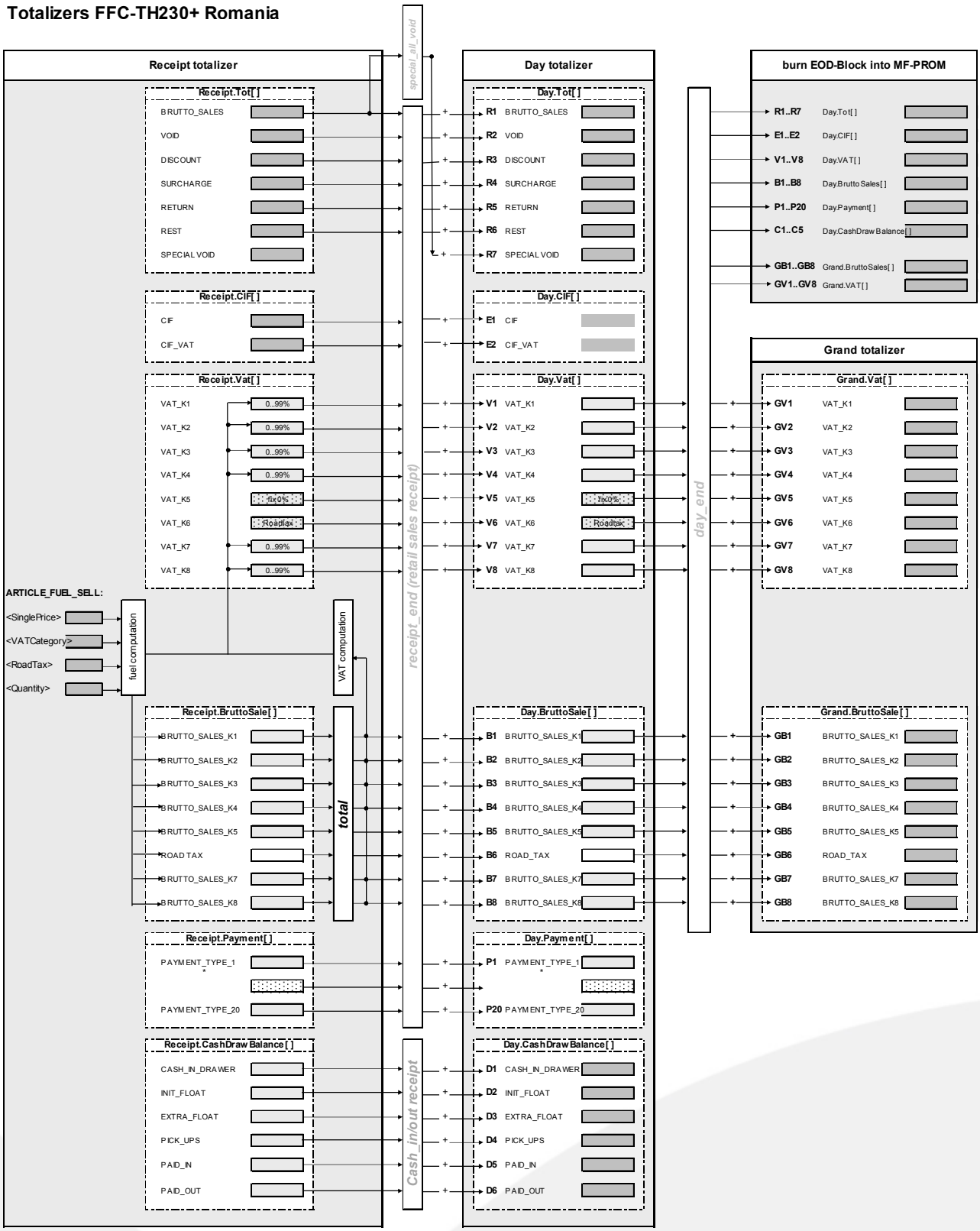
5.5 Counters and totalizers

5.5.1 Totalizers

Totalizers FFC-TH230+ Romania



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For derived values, the system always uses one floating point more than it actually prints. All internal calculations are done with three decimals. For print-out the last two digits is rounded up/down according to the value of the third digit (+0,5 and third digit is truncated). All three digits are internally hold until the DAY_END command is issued. The application can obtain all values with the GET_ commands (0

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5.5.2 Counters

Receipt event counters

Daily Event Counter Type	Commands
`Receipt.EventCnt[DISCOUNT]`	ARTICLE_DISCOUNT ARTICLE_DISCOUNT_VOID SUBTOTAL_DISCOUNT SUBTOTAL_DISCOUNT_VOID
`Receipt.EventCnt[SURCHARGE]`	ARTICLE_SURCHARGE ARTICLE_SURCHARGE_VOID SUBTOTAL_SURCHARGE SUBTOTAL_SURCHARGE_VOID
`Receipt.EventCnt[RETURN]`	ARTICLE_RETURN ARTICLE_RETURN_VOID
`Receipt.EventCnt[VOID]`	ARTICLE_VOID ARTICLE_FUEL_SELL_VOID ARTICLE_TAX_VOID
`Receipt.EventCnt[SPECIAL_VOID]`	SPECIAL_ALL_VOID
`Receipt.EventCnt[CIF]`	
`Receipt.EventCnt[CASH_IN_INIT_FLOAT]`	SPECIAL_CASH_IN
`Receipt.EventCnt[CASH_IN_EXTRA_FLOAT]`	SPECIAL_CASH_IN
`Receipt.EventCnt[CASH_IN_PAID_IN]`	SPECIAL_CASH_IN
`Receipt.EventCnt[CASH_OUT_PICK_UP]`	SPECIAL_CASH_INOUT
`Receipt.EventCnt[CASH_OUT_PAID_OUT]`	SPECIAL_CASH_IN

Daily event counters

Daily Event Counter Type	Commands
`Day.EventCnt[DISCOUNT]`	RECEIPT_END RECEIPT_END_MFLINE
`Day.EventCnt[SURCHARGE]`	RECEIPT_END RECEIPT_END_MFLINE
`Day.EventCnt[RETURN]`	RECEIPT_END RECEIPT_END_MFLINE

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Day.EventCnt[VOID]	RECEIPT_END RECEIPT_END_MFLINE
Day.EventCnt[SPECIAL_VOID]	SPECIAL_ALL_VOID
Day.EventCnt[CIF]	RECEIPT_END * RECEIPT_END_MFLINE *
Day.EventCnt[CASH_IN_INIT_FLOAT]	SPECIAL_CASH_IN
Day.EventCnt[CASH_IN_EXTRA_FLOAT]	SPECIAL_CASH_IN
Day.EventCnt[CASH_IN_PAID_IN]	SPECIAL_CASH_IN
Day.EventCnt[CASH_OUT_PICK_UP]	SPECIAL_CASH_OUT
Day.EventCnt[CASH_OUT_PAID_OUT]	SPECIAL_CASH_OUT

* Only if parameter <TaxID> is given

These counters are available via GET command: 7.15.15 GET_DAY_EVENT_CNT

Daily receipt type counters

Daily Receipt Type Counter	Commands	Comment
Day.ReceiptCnt [RETAIL_SALES_RECEIPT]	RECEIPT_BEGIN SPECIAL_ALL_VOID	
Day.ReceiptCnt [GENERIC_RECEIPT]	RECEIPT_BEGIN SPECIAL_ALL_VOID	
Day.ReceiptCnt [CASH_IN_RECEIPT]	RECEIPT_BEGIN	
Day.ReceiptCnt [CASH_OUT_RECEIPT]	RECEIPT_BEGIN	
Day.ReceiptCnt [FISCAL_REPORT]	REPORT_BLOCK2BLOCK REPORT_DATE2DATE REPORT_EOD2EOD REPORT_TOTAL_DATE2DATE	
Day.ReceiptCnt [DAY_END_REPORT]	DAY_END	Also when the day ends automatically (TKD reset)
Day.ReceiptCnt [DAY_X_REPORT]	DAY_XREPORT	
Day.ReceiptCnt [INFO_TICKET]	-	Not implemented yet
Day.ReceiptCnt [REPRINT]	EJ_REPRINT_...	

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Day.ReceiptCnt [RETAIL_SALES_RECEIPT_WITH_TAX_ID]	RECEIPT_BEGIN	
Day.ReceiptCnt [PAYMENT_CORRECTION]	RECEIPT_BEGIN	

These counters are available via GET command: 7.15.13 GET_DAY_RECEIPT_TYPE_CNT

Daily receipt group counters

Daily Receipt Group Counter	Commands	Comment
Day.ReceiptGroupCnt [LEGAL_RECEIPTS]	RECEIPT_BEGIN, DAY_BEGIN	
Day.ReceiptGroupCnt [ILLEGAL_RECEIPTS]	RECEIPT_BEGIN SPECIAL_ALL_VOID	
Day.ReceiptGroupCnt [EJ_TICKETS]	RECEIPT_BEGIN, DAY_END	
Day.ReceiptGroupCnt [INFO_RECEIPTS]		Not implemented yet

These counters are available via GET command: 7.15.14 GET_DAY_RECEIPT_GROUP_CNT

Grand receipt type counter

Daily Receipt Type Counter	Commands	Comment
stGrand.ulGrandFiscalReceiptCnt	RECEIPT_BEGIN	

This counter is available via GET command: 7.15.18 GET_GRAND_FISCAL_CNT

5.5.3 Size

Size overview for counters and totalizers

All counters compiled at the day stage have a maximum value of **65.535**.

The grand fiscal counter is limited to **4.294.967.295.**

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All totalizers on receipt level are limited to **4.294.967.295** (in new Lei 42.949.672,95).

The fiscal limitation for MAX_SINGLE_PRICE and MAX_RECEIPT_TOTAL is **999.999.999** (in new Lei 9.999.999,99).

All totalizers in day level are limited to **4.294.967.295** (in new Lei 42.949.672,95).

The grand fiscal totalizer limitation of 2^{64} is not reachable. The theoretical value is the maximum amount of EOD blocks multiply by maximal day total: 7.895 * 4.294.967.295 = 33.908.766.794.025.

5.6 Payment list handling

The MFC has the possibility to handle up to 20 payment types inside a payment list. The list contains predefined payment types in the entries from 0 to 8. After MFC_RECEIPT_TOTAL is set during a fiscal receipt it is possible to pay using a payment type by specifying the *<PaymentType> and <PaymentValue>*.
All elements of the payment list that have payment activities during the fiscal period are printed in the Z and X reports.

Each entry of the payment list contains of the following fields:

Parameter	Value	Description

ucEntryValid	unsigned character	Keeps the combination of the following flags:
		`PS_INVALID = 00H`
		`PS_VALID = 01H`
		`PS_PRE_DEFINED = 02H`

ucEntryUsed	unsigned character	Indicates if the payment entry is used:
		`entry not used = FALSE`
		`entry used = TRUE`

ucPaymentCategory	unsigned character	Define the payment type category:
		`PC_LOCAL_CASH = 00H`
		`PC_FOREIGN_CASH = 01H`
		`PC_NON_CASH = 02H`

pcPaymentName	string	Name of payment category.

All entries of the structure are cleared to the default values when the serial number block has been burned successfully or when a TKD reset has been successfully executed.

The default values are zeroes except the following fields for entries from 00…08:

Entry 00 = PT_CASH
ucPaymentStatus = PS_PRE_DEFINED
ucPaymentCategory = PC_LOCAL_CASH
pcPaymentName = "CASH"

Entry 01 = PT_CARD
ucPaymentStatus = PS_ PRE_DEFINED
ucPaymentCategory = PC_NON_CASH
pcPaymentName = "CARD"

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Entry 02	= PT_FOOD
ucPaymentStatus	= PS_PRE_DEFINED
ucPaymentCategory	= PC_NON_CASH
pcPaymentName	= "TICHETE DE MASA"
Entry 03	= PT_VOUCHER
ucPaymentStatus	= PS_PRE_DEFINED
ucPaymentCategory	= PC_NON_CASH
pcPaymentName	= "VOUCHER"
Entry 04	= PT_OTHER
ucPaymentStatus	= PS_PRE_DEFINED
ucPaymentCategory	= PC_NON_CASH
pcPaymentName	= "BON/TICHET VALORIC"
Entry 05	= PT_OTHER
ucPaymentStatus	= PS_PRE_DEFINED
ucPaymentCategory	= PC_NON_CASH
pcPaymentName	= "CREDIT"
Entry 06	= PT_OTHER
ucPaymentStatus	= PS_PRE_DEFINED
ucPaymentCategory	= PC_NON_CASH
pcPaymentName	= "AVANS IN NUMERAR"
Entry 07	= PT_OTHER
ucPaymentStatus	= PS_PRE_DEFINED
ucPaymentCategory	= PC_NON_CASH
pcPaymentName	= "METODE MODERNE DE PLATA"
Entry 08	= PT_OTHER
ucPaymentStatus	= PS_PRE_DEFINED
ucPaymentCategory	= PC_NON_CASH
pcPaymentName	= "ALTE METODE DE PLATA"

The first nine payment list entries are not reprogrammable!

The following new commands are available to handle the payment list:

SET_PAYMENT_LIST:	Set the entry information of a payment category.
SET_PAYMENT_LIST_VOID:	Delete the information of the given entry.
GET_PAYMENT_LIST:	Return the payment category information for a given entry.

5.7 Logical block types

5.7.1 Overview

As described in chapter 3.4 “Block handling (logical and physical blocks)”, the data inside the MFMEM is organized in logical blocks. The following chapter gives information about the content of the different block types used.

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Block Type	Name	Description
0	SNO	Serial number block
1	EOD	End of day block
2	TKD	Technical service block (TKD block)
3	EJ_Change	EJ medium initialization block
4	TH	Ticket header change block
5	VAT	VAT rate change block
6	CRY	Currency change block
7		*reserved*
8	FW	Firmware change block

Note: Some block header data is defined by Wincor Nixdorf basic firmware (marked in gray in the table below). This data cannot be modified by country specific firmware.

5.7.2 SNO block

Address (hex)*	Data	Address (dec)*	* This address is relative to the beginning of the fiscal memory block.
FFh	XOR checksum	255	
41h..FEh	Not used	65..254	
2Ch..40h	Printer Serial Number	44..64	20 bytes + string terminator (e.g. “97AX522077”)
1Eh..2Bh	CIF	30..43	13 bytes + string terminator (e.g. "1234567890123")
13h..1Dh	Fiscal Serial Number	19..29	10 bytes + string terminator (e.g. "0102030405")
0Fh..12h	Firmware version (version / country / subversion / CRC)	15..18	4 bytes ‘011400’
00h..0Eh	Block header (see chapter 3.4)	0..14	

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5.7.3 EOD block

EOD master block:

Address (hex)*	Data	Address (dec)*	
FF	XOR checksum	255	
FA	Day cash draw balance type, part 1	250	6 x 4 bytes
AA	Day payment totalizer	170	20 x 4 bytes
94	Day event counter	148	11 x 2 bytes
8C	Day receipt group counter	140	4 x 2 bytes
74	Day receipt type counter	116	12 x 2 bytes
54	Day VAT amount of VAT category A..H	84	8 x 4 bytes
34	Day brutto sales of the VAT category A..H	52	8 x 4 bytes
2C	Day statistic	44	2 x 4 bytes
10	Day totalizer	16	7 x 4 bytes
0F	Currency byte	15	
00	Block header (see chapter 3.4)	0	

* This address is relative to the beginning of the fiscal memory block.

EOD slave block:

Address (hex)*	Data	Address (dec)*	
FF	XOR checksum	255	
DF	Not used	223	
DE	Working Mode	222	1 byte
9E	Grand VAT amount of the VAT category A..H	158	stGrand.aulGrandVATAmount[A..H] 8 x 8 bytes = 64 bytes
5E	Grand brutto sales of the VAT category A..H	94	stGrand.aulGrandBruttoSales[A..H] 8 x 8 bytes = 64 bytes
5A	Grand legal receipt counter	90	stGrand.ulGrandGroupCnt (4 byte)
4A	Day VAT rates	74	8 x 2 byte
49	Day VAT content	73	1 byte
34	EJ unique media identification	52	21 bytes
14	Session checksum	20	32 bytes
01	Day cash draw balance type, part 2	1	
00	Block status	0	Block status (Slave block) = 0001 0111

* This address is relative to the beginning of the fiscal memory block.

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5.7.4 TKD block

Address (hex)*	Data	Address (dec)*	* This address is relative to the beginning of the fiscal memory block.
FF	XOR checksum	255	
35	Not used	53	
34	Checksum	52	1 byte
33	Processed flag	51	1 byte
2F	Number of line in which error has been set	47	4 byte
1B	Name of file in which the error has been set	27	20 chars
14	Day, month, year, hour, minute, second of blockade error	20	4 bytes (date) + 3 bytes (time)
13	blockade error	19	1 byte
0F	Firmware version (version / country / subversion / CRC)	15	
00	Block header (see chapter 3.4)	0	

5.7.5 EJ change block

Address (hex)*	Data	Address (dec)*	* This address is relative to the beginning of the fiscal memory block.
FF	XOR checksum	255	
2A	Not used	42	
28	EJ Code	40	2 bytes – Number of EJ
13	EJ UMID	19	21 chars Identification of this EJ
0F	Firmware version (version / country / subversion / CRC)	15	
00	Block header (see chapter 3.4)	0	

5.7.6 TH block

TH master block:

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Address (hex)*	Data	Address (dec)*	* This address is relative to the beginning of the fiscal memory block.
FF	XOR checksum	255	
0F	Ticket header, part 1	15	Out of 402 bytes to be written in the Ticket Header (5 Header Lines + CIF ID), first 239 free bytes in master block and the rest in TH slave block.
00	Block header (see chapter 3.4)	0	

TH slave block:

Address (hex)*	Data	Address (dec)*	* This address is relative to the beginning of the fiscal memory block.
FF	XOR checksum	255	
A4	Not used	164	
01	Ticket header, part 2	1	
00	Block status	0	Block status (Slave block)

5.7.7 VAT block

Address (hex)*	Data	Address (dec)*	* This address is relative to the beginning of the fiscal memory block.
FF	XOR checksum	255	
B9	Not used	185	
B5	Grand fiscal receipt counter	181	4 byte
75	Grand VAT amount of VAT category A…H	117	8 x 8 bytes = 64 bytes
35	Grand brutto sales of the VAT category A…H	53	8 x 8 bytes = 64 bytes
25	VAT rate A…H – new value	37	8 x 2 bytes = 16 bytes
24	ucContentByte new	36	
14	VAT rate A…H – old value	20	8 x 2 bytes = 16 bytes
13	ucContentByte old	19	
0F	Firmware version (version / country / subversion / CRC)	15	
00	Block header (see chapter 3.4)	0	

5.7.8 CRY block

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Address (hex)*	Data	Address (dec)*	
FF	XOR checksum	255	
9C	Not used	156	
5C	Grand VAT amount of VAT category A…H	92	8 x 8 bytes = 64 bytes
1C	Grand Brutto Sales of VAT category A…H	28	8 x 8 bytes = 64 bytes
18	Grand fiscal receipt counter	24	4 byte
13	Currency label	19	5 bytes e.g. “EURO” (00h terminated string)
0F	Firmware version (version / country / subversion / CRC)	15	
00	Block header (see chapter 3.4)	0	

* This address is relative to the beginning of the fiscal memory block.

5.7.9 FW block

Address (hex)*	Data	Address (dec)*
FF	XOR checksum	255
17	Not used	23
13	new firmware version (version / country / subversion / CRC)	19
0F	old firmware version (version / country / subversion / CRC)	15
00	Block header (see chapter 3.4)	0

* This address is relative to the beginning of the fiscal memory block.

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5.7.10 Block restriction

The MF memory must be able to hold seven years worth of sales data (with max. 8 used VAT categories). Each year is calculated as having 360 working days, which gives a minimum capacity of 2690 sales data entries.

The PFM has space for 16.384 physical blocks. For service blocks (TKD block), VAT change entries (VAT block), EJ change blocks, TT and FW blocks or defect blocks a buffer of 694 blocks is provided. Note that each sales data entry (EOD block) uses **two** physical blocks:

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Max Blocks (guarantee) phys.	logical.	Block Type	Remarks	Error Reaction
15.690	7.845 (phys. / 2)	EOD	Block amount is guaranteed	If EOD block 7.785 is reached, the nearly full bit in the MFMEM status byte (see chapter 3.3) is active. If the absolute maximum block number is reached, the firmware responds with an EM_MFMEM_FULL error and sets the MFMEM to full.
200	200	EJ change		If the maximum block number is reached, the firmware responds with an EM_MFMEM_BL_LIMIT_OVERRUN error. A blockade error is not generated.
200	200	TKD		If the maximum block number is reached the firmware responds with an EM_TKD_RESET_LIMIT_OVERRUN error and sets the MFMEM to full.
30	30	VAT		If the maximum block number is reached the firmware does not execute additional SET_VATRATES commands and responds with an EM_MFMEM_BL_LIMIT_OVERRUN error. A blockade error is not generated.
1	1	SNO	Before the first fiscal operation the MFC has to be initialised with a serial number.	Attempting to store a second SNO block leads to an EM_CMD_WRONG_ORDER error.
100	100	Defect	If the first attempt to burn a block fails, the firmware marks the block as “defect” and retries the burning. After the third unsuccessful try to burn a block, the fiscal printer goes into a blockade state.	If the limit is reached the firmware issues an error EM_MFMEM_DEFECT and marks the entire MFMEM as defect.
32	16 (phys. / 2)	TH	These blocks are set with the set header command.	If the maximum block number is reached the firmware responds with an EM_MFMEM_BL_LIMIT_OVERRUN error. A blockade error is not generated.
30	30	CRY	These blocks are set with the currency change.	Attempting to store a second CRY block leads to an EM_CMD_WRONG_ORDER error.
1	1	HWDATA	First block is used for HW data	See chapter 3.2 Hardware data (HWDATA)
100	100	FW	These blocks are set with each firmware change.	If the maximum block number is reached, the firmware responds with an EM_MFMEM_BL_LIMIT_OVERRUN error. A blockade error is not generated.
16.384	**8.423**		16384 x 256 byte = 4Mb (PFM size)	If none EOD+TKD block is possible anymore, the firmware issues an EM_MFMEM_FULL error and sets the MFMEM status to full.

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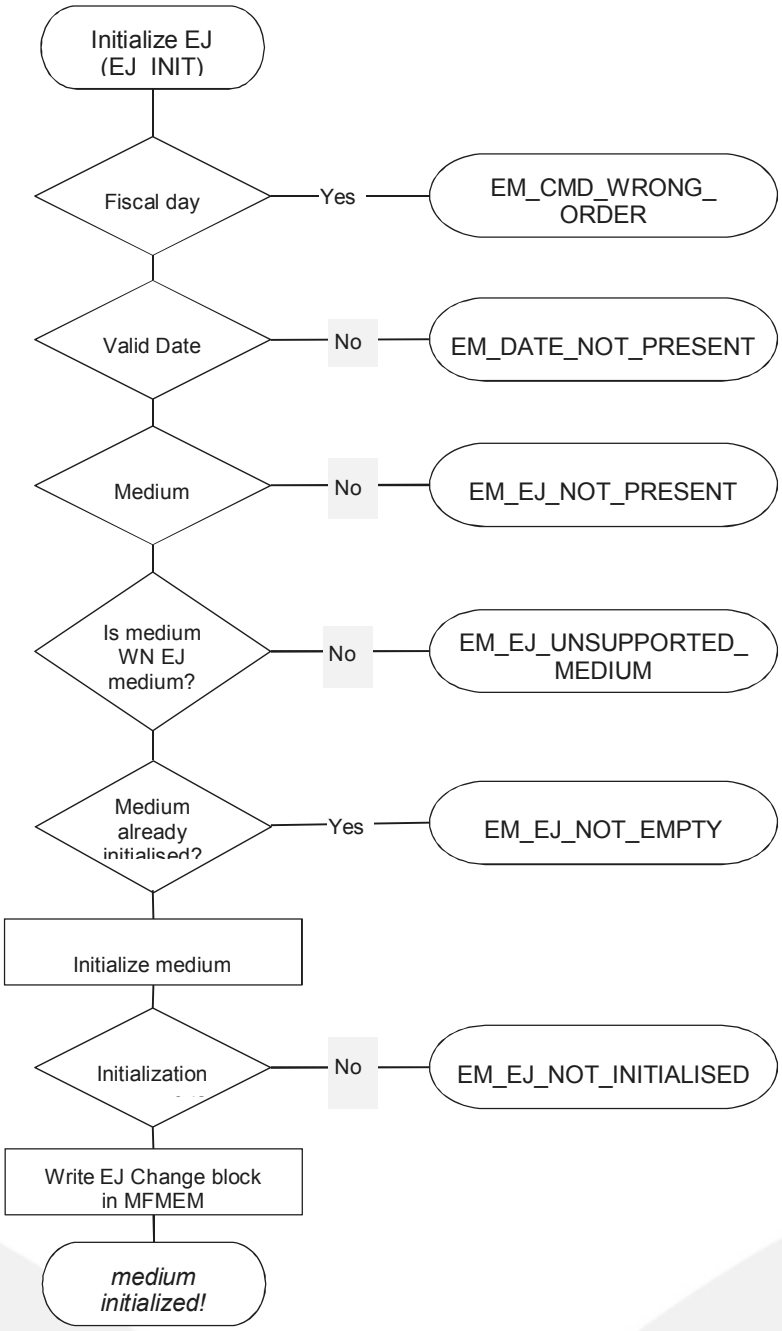
5.8 EJ medium handling

Before a new EJ medium can be used by the fiscal printer it has to be initialized. This can only be done in the same printer in which it will later be used. After the initialization, an EJ change block (refer to chapter 5.7.5 EJ change block for details) will be burned into the MFMEM and the EJ medium is marked as authorized for this printer.

5.8.1 EJ medium initialization (Flow chart)



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5.8.2 Electronic journal operation

At the beginning of an EJ session (DAY_BEGIN command):

The fiscal firmware initializes the SHA-2 message digest with 256 bit (see also chapter 4.3 Electronic journal organization).

During the EJ session:

The application continuously sends transaction data to be printed and recorded to MFC. The fiscal firmware simultaneously records the printed data in the EJ and recalculates the message digest based on the data currently being printed. The message digest is stored in the NV RAM of the MFC.

At the end of an EJ session (DAY_END command):

The application sends the DAY_END command to the MFC. The fiscal firmware completes the daily closure and sends the message digest (MD). After the completion of the EOD process the fiscal firmware opens the next (current) EJ file with system and hidden attribute. Parallel to this procedure, the MFC burns the message digest (MD) into the MFMEM EOD block.

After the current EJ file is closed (or parallel to its generation) the application is to read the complete file from the SD card and prepare a safety copy on the hard disk of the POS.

5.9 EJ data security

5.9.1 EJ file logical structure

The name of an EJ file on the SD card contains 8 characters, the first 3 characters are the last 3 digits of the fiscal memory identification number, and the next 5 characters are the EJ session number and has the file extension is EJ (78900001.EJ). The EJ file is a normal text file containing the information printed on the receipt station. All information written in the EJ file is represented in codepage PC852. Every line in the text file has the following record structure:

PrintedLineData<CR><LF>

Or

@Special EJ Records<TAB><CR><LF>

With:

- **PrintedLineData** being a variable length string containing characters with hex values >= 20H or space replacement TAB (09H) characters. The space replacement algorithm is based on the 9,17,25,33 tabulator columns. Trailing spaces are eliminated from this string. An empty string (just line feed) is not registered. The MF Logo will be replaced by the string ".MF".

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The EJ file has the following logical file structure:

-
- Ticket 1
- Ticket 2
- …
- Ticket n-1
- DAY_ZREPORT ticket n
- Tickets pointer table referencing ticket 1 to ticket n
- Session checksum**<CR><LF>**

The tickets pointer table contains n lines referencing ticket 1 to ticket n. Each line has the following structure:

|<P1>|<P2>|<P3>|<P4>|<P5>|<P6>|<P7>|<P8>|<P9>|**<CR><LF>**

Where:

<P1> - Session number 4 characters

<P2> - Ticket number 4 characters

<P3> - Ticket type 10 characters

<P4> - Ticket status 2 characters

<P5> - File offset in bytes of the ticket beginning

<P6> - File offset in bytes of the ticket ending

<P7> - Ticket start time in DD.MM.YYYY HH:MM:SS format

<P8> - Ticket end time in DD.MM.YYYY HH:MM:SS format

The session checksum is computed over the complete EJ file (excluding the session checksum of course) based on the standard SHA-2 algorithm with 256 bit. The checksum is stored in 64 ASCII bytes representing the checksum in hex format.

5.9.2 EJ verify utility

The EJVERIFY utility has two main functions:

1. The EJ file verification and data compress function.

- Execution syntax under **DOS**:

EJVERIFY [-Ooutputpath\] [drive:][inputpath\]ejfile.EJ

Where **ejfile** is the name of electronic journal with wildcards.

- Execution syntax under **Windows**:

WINEJVERIFY [-Ooutputpath\] [drive:][inputpath\]ejfile.EJ

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Where **ejfile** is the name of electronic journal with wildcards.

- Execution syntax under **Linux**:

ejverify [-Ooutputpath/] [inputpath/]ejfile1.EJ [inputpath/]ejfile2.EJ ...
or
ejverify [-Ooutputpath/] [inputpath/]ejfile.EJ

In the **first step**, the utility will verify the signature of the EJ file(s) defined by the command line parameter. An I/O error will stop the execution of the utility with an exit code greater than 2.
In the **second step**, the utility will create the compressed format of the EJ file(s) with an **EJC** extension defined by the command line parameter, in the directory defined by the **–O** switch. In parallel, a checksum file with the same name and the extension **CHK** will be generated in the directory defined by the **–O** switch. If the **–O** switch is missing the **EJC** and the **CHK** files are generated in the same directory as the original EJ file.

2. The EJC compressed file verification and data decompression function.

- Execution syntax under **DOS**:

EJVERIFY [-Ooutputpath\] [drive:][inputpath\]ejfile.EJC

Where **ejfile** is the name of electronic journal with wildcards.

- Execution syntax under **Windows**:

WINEJVERIFY [-Ooutputpath\] [drive:][inputpath\]ejfile.EJC

Where **ejfile** is the name of electronic journal with wildcards.

- Execution syntax under **Linux**:

ejverify [-Ooutputpath/] [inputpath/]ejfile1.EJC [inputpath/]ejfile2.EJC ...
or
ejverify [-Ooutputpath/] [inputpath/]ejfile.EJC

The utility will create the decompressed format of the EJC file(s) defined by the command line parameter with an **EJD** extension in the directory defined by the **–O** switch. If the **–O** switch is missing the **EJD** is generated in the same directory as the original EJC file.
After decompression the utility will execute the decompressed file signature verification. An I/O error stops the execution of the utility with an exit code greater than 2.

The EJVERIFY utility exit codes:

- ERRORLEVEL 0 - Signature OK
- ERRORLEVEL 1 - Signature error
- ERRORLEVEL 2 - File structure error
- ERRORLEVEL 3 - I/O error, execution stopped
- ERRORLEVEL 4 - File not found, execution stopped

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- ERRORLEVEL 5 - Command error, execution stopped
- ERRORLEVEL 6 - EJC file creation error, execution stopped
- ERRORLEVEL 7 - Compression error, execution stopped
- ERRORLEVEL 8 - EJD file creation error, execution stopped
- ERRORLEVEL 9 - Decompression error, execution stopped

6 PRINTER COMMANDS

The following table lists all TH230 ESC POS commands and their fiscal handling. Please refer to TH230+ Programmers Guide for more details.

i	Ignored.
a	Accepted.
b	Accepted, but only affects "non-fiscal mode" (settings will be buffered).
m	Accepted, but with special fiscal modifications.
r	Accepted, but with the specified restrictions.
p	Accepted in SET_HEADER(7.4.3) or FREEPRINT_NORMAL(7.11.2) .

ESC POS Command Sequence	Code (hex)	Description	NON_FISCAL_MODE	FISCAL_MODE				
				General Status	*restrictions*			
					DAY_OPEN	RECEIPT_OPEN	REPORT_OPEN	DOCU_OPEN
HT	0x09	*Horizontal tab*	**a**	**i**				
LF	0x0a	*Print and line feed*	**a**	**a**				
FF	0x0c	*Print and form feed*	**a**	**a**				
CR	0x0d	*Carriage return*	**i**	**i**				
DLE	0x10	*Clear printer*	**i**	**i**				
DLE EOT n	0x10 0x04	*Transmit real-time status*	**a**	**a**				
DLE ENQ n	0x10 0x05	*Real-time request to printer*	**a**	**a**				
DLE SO (fn=2)	0x10 0x0e	*Turn off the power*	**i**	**i**				

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ESC POS Command Sequence	Code (hex)	Description	NON_FISCAL_MODE	FISCAL_MODE				
				General Status	restrictions			
					DAY_OPEN	RECEIPT_OPEN	REPORT_OPEN	DOCU_OPEN
DC1	0x11	*Print raster monochrome graphics*	i	i				
DC2	0x12	*Select double width characters*	a	b				
DC3	0x13	*Select single width characters*	a	b				
DC4	0x14	*Feed n print lines*	i	i				
NAK	0x15	*Feed n dot rows*	i	i				
SYN	0x16	*Add n extra dot rows*	i	i				
ETB	0x17	*Print*	i	i				
CAN	0x18	*Cancel print data in page mode*	i	i				
EM	0x19	*Perform full cut*	i	i				
SUB	0x1a	*Perform partial cut*	i	i				
ESC BEL	0x1b 0x07	*Generate tone*	a	a				
ESC FF	0x1b 0x0c	*Print data in page mode*	i	i				
ESC DC2	0x1b 0x12	*Select 90 degree counter clockwise rotated print*	i	i				
ESC DC4	0x1b 0x14	*Set column*	i	i				
ESC SYN	0x1b 0x16	*Select pitch*	a	b				
ESC SP n	0x1b 0x20	*Set right side character spacing*	a	b				
ESC ! n	0x1b 0x21	*Select print mode*	a	b/p				
ESC $ nL nH	0x1b 0x24	*Set absolute print position*	a	i				
ESC % n	0x1b 0x25	*Select/cancel user-defined character set*	i	i				
ESC & y …	0x1b 0x26	*Define user-defined characters*	i	i				
ESC '	0x1b 0x27	*Write to user data storage*	i	i				

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ESC POS Command Sequence	Code (hex)	Description	NON_FISCAL_MODE	FISCAL_MODE			
				General Status	restrictions		
					DAY_OPEN	RECEIPT_OPEN	REPORT_OPEN	DOCU_OPEN
ESC * m …	0x1b 0x2a	*Select bit image mode*	i	i			
ESC - n	0x1b 0x2d	*Turn underline mode on/off*	a	b			
ESC .	0x1b 0x2e	*Print advanced raster graphics*	i	i			
ESC 2	0x1b 0x32	*Select default line spacing*	i	i			
ESC 3 n	0x1b 0x33	*Set line spacing*	a	b			
ESC 4	0x1b 0x34	*Read from user data storage*	i	i			
ESC :	0x1b 0x3a	*Copy character set from ROM to RAM*	i	i			
ESC =	0x1b 0x3d	*Select peripheral device*	a	a			
ESC ?	0x1b 0x3f	*Cancel user-defined characters*	i	i			
ESC @	0x1b 0x40	*Initialize printer*	a	m[1]			
ESC B M	0x1b 0x42 0x4d	*Download BMP logo*	r	r	i	i	
ESC D	0x1b 0x44	*Set horizontal tab position*	i	i			
ESC E	0x1b 0x45	*Turn emphasized mode on/off*	a	b			
ESC G	0x1b 0x47	*Turn double strike mode on/off*	a	b			
ESC I	0x1b 0x49	*Select or cancel italic print*	a	b			
ESC J	0x1b 0x4a	*Print and feed paper*	a	a			
ESC K	0x1b 0x4b	*Select single density graphics*	i	i			
ESC L	0x1b 0x4c	*Select page mode*	i	i			
ESC R	0x1b 0x52	*Select an international character set*	a	b			
ESC S	0x1b 0x53	*Select standard mode*	i	i			
ESC T	0x1b 0c54	*Select print direction in page mode*	i	i			

[1] After the initialization of the printer, the fiscal default print parameters will be set

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ESC POS Command Sequence	Code (hex)	Description	NON_FISCAL_MODE	FISCAL_MODE				
				General Status	restrictions			
					DAY_OPEN	RECEIPT_OPEN	REPORT_OPEN	DOCU_OPEN
ESC V	0x1b 0x56	*Turn 90^0 rotation mode on/off*	**a**	**b**				
ESC W	0x1b 0x57	*Set printing area in page mode*	**i**	**i**				
ESC Y	0x1b 0cx59	*Select double density graphics*	**i**	**i**				
ESC [! t	0x1B 0x5B 0x21	*Set control point*	**a**	**a**				
**ESC **	0x1b 0x5c	*Set relative print position*	**a**	**i**				
ESC a	0x1b 0x61	*Select justification*	**a**	**b/p**				
ESC c 4	0x1b 0x63 0x34	*Select paper sensor(s) to stop printing*	**i**	**i**				
ESC c 5	0x1b 0x63 0x35	*Enable/disable panel buttons*	**a**	**a**				
ESC d	0x1b 0x64	*Print and feed n lines*	**a**	**a**				
ESC i	0x1b 0x69	*Execute cut*	**m**[2]	**i**				
ESC l	0x1b 0x6c	*Start firmware upgrade mode (special handling, for more information see References: [R7] WN Enhanced Serial Protocol)*	**i**	**i**				
ESC m	0x1b 0x6d	*Execute partial cut*	**m**[2]	**i**				
ESC p	0x1b 0x70	*Generate pulse for cash drawer*	**a**	**a**				
ESC r	0x1b 0x72	*Set current colour*	**a**	**b/p**				
ESC t	0x1b 0x74	*Select character code table*	**a**	**b**				
ESC u	0x1b 0x75	*Transmit peripherals device status*	**a**	**a**				
ESC v	0x1b 0x76	*Transmit station sensor status*	**a**	**a**				
ESC w n 7	0x1b 0x77 0x6e 0x37	*Receipt shooting flush or printer synchronize*	**a**	**a**				
WSC w n 8	0x1b 0x77 0x6e 0x38	*Play melody from flash*	**a**	**a**				
ESC {	0x1b 0x7b	*Turn upside-down printing mode on/off*	**a**	**b**				

[2] The paper will always be feed to cutting position and the non-fiscal remark will be printed on the ticket before cutting

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ESC POS Command Sequence	Code (hex)	Description	NON_FISCAL_MODE	FISCAL_MODE				
				General Status	restrictions			
					DAY_OPEN	RECEIPT_OPEN	REPORT_OPEN	DOCU_OPEN
GS ETX	0x1d 0x03	*Real-time request to printer*	a	a				
GS EOT	0x1d 0x04	*Real-time status transmission*	a	a				
GS ENQ	0x1d 0x05	*Real-time request to printer*	a	a				
GS !	0x1d 0x21	*Select character size*	a	b				
GS "	0x1d 0x22	*Select memory type in which to save logos or user defined fonts*	a	a				
GS " U	0x1d 0x22 0x55	*Flash memory user sectors allocation*	a	a				
GS #	0x1d 0x23	*Select the current logo*	a	a				
GS $	0x1d 0x24	*Set absolute vertical print position in page mode*	i	i				
GS (A	0x1d 0x28 0x41	*Execute test print*	i	i				
GS (D	0x1d 0x28 0x44	*Enable/disable real-time command*	i	i				
GS (E	0x1d 0x28 0x45	*Customize NV RAM memory area*	r	r	i	i		
GS (F	0x1d 0x28 0x46	*Set adjustment values*	i	i				
GS (L fn ≠ 69	0x1d 0x28 0x4c	*Select graphics data*	a	a				
GS (L fn = 69	0x1d 0x28 0x4c	*Print graphics data*	aa	r/p	i	i		
GS (k cn = 0 fn = 0	0x1d 0x28 0x6b	*Erase symbol storage area*	a	a				
GS (k cn = 49 fn = 65	0x1d 0x28 0x6b	*Select the model*	a	a				
GS (k cn = 49 fn = 67	0x1d 0x28 0x6b	*Set the size of module*	a	a				
GS (k cn = 49 fn = 69	0x1d 0x28 0x6b	*Select the error correction level*	a	a				
GS (k cn = 49 fn = 80	0x1d 0x28 0x6b	*Store the data into the symbol storage area*	a	a				
GS (k cn = 49 fn = 81	0x1d 0x28 0x6b	*Print the symbol data in the symbol storage area*	i	m	a	iai		
GS *	0x1d 0x2a	*Define downloaded bit image*	i	i				

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ESC POS Command Sequence	Code (hex)	Description	NON_FISCAL_MODE	FISCAL_MODE				
				General Status	restrictions			
					DAY_OPEN	RECEIPT_OPEN	REPORT_OPEN	DOCU_OPEN
GS /	0x1d 0x2f	*Print downloaded bit image*	**a**	**p**				
GS :	0x1d 0x3a	*Start/end macro definition*	**i**	**i**				
GS @	0x1d 0x40	*Erase user flash sector*	**a**	**a**				
GS B	0x1d 0x42	*Turn white/black reverse printing mode on/off*	**a**	**b**				
GS H	0x1d 0x48	*Select printing position for HRI characters*	**a**	**b/p**				
GS I	0x1d 0x49	*Transmit printer ID*	**a**	**a**				
GS L	0x1d 0x4c	*Set left margin*	**a**	**b**				
GS P	0x1d 0x50	*Set horizontal and vertical motion units*	**a**	**b**				
GS V	0x1d 0x56	*Select cut mode and cut paper*	**m**[3]	**i**				
GS W	0x1d 0x57	*Set printing area width*	**a**	**b**				
**GS **	0x1d 0x5c	*Set relative vertical print position in page mode*	**i**	**i**				
GS^	0x1d 0x5e	*Execute macro*	**i**	**i**				
GS a	0x1d 0x61	*Enable/Disable automatic status back (ASB)*	**a**	**a**				
GS b	0x1d 0x62	*Turns smoothing mode on/off*	**i**	**i**				
GS f	0x1d 0x66	*Select font for HRI characters*	**a**	**b/p**				
GS g 0	0x1d 0x67 0x30	*Initialize maintenance counter*	**a**	**a**				
GS g 2	0x1d 0x67 0x32	*Transmit maintenance counter value*	**a**	**a**				
GS h	0x1d 0x68	*Select bar code height*	**a**	**b**				
GS k	0x1d 0x6b	*Print bar code*	**a**	**p**				
GS p	0x1d 0x70	*Select PDF 417 parameters*	**a**	**b/p**				
GS r	0x1d 0x72	*Transmit status*	**a**	**a**				

[3] The paper will always be feed to cutting position and the non-fiscal remark will be printed on the ticket before cutting

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ESC POS Command Sequence	Code (hex)	Description	NON_FISCAL_MODE	FISCAL_MODE				
				General Status	restrictions			
					DAY_OPEN	RECEIPT_OPEN	REPORT_OPEN	DOCU_OPEN
GS w	0x1d 0x77	*Set bar code width*	**a**	**p**				
GS 0x81	0x1d 0x81	*Set paper type*	**a**	**r**	**a**	**i**		
GS 0x82	0x1d 0x82	*Print raster monochrome graphics*	**i**	**i**				
GS 0x83	0x1d 0x83	*Print raster colour graphics*	**i**	**i**				
GS 0x84	0x1d 0x84	*Download logo image*	**i**	**i**				
GS 0x85	0x1d 0x85	*Reverse colour text mode (two colour)*	**i**	**i**				
GS 0x86	0x1d 0x8d	*Monochrome shade mode*	**i**	**i**				
GS 0x87	0x1d 0x87	*Colour shade mode*	**i**	**i**				
GS 0x89	0x1d 0x89	*Logo print with colour plane swap*	**i**	**i**				
GS 0x8b	0x1d 0x8b	*Apply shading to logo*	**i**	**i**				
GS 0x8c	0x1d 0x8c	*Merge watermark mode*	**i**	**i**				
GS 0x8d	0x1d 0x8d	*Text strike through mode*	**i**	**i**				
GS 0x8e	0x1d 0x8e	*Download paper type description*	**i**	**i**				
GS 0x8f	0x1d 0x8f	*Return paper type description*	**i**	**i**				
GS 0x90	0x1d 0x90	*Form and merge real-time surround graphic*	**i**	**i**				
GS 0x91	0x1d 0x91	*Save graphics buffer as logo*	**i**	**i**				
GS 0x92	0x1d 0x92	*Background logo print mode*	**i**	**i**				
GS 0x97	0x1d 0x97	*User storage status*	**i**	**i**				
GS 0x99	0x1d 0x99	*Apply margin message mode*	**i**	**i**				
GS 0x9a	0x1d 0x9a	*Shade and store logo*	**i**	**i**				
GS 0x9b	0x1d 0x9b	*Logo print with knife cut*	**i**	**i**				
GS 0xa0	0x1d 0xa0	*Set temporary max target speed*	**a**	**a**				

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ESC POS Command Sequence	Code (hex)	Description	NON_FISCAL_MODE	FISCAL_MODE			
				General Status	restrictions		
					DAY_OPEN	RECEIPT_OPEN	REPORT_OPEN	DOCU_OPEN
GS 0xff	0x1d 0xff	*Reset firmware*	i	i				
US EOT	0x1f 0x04	*Convert 6 dots/mm bitmap to 8 dots/mm bitmap*	i	i				
US ENQ	0x1f 0x05	*Select superscript or subscript modes*	a	b				
US t	0x1f 0x74	*Print test form*	i	i				
US V	0x1f 0x56	*Send printer software version*	i	i				
US z	0x1f 0x7a	*Real-time commands disabled*	a	a				
US {	0x1f 0x7b	*Enable constant speed logos*	i	i				

7 FISCAL COMMANDS

7.1 General rules

The firmware parses the command sequence sent by the application and will respond with an error message from the 'EM_PARA_...' series of errors in case of a parameter related problem. To avoid these programming errors and in order to help application developers to understand how the firmware translates this data, the following rules must be considered when passing parameters:

1. Undersize text parameter will be right-padded with blanks at print time.
2. Leading zeroes in numeric parameters are rejected with an EM_PARA_WRONG_FORMAT error.
3. Decimal points and signs in numeric parameters are always implicit – they should not be sent.
4. Date and time parameters are always checked for correctness and if found wrong will be rejected with an EM_PARA_DATE_WRONG_FORMAT error.
5. From/to parameters are logically checked and if found wrong will be rejected with an EM_PARA_RANGE_ERROR or (for dates) EM_PARA_DATE_WRONG_FORMAT error.
6. Appropriate errors will be generated if the parameter supplied is invalid or unknown.
7. If multiple instances of the same parameter are present in the command sequence, the system accepts the last in the sequence and checks the validity of only that one.

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7.1.1 Parameters

In addition to the above rules, parameters that must always be part of a command sequence need to follow the rules below:

1. At least one byte must be non-blank (for text values) or greater than zero.
2. The parameter's length cannot be greater than the maximum allowable.
3. Parameter specific content rules must be followed (if applicable, they are given in their corresponding command description).
4. Optional Parameter: The inclusion of some optional parameters in certain commands is left to the discretion of the application.

7.2 Notation

Each valid control sequence to the MF controller starts with ESC 80H and ends with ESC 83H. Between these start and end markers there are the characters which define the function of the sequence. Some control sequences contain additional start marker for sub sequences (ESC 81H and ESC 82H). The MF controller only performs the requested command if the complete character string is valid.

The following notation is used in the document:

Decimal	123	e.g.: 123
Hexadecimal	80H	e.g.: 80H
Parameter Field	<..>	*<ParameterField>*
ASCII	"A"	20H…7FH
ESC	1BH	
MFB	80H	start sequence
MFB1	81H	mandatory parameter
MFB2	82H	optional parameter
MFE	83H	stop sequence

The description of the commands includes the following information:

Function	Function of the command.
Code	Structure of the command.
Example (optional)	A typical example is given. The sequence sent is between quotation marks.
Condition	Condition when the command is allowed.
Computation	Changes of the fiscal totalizers and counters.

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(optional)	
Note (optional)	Additional information.
Reference (optional)	Additional information.
Error Msg	The fiscal error message which is sent in case of a fiscal error.
Printout (optional)	Printout of the command.
Update	Describes when values are set and available.

7.2.1 Restricted Text

There are no restricted texts in the Romanian implementation.

When <*ArticleName*>, <*QuantityUnit*>, <*TaxName*>, <*PrintLine*>, <*PrintLinePre*> or <*PrintLinePost*> are listed, they will replace any non-printing character (00H...1FH, 7Fh) with blanks.

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7.3 INIT commands

All INIT_ commands may return error messages from the set of 8.2 FM error codes.

7.3.1 INIT_HWDATA

Function: The hardware data will be stored in the fiscal memory as the first initialisation step.

Code: **ESC MFB s a ESC MFB1 f n <*HardwareData*> ESC MFE**

Code old: ESC MFB K A <*HardwareData*> ESC MFE

It is not recommended to use the old command syntax for new implementations. *This syntax is obsolete and may be not supported in the future.*

Note: After successful execution, the red green LED stays permanently on (please refer to chapter 12.1 Printer and MF LED's)

Reference: The structure and the elements of the hardware data are described in chapter 3.2 Hardware data (HWDATA).

HWDATA for FFC/TH230+:

ESC MFB s a ESC MFB1 f n 001;000;000;000;001;000;064;000;016;000;001;003;007;015;031;063;127;255; 061;001; ESC MFE

Error Msg: EM_CMD_WRONG_ORDER

7.3.2 INIT_PRE_FORMAT

Function: The fiscal memory (PFM) will be formatted partly in production line.
The third of all blocks are formatted.

Code: **ESC MFB s f ESC MFE**

Code old: ESC MFB K B <*BlockSpacing*>;<*FromValue*> ESC MFE

Parameter	Value	Format
<*BlockSpacing*>	3	ASCII Only value '3' would be accepted.
<*FromValue*>	0	ASCII Only value '0' would be accepted.

Example for old code: <u>Formatting of 33%:</u> "ESC MFB K B 3 ; 0 ESC MFE"

It is not recommended to use the old command syntax for new implementations. *This syntax is obsolete and may be not supported in the future.*

Note: Before the fiscal memory (PFM) can be formatted, the HWDATA must be stored inside the memory.

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After the MFMEM is fully formatted, the MFC will clear the validity of the internal date and time to force the date and time setting before the SNO block is stored.

While the format process is active, the green and red LEDs are blinking.

Error Msg: EM_CMD_WRONG_ORDER
EM_HWDATA_NOT_PRESENT
Error messages from the set of 8.2 FM error codes

7.3.3 INIT_FULL_FORMAT

Function: The fiscal memory (PFM) will be full formatted at the end-user.

Code: **ESC MFB s b ESC MFE**

Code old: ESC MFB K B *<BlockSpacing>*;*<FromValue>* ESC MFE

Parameter	Value	Format
<BlockSpacing>	1	ASCII Only value '1' would be accepted.
<FromValue>	0	ASCII Only value '0' would be accepted.

Example for old code: Formatting of 100%: "ESC MFB K B 1 ; 0 ESC MFE"

It is not recommended to use the old command syntax for new implementations.
This syntax is obsolete and may be not supported in the future.

Note: Before the fiscal memory (PFM) can be formatted, the HWDATA must be stored inside the memory.

After the fiscal memory (PFM) is fully formatted, the MFC will clear the validity of the internal date and time to force the date and time setting before the SNO block is stored.

While the format process is active, the green and red LEDs are blinking.

Error Msg: EM_CMD_WRONG_ORDER
EM_HWDATA_NOT_PRESENT
Error messages from the set of 8.2 FM error codes

7.3.4 INIT_WRITE_SNO

Function: The SNO block is stored in the fiscal memory as the first logical block. The SNO block contains national information e.g. fiscal serial number (fiscal ID), the CIF number and delivery date. This command is valid only once during the MFMEM lifetime!

Code:
ESC MFB s c
ESC MFB1 d u *<FiscalID>*
ESC MFB1 d s *<CIF>*
ESC MFB1 d p *<PrinterSerialNumber>*
ESC MFE

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Parameter	Value	Format	
<FiscalID>	30 H ≤ *n* ≤ 39 H	10 bytes (digits)	**F**
<CIF>	30 H ≤ *n* ≤ 39 H	3-13 bytes (digits)	**C**
<PrinterSerialNumber>	20 H ≤ *n* ≤ FF H	Up to 20 bytes (alphanumeric characters)	**P**

Example: ESC MFB s c ESC MFB1 d u 0102030405 ESC MFB2 d o RO MFB1 d s 12345678901234 ESC MFE

Note: On successful execution the MFC_STATUS is changed to Fiscal Mode

Note: It is not possible to change one of the parameters *<FiscalID>* or *<CIF>* during the fiscal period. If this has to be done due to fiscal reason it is absolutely necessary to replace the MFMEM by a new one and start the complete initialisation routine again.

Please refer to chapter 5.7.1 SNO block for the SNO block format.

All printer statistic data are initialised to zero.

Error Msg: EM_DATE_NOT_PRESENT
EM_CMD_WRONG_ORDER
EM_PARA_MISSING
EM_PARA_WRONG_FORMAT

Printout: Administrative receipt:

```
****  BON NEFISCAL  ****  BON NEFISCAL  ****

          INIT_WRITE_SNO

  FISCAL ID:                FFFFFFFFFF
  CIF  ID:                  II
  CIF:                      CCCCCCCCCCCCC
  PRINTER SERIAL No.:       PPPPPPPPPPPPPPPPP

  DATA:16/05/2015     ORA:08-11-06
****  BON NEFISCAL  ****  BON NEFISCAL  ****
```

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7.4 SET commands

All commands are available in fiscal mode (5.1 Fiscal Status) only. The fiscal day may not be open.

7.4.1 SET_DATE_TIME

Function: With this command the date and time of the real-time clock will be set.

Code: **ESC MFB h e ESC MFB1 e k <*DateTime*> ESC MFE**

Parameter	Value	Format
<DateTime>	30h ≤ char ≤ 39h	ASCII "DD.MM.YYYY hh:mm:ss" DD = Day, MM = Month, YYYY = Year, hh = hours, mm = minutes, ss = seconds

Example: ESC MFB h e ESC MFB1 e k "02.08.2015 13:26:48" ESC MFE

Code old: *ESC MFB I <TimeDate> ESC MFE*

Parameter	*Value*	*Format*
<TimeDate>	*30h ≤ char ≤ 39h*	*ASCII "ss,mm,hh;DD,MM,YY"* *ss = seconds, mm = minutes, hh = hour,* *DD = Day, MM = Month, YY = Year*

Example old: *ESC MFB I "48,26,13;02,08,12" ESC MFE*

It is not recommended to use the old command syntax for new implementations.
This syntax is obsolete and may be not supported in the future.

Condition:

MFC Status byte 1

MFC_ TKD_RESET	MFC_ RECEIPT_ TOTAL	MFC_ DOCU_OPEN	MFC_ REPORT_ OPEN	MFC_ RECEIPT_ OPEN	MFC_ DAY_OPEN	MFC_ TRAINING_ MODE	MFC_ BLOCKADE
x	**x**	**x**	**x**	**x**	**0**	**x**	**x**

MFC Status byte 2

not used	not used	EJ_DATA_ PENDING	EJ_FULL	EJ_NEAR_ FULL	MFC_ CMD_ INTERRUPTED	MFC_ PRINTOUT_ INTERRUPTED	MFC_ FISCAL_ MODE
x	**x**	**x**	**x**	**x**	**x**	**x**	**x**

Note: The date and time must be equal or greater than the date and time of the last written block in the fiscal memory (PFM).

In case of a date entry, which is more than **two days** later to the current valid date, the fiscal device has to be secured against any casual error of such entry. In this case the command will be refused with an error 62 EM_DATE_NOT_ALLOWED.

The "two days" rule **does not apply** when the printer is in **blockade mode**

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(MFC_BLOCKADE == 1).

Error Msg:

EM_DATE_NOT_ALLOWED
EM_DATE_NOT_PRESENT
EM_SNO_COMPARE_ERROR
EM_SNO_NOT_PRESENT
EM_CMD_WRONG_ORDER
EM_PARA_DATE_WRONG_FORMAT
EM_MFMEM_NOT_CONNECTED
EM_PARA_DATE_WRONG_FORMAT
EM_RAM_CHECKSUM_ERROR
Error messages from the set of 8.2 FM error codes

Printout: Administrative receipt with optional header lines:

```
******  W I N C O R - N I X D O R F      *******
        receipt header line 2
        receipt header line 3
        receipt header line 4
   Cod Fiscal C.I.F.:1234567890123 4

****   BON NEFISCAL   ****   BON NEFISCAL   ****

          S E T _ D A T E _ T I M E

******          NOUA DATA SI ORA          ******

******          ZZ.LL.AAAA  HH:MM:SS      ******
******          16.05.2015  17:57:56      ******

     DATA: 16/05/2015     ORA:17-57-56
****   BON NEFISCAL   ****   BON NEFISCAL   ****
```

7.4.2 SET_VATRATES

Function: With this command the application specifies a new set of VAT percentages that will be used in calculating VAT amounts.

Due to previous inconsistencies between the value sent for the VAT Rates (0-7) and the naming, all VAT categories are now referenced with the category character instead of numbering.

Code:

ESC MFB h a
ESC MFB1 c b <*VATRateA*>
ESC MFB1 c c <*VATRateB*>
ESC MFB1 c d <*VATRateC*>
ESC MFB1 c e <*VATRateD*>
ESC MFB1 c f <*VATRateE*>

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ESC MFB1 c g <*VATRateF*>
ESC MFB1 c h <*VATRateG*>
ESC MFB1 c i <*VATRateH*> ESC MFE

Parameter	Value	Format
<VATRateA>	30H ≤ *n* ≤ 39H	4 digits, category " A "
<VATRateB>	30H ≤ *n* ≤ 39H	4 digits, category " B "
<VATRateC>	30H ≤ *n* ≤ 39H	4 digits, category " C "
<VATRateD>	30H ≤ *n* ≤ 39H	4 digits, category " D "
<VATRateE>	always zero	4 digits, category " E "
<VATRateF>	always zero	4 digits, category " F "
<VATRateG>	30H ≤ *n* ≤ 39H	4 digits, category " G "
<VATRateH>	30H ≤ *n* ≤ 39H	4 digits, category " H "

Example:

<VATRate_A> = 24%, <VATRate_B> = 9%, <VATRate_C> = 7,5%, <VATRate_D> = 0%, <VATRate_E> = 0%, <VATRate_F> = 0%, <VATRate_G> and <VATRate_H> unused.

ESC MFB h a ESC MFB1 c b 2400 ESC MFB1 c c 0900 ESC MFB1 c d 0750 ESC MFB1 c e 0000 MFB1 c f 0000 MFB1 c g 0000 ESC MFB1 c h 9999 ESC MFB1 c i 9999 ESC MFE

Condition:

MFC Status byte 1

MFC_ TKD_RESET	MFC_ RECEIPT_ TOTAL	MFC_ DOCU_OPEN	MFC_ REPORT_ OPEN	MFC_ RECEIPT_ OPEN	MFC_ DAY_OPEN	MFC_ TRAINING_ MODE	MFC_ BLOCKADE
x	**x**	**x**	**x**	**x**	**0**	**x**	**0**

MFC Status byte 2

not used	not used	EJ_DATA_ PENDING	EJ_FULL	EJ_NEAR_ FULL	MFC_ CMD_ INTERRUPTED	MFC_ PRINTOUT_ INTERRUPTED	MFC_ FISCAL_ MODE
x	**x**	**x**	**x**	**x**	**x**	**x**	**1**

Note:

1. A VAT rate change block is burned into the MF memory with the supplied data (if different from the last one written).
2. The MFC only accepts a certain number of changes to the VAT block. The number is defined in 5.7.10 Block restriction. After the defined limit is reached, the command is refused with the error 2B EM_MFMEM_BL_LIMIT_OVERRUN. 7.15.19 GET_MFMEM_CNT can be used to find the number of TH blocks already written.
3. All 8 VAT rates must be supplied.
4. The VAT E and F are used internally and cannot be changed. Any values different from zero received in the SET_VATRATES commands will be rejected with EM_PARA_WRONG.
5. VAT rates which are not used by the application must be set to 99,99%. The MF controller does not use those categories which have a VAT rate equal to 99,99%, in reports or in other commands.
6. The first 4 VAT categories are mandatory to be set **unequal** to 99,99%. Otherwise the command is rejected with EM_PARA_WRONG.

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7. VAT G and H are only printed only if the VAT category is active.
8. Command is ignored if all the new vat rates are identical to all the old vat rates.

Please refer to chapter 5.7.2 for the VAT block format.

Error Msg:

EM_PARA_WRONG
EM_PARA_WRONG_FORMAT
EM_PARA_MISSING
EM_MFMEM_BL_LIMIT_OVERRUN
EM_CMD_WRONG_ORDER

Error messages from the set of 8.2 FM error codes.

Error messages from the set of 8.4 Supplementary checks error codes.

Printout: Administrative receipt with optional header lines:

```
******  W I N C O R - N I X D O R F      ******
        receipt header line 2
        receipt header line 3
        receipt header line 4
     Cod Fiscal C.I.F.:1234567890123 4

****   BON NEFISCAL   ****   BON NEFISCAL   ****

        S E T _ V A T R A T E S

******  NUMĂR DE MODIFICĂRI COTE DE TVA  ******
******  MAI SUNT PERMISE: 29             ******

      TVA_A - veche  0,00% noua 21,00%
      TVA_B - veche  0,00% noua  9,00%
      TVA_C - veche  0,00% noua  7,50%
      TVA_D - veche  0,00% noua  0,00%
      TVA_E - veche  0,00% noua  0,00%
      TVA_F - veche  0,00% noua  0,00%

      DATA:16/05/2015   ORA:08-11-06
****   BON NEFISCAL   ****   BON NEFISCAL   ****
```

The lines for TVA_G and TVA_H will only be printed if they are active (not equal to 99,99%)

7.4.3 SET_HEADER

Function: With this command the application specifies the header contents that will be printed within a legal receipt header.

Code:

ESC MFB h b
ESC MFB1 b l <*ReceiptHeaderLine1*>
ESC MFB1 b m <*ReceiptHeaderLine2*>
[ESC MFB2 b n <*ReceiptHeaderLine3*>]
[ESC MFB2 b o <*ReceiptHeaderLine4*>]
ESC MFB1 b p <*ReceiptHeaderLine5*>

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[ESC MFB2 d o <*CIF_ID*>]
ESC MFE

Parameter	Value	Format	
<ReceiptHeaderLine1> □ □ □ *<ReceiptHeaderLine4>*	20h* ≤ *n* ≤ FFh	Maximal numbers of printable characters per line: 44 Maximal numbers of printable characters and printer commands per line: 79 * See Application hint: for further details	
<ReceiptHeaderLine5>	20h* ≤ *n* ≤ FFh	Maximal numbers of printable characters per line: 25 No printer commands allowed	**D**
<CIF_ID>	20 H ≤ n ≤ FF H	2 bytes (alphanumeric characters) allowed range: a-z, A-Z, 0-9, Space(0x20)	**I**

Condition:

MFC Status byte 1

MFC_ TKD_RESET	MFC_ RECEIPT_ TOTAL	MFC_ DOCU_OPEN	MFC_ REPORT_ OPEN	MFC_ RECEIPT_ OPEN	MFC_ DAY_OPEN	MFC_ TRAINING_ MODE	MFC_ BLOCKADE
x	**x**	**x**	**x**	**x**	**0**	**x**	**0**

MFC Status byte 2

not used	not used	EJ_DATA_ PENDING	EJ_FULL	EJ_NEAR_ FULL	MFC_ CMD_ INTERRUPTED	MFC_ PRINTOUT_ INTERRUPTED	MFC_ FISCAL_ MODE
x	**x**	**x**	**x**	**x**	**x**	**x**	**1**

Note: **The POS application or service software tool is fully responsible for the ticket header layout!**

1. The firmware terminates every parameter *<ReceiptHeaderLine1..4>* with a CR and LF (exceptions: print line contains graphic and barcode commands)
2. If one of the parameter conditions is not fulfilled, the command is rejected with EM_PARA_WRONG_FORMAT.
3. A TH change block is burned into the MF memory with the supplied data (if different from the existing ones). Please refer to chapter 5.7.6 TH block for the TH change block format.
4. *<ReceiptHeaderLine5>* The CIF will be inserted automatically with the format “12345678901234 (see chapter 7.3.4 INIT_WRITE_SNO for more details)

D	D	D	D	D	D	D	D	D	D	D	D	D	D	D	D	D	D	D	D	D	D	D	D	I	I	1	2	3	4	5	6	7	8	9	0	1	2	3

5. If <ReceiptHeaderLineN> is filled with blanks the command will be refused with EM_PARA_WRONG_FORMAT.
6. The MFC only accepts a certain number of changes to the TH block. The number is defined in 5.7.10 Block restriction. After the defined limit is

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reached, the command is refused with the error 2B EM_MFMEM_BL_LIMIT_OVERRUN. 7.15.19 GET_MFMEM_CNT can be used to find the number of TH blocks already written.

Error Msg: EM_CMD_WRONG_ORDER
EM_PARA_MISSING
EM_PARA_WRONG
EM_PARA_WRONG_FORMAT
EM_MFMEM_BL_LIMIT_OVERRUN
Error messages from the set of 8.2 FM error codes.
Error messages from the set of 8.4 Supplementary checks error codes.

Printout: Administrative receipt with optional header lines:

```
******  W I N C O R - N I X D O R F      ******
              receipt header line 2
              receipt header line 3
              receipt header line 4
      Cod Fiscal C.I.F.: RO12345678901 2 3

****   BON NEFISCAL   ****   BON NEFISCAL   ****

              S E T _ H E A D E R

******  NUMĂRUL DE MODIFICĂRI DE ANTET  ******
******  MAI SUNT PERMISE: 29            ******

      DATA:16/05/2015    ORA:08-11-06
****   BON NEFISCAL   ****   BON NEFISCAL   ****
```

The following printer commands are allowed in the header definition:

(Please refer to the TH230+ Programmers Guide for further information about the printer commands)

<u>Print Attributes</u>

ESC !	Select printer mode
ESC a	Select justification
ESC I	Turn italic mode on/off
GS B	white/black reverse printing
US ENQ	Select superscript/subscript

<u>Graphic</u>

GS #	Select current logo
GS /	Print current logo
GS (L fn=69	Print the specified graphic

<u>Barcode</u>

GS H	Printing Position of HRI
GS f	Select Font for HRI

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GS h	Set barcode height
GS w	Set barcode width
GS k	Print bar code
GS p	Select PDF417 parameters
Color	
ESC r	Set current color

If the application prints a graphic, the previous line should be terminated by LF.

7.4.4 SET_CURRENCY_COUNTRY

Function: This command is used to inform the MFC that from this point on all turnovers, fiscal transactions and receipts are stored and calculated in other (formerly New Romanian Lei) currency.

Code: **ESC MFB h d**
ESC MFB1 i m <*CurrencyLabel*> ESC MFE

Parameter	Value	Format
<*CurrencyLabel*>	20 H ≤ *n* ≤ FF H	Valid (case-sensitive) values are: - “Lei” (default) - “Euro”

Condition:

MFC Status byte 1

MFC_ TKD_RESET	MFC_ RECEIPT_ TOTAL	MFC_ DOCU_OPEN	MFC_ REPORT_ OPEN	MFC_ RECEIPT_ OPEN	MFC_ DAY_OPEN	MFC_ TRAINING_ MODE	MFC_ BLOCKADE
x	x	x	x	x	0	x	0

MFC Status byte 2

not used	not used	EJ_DATA_ PENDING	EJ_FULL	EJ_NEAR_ FULL	MFC_ CMD_ INTERRUPTED	MFC_ PRINTOUT_ INTERRUPTED	MFC_ FISCAL_ MODE
x	x	x	x	x	x	x	1

Note:

1. The application is fully responsible for the exact moment of the transmission of this command.
2. This command is used to switch to another currency. The application has to provide a mechanism to force the operator to confirm by manager password the activity when using this command.
3. One Currency Change Block (CRY) is created and stored into the MFMEM. Please refer to chapter 5.7.8 CRY block for the CRY change block format.
4. The command is ignored (without error), if <*CurrencyLabel*> is the same as last
5. The currency mode can be read by GET_CURRENCY_COUNTRY.
6. The currency label is printed in front of the first article inside a fiscal receipt.

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7. The MFC only accepts a certain number of changes to the CRY block. The number is defined in 5.7.10 Block restriction. After the defined limit is reached, the command is refused with the error 2B EM_MFMEM_x_LIMIT_OVERRUN.
GET_MFMEM_CNT can be used to find the number of TH blocks already written.

Error Message: EM_CMD_WRONG_ORDER
EM_PARA_MISSING
EM_PARA_WRONG
EM_PARA_WRONG_FORMAT
EM_MFMEM_BL_LIMIT_OVERRUN
Error messages from the set of FM error codes (8.2).
Error messages from the set of Supplementary checks error codes (8.4).

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Printout: Administrative receipt with optional header lines:

```
******  W I N C O R - N I X D O R F      *******
                 receipt header line 2
                 receipt header line 3
                 receipt header line 4
          Cod Fiscal C.I.F.:1234567890123 4

****   BON NEFISCAL   ****   BON NEFISCAL    ****

                 S E T _ C U R R E N C Y

******  NUMĂRUL DE SCHIMBĂRI DE MONEDĂ  ******
******  CARE MAI SUNT PERMISE: 29       ******

   B O N   p e n t r u   D E N O M I N A R E

          ultimul bloc EURO cu  Lei: 10

                                          EURO
   TVA  (A)
          VÂNZĂRI BRUTE                 4.734,12
          VALOARE TVA                     882,91
   TVA  (B)
          VÂNZĂRI BRUTE              11.050,20
          VALOARE TVA                  2.188,06
   TVA  (C)
          VÂNZĂRI BRUTE                     0,00
          VALOARE TVA                       0,00
   TVA  (D)
          VÂNZĂRI BRUTE                     0,00
          VALOARE TVA                       0,00

   TOTAL VÂN. EXCEPTATE TVA                 0,00
   TOTAL VÂNZĂRI ALTE TAXE                  0,00

   TVA  (F)
          VÂNZĂRI BRUTE                     0,00
          VALOARE TVA                       0,00
   TVA  (G)
          VÂNZĂRI BRUTE                     0,00
          VALOARE TVA                       0,00
   TVA  (H)
          VÂNZĂRI BRUTE                     0,00
          VALOARE TVA                       0,00

   TOTAL GENERAL
       VÂNZĂRI TAXABILE                15.784,32
       VALOARE TVA                      3.070,97

          DATA:16/05/2015   ORA:08-11-06
****   BON NEFISCAL   ****   BON NEFISCAL    ****
```

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7.4.5 SET_PAYMENT_LIST

Function: This command sets the contents of one payment list entry.

Code:

ESC MFB h s
ESC MFB1 g a <*PaymentType*>
ESC MFB1 g e <*PaymentName*>
ESC MFB1 g f <*PaymentCategory*>
ESC MFE

Parameter	Value	Format
<PaymentType> Index of the entry in the payment list	30 H ≤ *n* ≤ 39 H	max. 2 bytes 9..19 0 – cash is fixed 1 – card is fixed 2 – food stamp is fixed 3 – voucher is fixed 4 – ticket value is fixed 5 – credit is fixed 6 – cash back is fixed 7 – modern media is fixed 8 – other is fixed
<***PaymentName***>	20 H ≤ *n* ≤ FF H	27 bytes
<PaymentCategory>	30 H ≤ *n* ≤ 39 H	1 bytes 0…2 0 = PC_LOCAL_CASH 1 = PC_FOREIGN_CASH 2 = PC_NON_CASH

Condition:

MFC Status byte 1

MFC_ TKD_RESET	MFC_ RECEIPT_ TOTAL_	MFC_ DOCU_OPEN	MFC_ REPORT_ OPEN_	MFC_ RECEIPT_ OPEN_	MFC_ DAY_OPEN	MFC_ TRAINING_ MODE	MFC_ BLOCKADE
X	**X**	**x**	**x**	**x**	**0**	**x**	**0**

MFC Status byte 2

not used	not used	EJ_DATA_ PENDING	EJ_FULL	EJ_NEAR_ FULL	MFC_ CMD_ INTERRUPTED	MFC_ PRINTOUT_ INTERRUPTED	MFC_ FISCAL_ MODE
X	**X**	**x**	**x**	**x**	**x**	**x**	**x**

Note: If a list entry is successfully stored the <EntryValid> byte is set:

```
astPaymentList[ucListIndex].stListEntry.ucEntryValid = TRUE
```

All fixed payment media entries can't be changed.

See chapter 7.10.1 PAYMENT_IN

The sequence of the valid list entries and possible double entries or gaps between the entries is organised by the application.

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A list entry can be changed or modified by multiple usage of SET_PAYMENT_LIST command.

Error Msg: EM_CMD_WRONG_ORDER
EM_PARA_WRONG_FORMAT
EM_PARA_MISSING
EM_PARA_WRONG

Error messages from the set of 8.4 Supplementary checks error codes.

Printout: Administrative receipt with optional header lines:

```
******  W I N C O R - N I X D O R F      ******
          rEceipt header line 2
          rEceipt header line 3
          rEceipt header line 4
     Cod FisCal C.I.F.:1234567890123 4

****   BON NEFISCAL   ****   BON NEFISCAL   ****

          S E T _ P A Y M E N T _ L I S T

TIPUL PLATII      :<PAYMENT TYPE>
CATEGORIA PLATII:<PAYMENT CATEGORY>
NUMELE PLATII     :<PAYMENT NAME>

      DATA:16/05/2015    ORA:08-11-06
****   BON NEFISCAL   ****   BON NEFISCAL   ****
```

7.4.6 SET_PAYMENT_LIST_VOID

Function: This command deletes one payment list entry.

Code: **ESC MFB h t**
ESC MFB1 g a <*PaymentType*>
ESC MFE

Parameter	Value	Format
<PaymentType> Index of the entry in the payment list	30 H ≤ *n* ≤ 39 H	max. 2 bytes 9..19

Condition:

MFC Status byte 1

MFC_ TKD_RESET	MFC_ RECEIPT_ TOTAL	MFC_ DOCU_OPEN	MFC_ REPORT_ OPEN	MFC_ RECEIPT_ OPEN	MFC_ DAY_OPEN	MFC_ TRAINING_ MODE	MFC_ BLOCKADE
x	**X**	**x**	**x**	**x**	**0**	**x**	**0**

MFC Status byte 2

not used	not used	EJ_DATA_ PENDING	EJ_FULL	EJ_NEAR_ FULL	MFC_ CMD_ INTERRUPTED	MFC_ PRINTOUT_ INTERRUPTED	MFC_ FISCAL_ MODE
x	**X**	**x**	**x**	**x**	**X**	**x**	**x**

Note: If a list entry is successfully deleted, then the <EntryValid> byte is reset:

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```
astPaymentList[ucListIndex].stListEntry.ucEntryValid = FALSE
```

All fixed payment media entries can't be changed.

The sequence of the valid list entries and possible double entries or gaps between the entries is organised by the application.

Error Msg: EM_PARA_WRONG_FORMAT
EM_PARA_MISSING
EM_PARA_WRONG
Error messages from the set of 8.4 Supplementary checks error codes.

7.4.7 SET_WORKING_MODE

Function: This command sets the working modes of the device.

Code: **ESC MFB h y**
ESC MFB1 i o <*WorkingMode*>
ESC MFE

Parameter	Value	Format
<WorkingMode> Mode entry	30 H ≤ *n* ≤ 33 H	max. 1 byte 0..3 0 DEFAULT Mode 1 TEST Mode 2 PROBA Mode 3 INITIERE Mode

Condition:

MFC Status byte 1

MFC_ TKD_RESET	MFC_ RECEIPT_ TOTAL	MFC_ DOCU_OPEN	MFC_ REPORT_ OPEN	MFC_ RECEIPT_ OPEN	MFC_ DAY_OPEN	MFC_ TRAINING_ MODE	MFC_ BLOCKADE
x	X	x	x	x	0	x	0

MFC Status byte 2

not used	not used	EJ_DATA_ PENDING	EJ_FULL	EJ_NEAR_ FULL	MFC_ CMD_ INTERRUPTED	MFC_ PRINTOUT_ INTERRUPTED	MFC_ FISCAL_ MODE
x	x	x	x	x	X	x	x

Note:

Error Msg: EM_PARA_WRONG_FORMAT
EM_PARA_MISSING
EM_PARA_WRONG
Error messages from the set of 8.4 Supplementary checks error codes.

7.4.8 SET_CUT_MODE

Function: This command sets the working modes of all cut commands.

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Code:

ESC MFB h 2
ESC MFB1 f f <*CutType*>
ESC MFE

Parameter	Value	Format
<CutType>	30 H , 31H	0 full cut 1 partial cut

Condition:

MFC Status byte 1

MFC_ TKD_RESET	MFC_ RECEIPT_ TOTAL	MFC_ DOCU_OPEN	MFC_ REPORT_ OPEN	MFC_ RECEIPT_ OPEN	MFC_ DAY_OPEN	MFC_ TRAINING_ MODE	MFC_ BLOCKADE
x	**x**	**x**	**x**	**x**	**0**	**x**	**0**

MFC Status byte 2

not used	not used	EJ_DATA_ PENDING	EJ_FULL	EJ_NEAR_ FULL	MFC_ CMD_ INTERRUPTED	MFC_ PRINTOUT_ INTERRUPTED	MFC_ FISCAL_ MODE
x	**x**	**x**	**x**	**x**	**x**	**x**	**x**

Note:

The cut type is stored in the non volatile RAM and is after power on further valid.

The default value after initialisation and after TKD-Reset is “full cut”.

Error Msg:

EM_PARA_WRONG_FORMAT
EM_PARA_MISSING
EM_PARA_WRONG

Error messages from the set of 8.4 Supplementary checks error codes.

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7.5 DAY commands

All DAY_ commands may return error messages from the set of 8.3 EJ error codes and from 8.4 Supplementary checks error codes.

7.5.1 DAY_BEGIN

Function: This command starts a fiscal day, enables fiscal commands (e. g. ARTICLE_SELL) and starts the EJ session.

Code: **ESC MFB a a ESC MFE**

Condition:

MFC Status byte 1

MFC_ TKD_RESET	MFC_ RECEIPT_ TOTAL_	MFC_ DOCU_OPEN	MFC_ REPORT_ OPEN_	MFC_ RECEIPT_ OPEN_	MFC_ DAY_OPEN	MFC_ TRAINING_ MODE	MFC_ BLOCKADE
x	x	x	x	x	0	x	0

MFC Status byte 2

not used	not used	EJ_DATA_ PENDING	EJ_FULL	EJ_NEAR_ FULL	MFC_ CMD_ INTERRUPTED	MFC_ PRINTOUT_ INTERRUPTED	MFC_ FISCAL_ MODE
x	x	x	0	0	x	x	1

Note: After the command is successfully executed the status bit MFC_DAY_OPEN is set.

The command is only available if a header is defined (SET_HEADER) and the VAT rates are stored in the MFMEM (SET_VATRATES).

If at the time of issuing this command the PAPER NEAR END indication is on, the day will not be opened. The command will be rejected with an error EM_PRT_PAPER.

The command will be rejected with an error, if the Tax Terminal is not enabled (paired), any Tax Terminal component is not available or the customer display is not connected.

Error Msg:
EM_CMD_WRONG_ORDER
EM_UPDATE_JUMPER_SET
EM_VAT_NOT_PRESENT
EM_TH_NOT_PRESENT
EM_MFMEM_FULL
EM_PRT_PAPER
EM_TAX_TERMINAL_NOT_PRESENT
EM_TAX_TERMINAL_NOT_PAIRED
EM_TAX_TERMINAL_WRONG
EM_SC_DEFECT

7.5.2 DAY_END

Function: This command prints a complete Z report on the receipt (see printout), closes the fiscal day and ends the EJ session. The EJ session checksum is stored in the EOD block, automatically.

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Code: **ESC MFB a b ESC MFE**

Condition:

MFC Status byte 1

MFC_ TKD_RESET	MFC_ RECEIPT_ TOTAL	MFC_ DOCU_OPEN	MFC_ REPORT_ OPEN	MFC_ RECEIPT_ OPEN	MFC_ DAY_OPEN	MFC_ TRAINING__ MODE	MFC_ BLOCKADE
x	x	0	0	0	1	x	0

MFC Status byte 2

not used	not used	EJ_DATA_ PENDING	EJ_FULL	EJ_NEAR_ FULL	MFC_ CMD_ INTERRUPTED	MFC_ PRINTOUT_ INTERRUPTED	MFC_ FISCAL_ MODE
x	x	0	x	x	0	0	x

Computation: After this command the grand totalizers, the receipt totalizer, the EOD block counters and the numbers of fiscal tickets will be stored inside a new EOD block. See chapter 5.7.3 EOD block.

The FM block counter MFMEMCnt[BL_EOD] will be incremented (see chapter 7.15.19 GET_MFMEM_CNT).

The counter Day.ReceiptGroupCnt[EJ_TICKETS] will be incremented.

After the command is successfully executed the status bit MFC_OPEN is cleared and all day totalizers will be set to zero.

Note: Unlike the X report (see 'Note' in 7.5.3 DAY_XREPORT), the amounts that appear on the Z report reflect the actual totalizers inside the CMOS and are also stored in the MF Memory.

If at the time of issuing this command the PAPER NEAR END indication is on, the report will not be printed. The command will be rejected with an error EM_PRT_PAPER.

At the end of the Z report additional information about TKD resets since the last Z report and information about the condition of the MFMEM will appear (MFMEM nearly-full or full).

Before printing the MF Line the EOD block is stored and all MF memory counters and pointers are updated.

In case of an interruption during printing, the MFC_CMD_INTERRUPTED bit will be set and the application can restart this interrupted command by send a SPECIAL_CMD_RESTART command (see chapter 7.12.6 SPECIAL_CMD_RESTART).

Once a new DAY_END has been received, every power fail will force the printing of **a new Z report** after next power on (without cutting the previously interrupted Z report). The interruption is denoted by the 'CADERE TENSIUNE' text at the current stop-point.

Not used VAT categories (see chapter 7.4.2 SET_VATRATES) are not printed.

If memory reach less than 60 EOD blocks available on Z report must be printed a message "**`XX rapoarte zilnice pana la memorie plina`**" that means "**`XX daily Z report until memory full`**". Where XX is the number of free EOD blocks still available.

If memory is full, on the last Z report must be printed a message "**`Memoria fiscala este plina`**" that means "**`memory full`**".

Error Msg: EM_CMD_WRONG_ORDER

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EM_PARA_WRONG
EM_SC_DEFECT
EM_TAX_TERMINAL_NOT_PRESENT
EM_TAX_TERMINAL_WRONG

Printout: See 7.5.4 Z/X Report

7.5.3 DAY_XREPORT

Function: With this command the application requests an X report. This report is either printed on the receipt station or is held ready as soft copy to be fetched by the application.

Code: **ESC MFB a c [ESC MFB2 f a <*Station*>] ESC MFE**

Parameter	Value	Format
<***Station***>	30 H ≤ *n* ≤ 39 H	"2" – Receipt (Default) "16" – Host – For EJ soft copy.

Condition:

MFC Status byte 1

MFC_ TKD_RESET	MFC_ RECEIPT_ TOTAL	MFC_ DOCU_OPEN	MFC_ REPORT_ OPEN	MFC_ RECEIPT_ OPEN	MFC_ DAY_OPEN	MFC_ TRAINING_ MODE	MFC_ BLOCKADE
x	**x**	**x**	**0**	**0**	**1**	**x**	**0**

MFC Status byte 2

not used	not used	EJ_DATA_ PENDING	EJ_FULL	EJ_NEAR_ FULL	MFC_ CMD_ INTERRUPTED	MFC_ PRINTOUT_ INTERRUPTED	MFC_ FISCAL_ MODE
x	**x**	**x**	**x**	**x**	**0**	**x**	**x**

Computation: No calculations are performed.

Note: This command sets the REPORT_OPEN bit.

In case of power-fail while printing the application needs to issue a SPECIAL_ALL_VOID command in order to resume normal operations.

If at the time of issuing this command the PAPER NEAR END indication is on, the report will not be printed.

Error Msg: EM_CMD_WRONG_ORDER

Printout: See 7.5.4 Z/X Report

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7.5.4 Z/X Report

```
012345678901234567890123456789012345678901234567890123
```

***** W I N C O R - N I X D O R F *****	Header Line 1
receipt header line 2	Header Line 2
receipt header line 3	Header Line 3
receipt header line 4	Header Line 4
Cod Fiscal C.I.F.: RO1234567890123	Header Line 5

…

R A P O R T F I S C A L Z I L N I C - Z	Z report

or

R A P O R T I N T E R M E D I A R (R A P O R T - X)	*X report*

…

	Lei	Currency label	
TOTAL VÂNZĂRI BRUTE	3.258,74	Daily Total	Day.Tot[BRUTTO_SALES]
Nr. DISCOUNT	12	Number of Discounts	Day.EventCnt[DISCOUNT]
TOTAL DISCOUNT	120,00	Total of Discount	Day.Tot[DISCOUNT]
Nr. MAJORĂRI	10	Number of Surcharge	Day.EventCnt[SURCHARGE]
TOTAL MAJORĂRI	75,50	Total of Surcharge	Day.Tot[SURCHARGE]
Nr. CORECŢII	254	Number of Returns	Day.EventCnt[RETURN]
TOTAL CORECŢII	4.102,33	Total of Return	Day.Tot[RETURN]
Nr. ANULĂRI	31	Number of Voids	Day.EventCnt[VOID+SP.VOID]
TOTAL SUME ANULATE	820,30	Total of Voids	Day.Tot[VOID+SP.VOID]
Nr. BONURI ANULATE	0	Number of Ticket Void	Day.EventCnt[SP.VOID]
Nr. ARTICOLE ANULATE	0	Number of Article Void	Day.EventCnt[VOID]
TOTAL ARTICOLE ANULATE	0,00	Total of Article Void	Day.Tot[VOID]

TVA (A) 00,00%		VAT rate A	
VÂNZĂRI BRUTE	1.710,73	Total of VAT A	Day.BruttoSales[VAT_A]
VALOARE TVA	0,00	VAT amount of VAT A	Day.VAT[VAT_A]

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TVA (B) 19,00%		VAT rate B	
VÂNZĂRI BRUTE	11,88	Total of VAT B	Day.BruttoSales[VAT_B]
VALOARE TVA	1,90	VAT amount of VAT B	Day.VAT[VAT_B]
TVA (C) 07,50%		VAT rate C	
VÂNZĂRI BRUTE	16,92	Total of VAT C	Day.BruttoSales[VAT_C]
VALOARE TVA	1,18	VAT amount of VAT C	Day.VAT[VAT_C]
TVA (D) 32,25%		VAT rate D	
VÂNZĂRI BRUTE	11,99	Total of VAT D	Day.BruttoSales[VAT_D]
VALOARE TVA	2,92	VAT amount of VAT D	Day.VAT[VAT_D]
TOTAL VÂN. EXCEPTATE TVA	36,02	Total of VAT E = 0%	Day.BruttoSales[VAT_E]
TOTAL VÂNZĂRI ALTE TAXE	1.471,20	Total of Road Tax	Day.VAT[ROAD_TAX(VAT_F)]
TVA (G) 02,50%		VAT rate G	
VÂNZĂRI BRUTE	2.000,00	Total of VAT G	**Summary for VAT G and H is printed only if the VAT categories are switched on. See chapter 7.4.2 SET_VATRATES.**
VALOARE TVA	50,00	VAT amount of VAT G	
TVA (H) 19,00%		VAT rate H	
VÂNZĂRI BRUTE	25,00	Total of VAT H	
VALOARE TVA	5,05	VAT amount of VAT H	
TOTAL GENERAL			
VÂNZĂRI BRUTE	5.247,72	Sum of brutto sales	
VÂNZĂRI TAXABILE	3.776,52	Sum of taxable brutto sales (VAT A, B, C, D, G, H)	
VALOARE TVA	61,05	Sum of VAT amounts (VAT A, B, C, D, G, H)	

CONTOR ZILE:			
RAPORT Z NUMĂRUL	9	Z report: EOD number incl. open day	
MODIFICĂRI COTĂ TVA	1	VAT rate changes	
MODIFICĂRI ANTET	1	Ticket header changes	
NUMAR RESET TKD	1	Number of TKD resets	
MODIFICĂRI MONEDĂ	0	*Number of CRY changes*	
MODIFICĂRI JURNAL ELECTRONIC	3	*Number of EJ change blocks*	
NR. BONURI FISCALE	22		

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NR. BONURI NE-FISCALE	14	*Number of fiscal receipts*	
----------		*Number of illegal receipts*	
NR. BONURI CU CIF	8		
TOTAL VÂNZĂRI CU CIF	580,24	*Number of Retail Sales Receipt with Customer Tax ID (CIF)*	
VALOARE TVA CIF	61,05	*Total of CIF receipts*	Day.Statistic[CIF]
		VAT amount of CIF	Day.VAT[CIF]
ID JURNAL ELECTRONIC 01234567890123456789		*Electronic Journal UMID*	
DATA INITIALIZARII 09.10.2017 17:19:50		*Initialization Date Time*	
NR. MEDIU SD CARD	1	*Medium Number*	
NR. FISIER EJ	7	*Number of EJ Sessions*	

NR. SOLDURI INIŢIALE	0	*No. of Initial Float operations*	
SOLD INIŢIAL	100,00	*Value of initial Float*	
NR. ALIMENTARI EXTRA	0	*No. of extra Float operations*	
ALIMENTARE EXTRA	500,00	*Value of extra Float*	
NR. REMITERI	0	*No of Pick-up operations*	
REMITERE	300,00	*Value of Pick-Ups*	
NR. ÎNCASĂRI	0	*No. of Paid In operations*	
ÎNCASĂRI	14.189,56	*Value of Paid In*	
NR. PLAŢI	0	*No of Paid Out operations*	
PLĂŢI	10.113,04	*Value of Paid Out*	

CASH	2.820,70	*value of the cash payment - REST + CASH_IN - CASH_OUT*	Day.Payment[CASH]-Day.Tot[REST]
CASH ÎN SERTAR	2.010,00		Day.CashDrawBalance[CHASH_IN_DRAW]
CARD	995,82	*Value of payment Card*	
TICHETE DE MASA	0,00	*Value of payment Food Stamp*	

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VOUCHER	0,00	
BON/TICHET VALORIC	0,00	
CREDIT	0,00	*Value of payment Voucher*
AVANS IN NUMERAR	0,00	
METODE MODERNE de PLATA	0,00	
<PaymentName 9>	5,82	*Value of payment Other*
. . .		*Value of type 9 payment*
. . .		
<PaymentName 19>	5,00	*Value of type 19 payment*
LINII LIBERE IN JURNAL:	**2.654.234**	*Information if avail. print lines < 0.5% of User Capacity*
**** BLOC TKD NR. 1 03.04.2008 15:31:27 ****		*Information if a TKD reset happened after previous EOD*

…

EJTRZ: 00037	Z report
DATA:03/04/2008 ORA:15-42-23	
RL BU47101234567	
R A P O R T F I S C A L	
562e8372 3047a912 feaa55d7 1c11a362 51fa1b06 6156b4cf 0534d5a1 2abd1994	*EJ session checksum (SHA-2 32 bytes / 64 characters)*
XX rapoarte zilnice pana la memorie plina	*Printed if MFMEM near full (XX <= 60)*
Memoria fiscala este plina	*Printed if MFMEM full*

or

DATA:03/04/2008 ORA:15-42-23	*X report*
RL BU47101234567	
R A P O R T I N T E R M E D I A R	

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7.6 RECEIPT commands

All RECEIPT_ commands may return error messages from the set of 8.4 Supplementary checks error codes.

7.6.1 RECEIPT_BEGIN

Function: With this command the application specifies the type of the receipt and starts an EJ ticket.

Code:

ESC MFB b a
ESC MFB1 f b <*ReceiptType*> ESC MFE

Parameter	Value	Format
<ReceiptType>	31 H ≤ *n* ≤ 36 H	see list of receipt types (5.4.1)

Condition:

MFC Status byte 1

MFC_ TKD_RESET	MFC_ RECEIPT_ TOTAL	MFC_ DOCU_OPEN	MFC_ REPORT_ OPEN	MFC_ RECEIPT_ OPEN	MFC_ DAY_OPEN	MFC_ TRAINING_ MODE	MFC_ BLOCKADE
0	**x**	**0**	**0**	**0**	**1**	**x**	**0**

MFC Status byte 2

not used	not used	EJ_DATA_ PENDING	EJ_FULL	EJ_NEAR_ FULL	MFC_ CMD_ INTERRUPTED	MFC_ PRINTOUT_ INTERRUPTED	MFC_ FISCAL_ MODE
x	**x**	**x**	**0**	**0**	**0**	**0**	**x**

Computation: The according receipt type and receipt group counters will be incremented (see chapter 5.5 Counters and totalizers)

Day.ReceiptCnt[<*ReceiptType*>] += 1

If <*ReceiptType*> = RETAIL_SALES_RECEIPT|
RETAIL_SALES_RECEIPT_WITH_TAX_ID

Day.ReceiptGroupCnt[LEGAL_RECEIPT] += 1

Day.ReceiptGroupCnt[EJ_TICKET] += 1
stGrand.ulGrandLegalReceiptCnt += 1

If <*ReceiptType*> = GENERIC_RECEIPT or
CASH_IN_RECEIPT or
CASH_OUT_RECEIPT

Day.ReceiptGroupCnt[ILLEGAL_RECEIPT] += 1

Note: The MFC deletes all receipt totalizers.

After the RECEIPT_OPEN command is successfully executed the status bit MFC_RECEIPT_OPEN will be set.

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The command will be rejected with an error, if the Tax Terminal is not enabled (paired), any Tax Terminal component is not available or the customer display is not connected.

24 hour rule: This rule shall ensure that RETAIL_SALES_RECEIPT and RETAIL_SALES_RECEIPT_WITH_TAX_ID's are allowed in one fiscal day only within a window of 24 hours. The start time is available with command 7.15.34 GET_DATE_FIRST_RECEIPT. If 24 hours after the first RETAIL_SALES_RECEIPT have passed by no further RECEIPT_BEGIN (with receipt type=RETAIL_SALES_RECEIPT) is allowed. In this case the command is refused with error messages 0x86 EM_DAY_END_REQUIRED. For a new RETAIL_SALES_RECEIPT a DAY_END/DAY_BEGIN is necessary.

Error Msg:

EM_CMD_WRONG_ORDER
EM_DAY_END_REQUIRED
EM_PARA_WRONG
EM_PARA_MISSING
EM_PARA_RANGE_ERROR
EM_TAX_TERMINAL_NOT_PRESENT
EM_TAX_TERMINAL_NOT_PAIRED
EM_TAX_TERMINAL_WRONG
EM_SC_DEFECT
Error messages from the set of 8.3 EJ error codes.

Printout:

If successful, the RECEIPT_BEGIN command produces the ticket header (defined by a previous SET_HEADER command)

If <ReceiptType> = RETAIL_SALES_RECEIPT:

XXXX = " Lei" if the CRY Block doesn't exist; otherwise
XXXX = <*CurrencyLabel*> defined by SET_CURRENCY_COUNTRY (chapter 7.4.4)

If <ReceiptType> = GENERIC_RECEIPT and CASH_IN/_OUT_RECEIPT:

```
*****  BON  NEFISCAL  ******  BON  NEFISCAL  *****
```

7.6.2 RECEIPT_END

Function:

The command ends the receipt by printing the footer and cuts the ticket. Depending on the receipt type specified by RECEIPT_BEGIN, the ticket is ended as "Legal" or "Illegal". The EJ ticket will be closed.

Code:

ESC MFB b b [ESC MFB2 b h <TaxID>] ESC MFE

Parameter	Value	Format	
<TaxID>	20 H ≤ *n* ≤ FF H	15 bytes	T

Condition:

MFC Status byte 1

MFC_ TKD_RESET	MFC_ RECEIPT_ TOTAL	MFC_ DOCU_OPEN	MFC_ REPORT_ OPEN	MFC_ RECEIPT_ OPEN	MFC_ DAY_OPEN	MFC_ TRAINING_ MODE	MFC_ BLOCKADE
0	**x**	**x**	**x**	**1**	**x**	**x**	**0**

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MFC Status byte 2

not used	not used	EJ_DATA_ PENDING	EJ_FULL	EJ_NEAR_ FULL	MFC_ CMD_ INTERRUPTED	MFC_ PRINTOUT_ INTERRUPTED	MFC_ FISCAL_ MODE
x	x	x	x	x	0	0	x

Computation: If the receipt to be closed is RETAIL_SALES_RECEIPT or RETAIL_SALES_RECEIPT_WITH_TAX_ID the following will be done:

Receipt.Statistic[CIF] += Receipt.Tot[BRUTTO_SALES]*
Receipt.Statistic[CIF_VAT] += ∑Receipt.VAT[VAT_K(A..E, G, H)]
Day.Tot[BRUTTO_SALES] += Receipt.Tot[BRUTTO_SALES]
Day.Tot[DISCOUNT] += Receipt.Tot[DISCOUNT]
Day.Tot[SURCHARGE] += Receipt.Tot[SURCHARGE]
Day.Tot[RETURN] += Receipt.Tot[RETURN]
Day.Tot[VOID] += Receipt.Tot[VOID]
Day.Tot[REST] += Receipt.Tot[REST]
Day.Statistic[CIF] += Receipt.Statistic[CIF]*
Day.Statistic[CIF_VAT] += Receipt.Statistic[CIF_VAT]*
Day.EventCnt[<*EventCounterType*>] += Receipt.EventCnt[<*EventCounterType*>]
Day.VAT[VAT_K(A...H)] += Receipt.VAT[VAT_K(A...H)]**
Day.Payment[<*PaymentType*>] += Receipt.Payment[<*PaymentType*>]

Day.CashDrawBalance[CASH_IN_DRAWER] += Receipt.Payment[CASH]

Day.CashDrawBalance[CASH_IN_DRAWER] -= Receipt.Payment[CASH_BACK]

* if the <*TaxID*> is given
** VAT_E fix = 0; VAT_F = Road tax; VAT_G and VAT_H optional
Day.EventCnt[CIF] += 1*

If the receipt to be closed is CASH_IN/OUT the following will be done:
Day.CashDrawBalance[<*CashDrawBalanceType*>] += Receipt.CashDrawBalance[<*CashDrawBalanceType*>]

If the receipt to be closed is PAYMENT_CORRECTION the following will be done:
Day.Payment[<*PaymentType*>] += Receipt.Payment[<*PaymentType*>]

Note: For receipt types RETAIL_SALES_RECEIPT and RETAIL_SALES_RECEIPT _WITH_TAX_ID the status MFC_RECEIPT_TOTAL must be set. Otherwise the command will be rejected.

For receipt type PAYMENT_CORRECTION the sum of Receipt.Payment[<*all PaymentTypes*>] must be zero. Otherwise the command will be rejected.

After the command is successfully executed the status bits MFC_RECEIPT_OPEN and MFC_RECEIPT_TOTAL will be cleared.

Error Msg:
EM_PAYMENT_TOTAL_WRONG
EM_CMD_WRONG_ORDER
EM_PARA_WRONG
EM_PARA_WRONG_FORMAT
EM_TAX_TERMINAL_NOT_PRESENT
EM_TAX_TERMINAL_WRONG

Printout: If <ReceiptType> = RETAIL_SALES_RECEIPT:

YYYY = EjTrackNumber of the current receipt

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```
C.I.F. Client: TTTTTTTTTTTTTTTT
EJTRZ: YYYYY
ID UNIC: 1234567890201706261255550001000 1
BF. 12345    DATA: 02/04/2015    ORA:11-06-24
             RL    1234567890
             B O N  F I S C A L
```

If <ReceiptType> = GENERIC_RECEIPT or CASH_IN/_OUT_RECEIPT:

```
EJTRZ: YYYYY
ID UNIC: 1234567890201706261255550001000 1
BNF.12345    DATA:02/04/2015    ORA:11-06-24
             RL    1234567890
***** BON NEFISCAL ****** BON NEFISCAL *****
```

Fiscal receipt layout **without** Customer CIF:

```
****** W I N C O R - N I X D O R F    ******
             receipt header line 2
             receipt header line 3
             receipt header line 4
       Cod Fiscal C.I.F.:1234567890123 4
             B O N  F I S C A L
                                     Lei
QQQQQ,QQQ MMM x V.VVV.VVV,VV
DDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDX.XXX.XXX,XX C
QQQQQ,QQQ MMM x V.VVV.VVV,VV
DDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDX.XXX.XXX,XX E
                                     ----------
 T O T A L          V . V V V . V V V , V V

TVA C    RR,RR%                      X.XXX.XXX,XX
TOT. VAN.EXCEPTATE TVA E             X.XXX.XXX,XX

CASH                                 X.XXX.XXX,XX

REST                                 X.XXX.XXX,XX

EJTRZ: 00005
ID UNIC: 1234567890201706261255550001000 1
BF. 12345    DATA: 02/04/2015    ORA:11-06-24
             RL    1234567890
             B O N  F I S C A L
```

Fiscal receipt layout **with** Customer CIF:

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```
*****  W I N C O R - N I X D O R F    ******
                 receipt header line 2
                 receipt header line 3
                 receipt header line 4
          Cod Fiscal C.I.F.:1234567890123 4
                 B O N   F I S C A L
                                        Lei
QQQQQQ,QQQ  MMM  x  V.VVV.VVV,VV
DDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDX.XXX.XXX,XX  C
QQQQQQ,QQQ  MMM  x  V.VVV.VVV,VV
DDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDX.XXX.XXX,XX  E
                                    ----------
 T O T A L                V . V V V . V V V , V V

TVA C    RR,RR%                          X.XXX.XXX,XX
TOT. VAN.EXCEPTATE TVA E                 X.XXX.XXX,XX

CASH                                     X.XXX.XXX,XX

REST                                     X.XXX.XXX,XX

C.I.F. Client: RO1234567890123

EJTRZ: 00005
ID UNIC: 12345678902017062612555500010001
BF. 12345    DATA:02/04/2015    ORA:11-06-24
             RL   1234567890
             B O N   F I S C A L
```

Non-fiscal receipt layout:

```
*****  W I N C O R - N I X D O R F    ******
                 receipt header line 2
                 receipt header line 3
                 receipt header line 4
          Cod Fiscal C.I.F.:1234567890123 4

***** BON NEFISCAL ****** BON NEFISCAL *****

             COUPON de CUMPARATURI

   ATI PRIMIT UN CUPON IN VALOARE DE 30 Lei

EJTRZ: 00005
ID UNIC: 12345678902017062612555500010001
BNF.12345    DATA:02/04/2015    ORA:11-06-24
             RL   1234567890
***** BON NEFISCAL ****** BON NEFISCAL *****
```

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7.6.3 RECEIPT_END_MFLINE

Function: The command ends the receipt by printing the footer only. **The receipt is not cut.**
The EJ ticket will be closed.

Code: **ESC MFB b d [ESC MFB2 b h <TaxID>] ESC MFE**

Parameter	Value	Format	
<TaxID>	20 H ≤ *n* ≤ FF H	15 bytes	T

Condition:

MFC Status byte 1

MFC_ TKD_RESET	MFC_ RECEIPT_ TOTAL	MFC_ DOCU_OPEN	MFC_ REPORT_ OPEN	MFC_ RECEIPT_ OPEN	MFC_ DAY_OPEN	MFC_ TRAINING_ MODE	MFC_ BLOCKADE
x	**1**	**x**	**x**	**1**	**x**	**x**	**x**

MFC Status byte 2

not used	not used	EJ_DATA_ PENDING	EJ_FULL	EJ_NEAR_ FULL	MFC_ CMD_ INTERRUPTED	MFC_ PRINTOUT_ INTERRUPTED	MFC_ FISCAL_ MODE
x	**x**	**x**	**x**	**x**	**x**	**0**	**x**

Computation: The MFC ensures that the value specified on this receipt is transferred to the appropriate day totalizer. See chapter 7.6.2 RECEIPT_END.

Note: This command can be used to print additional text, barcode or graphic behind the fiscal footer.

After the command is successfully executed the status bits MFC_RECEIPT_OPEN and MFC_RECEIPT_TOTAL will be cleared.

Error Msg: EM_PAYMENT_TOTAL_WRONG
EM_CMD_WRONG_ORDER
EM_PARA_WRONG
EM_PARA_WRONG_FORMAT

Printout: The printout is the same as with RECEIPT_END, but the ticket will not be cut.

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7.7 ARTICLE commands

All ARTICLE_ commands may return error messages from the set of 8.4 Supplementary checks error codes.

7.7.1 ARTICLE_SELL

Function: This command prints one sales line on the receipt and stores this line inside the EJ. There is also the opportunity to print an additional line immediately after the article line. All printed values update the CMOS totalizers (see 5.5 Counters and totalizers).

Code1:

ESC MFB f a
ESC MFB1 b a <*ArticleName*>
ESC MFB1 a a <*SinglePrice*>
ESC MFB1 d d <*Quantity*>
ESC MFB1 c a <*VATCategory*>
ESC MFB1 b j <QuantityUnit>
ESC MFB1 d j <ArticleCategory>
[ESC MFB2 b e <*PrintLinePost*>] ESC MFE

Parameter	Value	Format	
<ArticleName>	00 H ≤ *n* ≤ FF H	max. 29 bytes	**D**
<SinglePrice> Price of the printed line	30 H ≤ *n* ≤ 39 H	max. 9 bytes 1..9.999.999,99	**V**
<Quantity> Number of goods	30 H ≤ *n* ≤ 39 H	max. 9 bytes 1..999.999,999	**Q**
<VATCategory>	30 H ≤ *n* ≤ 34 H, 36H, 37H	1 byte	**C**
<QuantityUnit>	00 H ≤ *n* ≤ FF H	max. 7 bytes	**M**
<ArticleCategory>	30 H ≤ *n* ≤ 39 H	Max. 7 bytes	
<PrintLinePost> Application specific information which is printed after the article line.	00 H ≤ *n* ≤ FF H (00h-1Fh and 7Fh are replaced with blanks)	35 bytes	**i**
Article Price calculated by MFC			**X**

Code2:

ESC MFB f a
ESC MFB1 b a <*ArticleName*>
ESC MFB1 a a <*SinglePrice*>
ESC MFB1 d d <*Quantity*>
ESC MFB1 i n <*QuantityBox*>
ESC MFB1 c a <*VATCategory*>

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ESC MFB1 b j <QuantityUnit>
ESC MFB1 d j <ArticleCategory>
[ESC MFB2 b e *<PrintLinePost>*] ESC MFE

Parameter	Value	Format	
<ArticleName>	00 H ≤ *n* ≤ FF H (00h-1Fh and 7Fh are replaced with blanks)	max. 29 bytes	**D**
<SinglePrice> Price of the printed line	30 H ≤ *n* ≤ 39 H	max. 9 bytes 1..9.999.999,99	**V**
<Quantity>	30 H ≤ *n* ≤ 39 H	max. 3 bytes 1..999	**Q**
<QuantityBox> Number of goods in box	30 H ≤ *n* ≤ 39 H	max. 4 bytes 1..9999	**B**
<VATCategory>	30 H ≤ *n* ≤ 34 H, 36H, 37H	1 byte	**C**
<QuantityUnit>	00 H ≤ *n* ≤ FF H	max. 7 bytes	**M**
<ArticleCategory>	30 H ≤ *n* ≤ 39 H	Max. 7 bytes	
<PrintLinePost> Application specific information which is printed after the article line.	00 H ≤ *n* ≤ FF H (00h-1Fh and 7Fh are replaced with blanks)	35 bytes	**i**
Article Price calculated by MFC			**X**

Condition:

MFC Status byte 1

MFC_ TKD_RESET	MFC_ RECEIPT_ TOTAL	MFC_ DOCU_OPEN	MFC_ REPORT_ OPEN	MFC_ RECEIPT_ OPEN	MFC_ DAY_OPEN	MFC_ TRAINING_ MODE	MFC_ BLOCKADE
x	**0**	**x**	**x**	**1**	**x**	**x**	**0**

MFC Status byte 2

not used	not used	EJ_DATA_ PENDING	EJ_FULL	EJ_NEAR_ FULL	MFC_ CMD_ INTERRUPTED	MFC_ PRINTOUT_ INTERRUPTED	MFC_ FISCAL_ MODE
x	**x**	**x**	**x**	**x**	**x**	**0**	**x**

Only allowed in receipt types RETAIL_SALES_RECEIPT and RETAIL_SALES_RECEIPT_WITH_TAX_ID.

Computation:

Code1: *<Quantity>* * *<SinglePrice>* = *ArticlePrice*
Code2: *<Quantity>* * *<SinglePrice>* / *<QuantityBox>* = *ArticlePrice*
<SinglePrice> ≙ price per box

If the calculated value *ArticlePrice* is bigger than 9.999.999,99, the ARTICLE_SELL command is rejected with the error message EM_RECEIPT_TOT_OVERFLOW.

Receipt.Tot[BRUTTO_SALES] += *ArticlePrice*
Receipt.BruttoSale [SALES_K*n*] += *ArticlePrice* *n* = *<VATCategory>*

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Error Msg:

EM_CMD_WRONG_ORDER
EM_PARA_WRONG_FORMAT
EM_PARA_WRONG
EM_PARA_MISSING
EM_EJ_FULL
EM_RECEIPT_TOT_OVERFLOW

Printout:

Code1:

```
QQQQQ,QQQ  MMMMM  x  V.VVV.VVV,VV
DDDDDDDDDDDDDDDDDDDDDDDDDDDDDDD  X.XXX.XXX,XX  C
iiiiiiiiiiiiiiiiiiiiiiiiiiiiiiii
```

Code2:

```
  QQQ/BBBB  MMMMM  x  V.VVV.VVV,VV
DDDDDDDDDDDDDDDDDDDDDDDDDDDDDDD  X.XXX.XXX,XX  C
iiiiiiiiiiiiiiiiiiiiiiiiiiiiiiii
```

If the **first** ARTICLE_SELL (code 1, 2) to print then currency information is automatically printed in front of the first ARTICLE_SELL print line.

XXXX = " Lei" if the CRY Block doesn't exist; otherwise
XXXX = <*CurrencyLabel*> defined by SET_CURRENCY_COUNTRY (see 7.4.4)

```
                                    XXXX
```

7.7.2 ARTICLE_VOID

Function: The application cancels a previous ARTICLE_SELL. The line is printed on the receipt and stored inside the EJ.

Code:

ESC MFB f b
ESC MFB1 b a <*ArticleName*>
ESC MFB1 a a <*SinglePrice*>
ESC MFB1 c a <VATCategory>
ESC MFB1 d j <ArticleCategory> ESC MFE

Parameter	Value	Format	
<ArticleName>	00 H ≤ *n* ≤ FF H (00h-1Fh and 7Fh are replaced with blanks)	max. 29 bytes	**D**
<SinglePrice> Price of the printed line	30 H ≤ *n* ≤ 39 H	max. 9 bytes 1..9.999.999,99	**V**
<VATCategory>	30 H ≤ *n* ≤ 34 H, 36H, 37H	1 byte	**C**
<ArticleCategory>	30 H ≤ *n* ≤ 39 H	Max. 7 bytes	

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Condition:

MFC Status byte 1

MFC_ TKD_RESET	MFC_ RECEIPT_ TOTAL	MFC_ DOCU_OPEN	MFC_ REPORT_ OPEN	MFC_ RECEIPT_ OPEN	MFC_ DAY_OPEN	MFC_ TRAINING_ MODE	MFC_ BLOCKADE
x	**0**	**x**	**x**	**1**	**x**	**x**	**0**

MFC Status byte 2

not used	not used	EJ_DATA_ PENDING	EJ_FULL	EJ_NEAR_ FULL	MFC_ CMD_ INTERRUPTED	MFC_ PRINTOUT_ INTERRUPTED	MFC_ FISCAL_ MODE
x	**x**	**x**	**x**	**x**	**x**	**0**	**x**

Only allowed in receipt types RETAIL_SALES_RECEIPT and RETAIL_SALES_RECEIPT_WITH_TAX_ID.

Computation:

Receipt.Tot[BRUTTO_SALES] -= <*SinglePrice*>
Receipt.BruttoSale [SALES_Kn] -= <*SinglePrice*>, *n* =<VATCategory>
Receipt.Tot[VOID] += <*SinglePrice*>
Receipt.EventCnt[VOID] += 1

Note:

If the value voided is bigger than the actual sum in concerning totalizer the command is ignored and an error EM_RECEIPT_TOT_UNDERFLOW is issued.

Error Msg:

EM_CMD_WRONG_ORDER
EM_PARA_WRONG_FORMAT
EM_PARA_WRONG
EM_PARA_MISSING
EM_RECEIPT_TOT_UNDERFLOW
EM_EJ_FULL

Printout:

```
STORNARE
DDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDD  V.VVV.VVV,VV-C
```

7.7.3 ARTICLE_DISCOUNT

Function:

This command specifies a discount amount to be applied, prints the sales line on the receipt and stores it inside the EJ.

Code:

ESC MFB f c
ESC MFB1 a c <*DiscountValue*>
ESC MFB1 c a <*VATCategory*>
ESC MFB1 d j <ArticleCategory>
[ESC MFB2 b d <*PrintLinePre*>] ESC MFE

Parameter	Value	Format	
<DiscountValue> Discount given by application.	30 H ≤ *n* ≤ 39 H	max. 9 bytes 1..9.999.999,99	**V**
<VATCategory> Index to VAT table	30 H ≤ *n* ≤ 34 H, 36H, 37H	1 byte	**C**
<ArticleCategory>	30 H ≤ *n* ≤ 39 H	Max. 7 bytes	

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<PrintLinePre> Application-specific information which is printed before the article line	00 H ≤ *n* ≤ FF H	max. 35 bytes	**I**

Condition:

MFC Status byte 1

MFC_ TKD_RESET	MFC_ RECEIPT_ TOTAL	MFC_ DOCU_OPEN	MFC_ REPORT_ OPEN	MFC_ RECEIPT_ OPEN	MFC_ DAY_OPEN	MFC_ TRAINING_ MODE	MFC_ BLOCKADE
x	**0**	**x**	**x**	**1**	**x**	**x**	**0**

MFC Status byte 2

not used	not used	EJ_DATA_ PENDING	EJ_FULL	EJ_NEAR_ FULL	MFC_ CMD_ INTERRUPTED	MFC_ PRINTOUT_ INTERRUPTED	MFC_ FISCAL_ MODE
x	**x**	**x**	**x**	**x**	**x**	**0**	**x**

Only allowed in receipt types RETAIL_SALES_RECEIPT and RETAIL_SALES_RECEIPT_WITH_TAX_ID.

Computation:

Receipt.Tot[BRUTTO_SALES] -= *<DiscountValue>*
Receipt.BruttoSale[SALES_Kn] -= *<DiscountValue>*, n = *<VATCategory>*
Receipt.Tot[DISCOUNT] += *<DiscountValue>*
Receipt.EventCnt[DISCOUNT] += 1

Error Msg:

EM_CMD_WRONG_ORDER
EM_PARA_WRONG_FORMAT
EM_PARA_WRONG
EM_PARA_MISSING
EM_RECEIPT_TOT_UNDERFLOW
EM_EJ_FULL

Printout:

The *<DiscountValue>* (V in the "Appearance Masks") is printed right-aligned followed by minus and the VAT category. If no *<PrintLinePre>* field is supplied by the application, nothing is printed. The same print mask is used also in the SUBTOTAL_DISCOUNT command.

```
IIII
```

Parameter	Value	Format	
<DiscountValue> Price of the printed line	30 H ≤ *n* ≤ 39 H first byte must be higher than 30 H	max. 9 bytes 1..9.999.999,99	**V**
<VATCategory> Index to VAT table	30 H ≤ *n* ≤ 34 H, 36H, 37H	1 byte	**C**
<ArticleCategory>	30 H ≤ *n* ≤ 39 H	Max. 7 bytes	

Condition:

FC Status byte 1

MFC_ TKD_RESET	MFC_ RECEIPT_ TOTAL	MFC_ DOCU_OPEN	MFC_ REPORT_ OPEN	MFC_ RECEIPT_ OPEN	MFC_ DAY_OPEN	MFC_ TRAINING_ MODE	MFC_ BLOCKADE
x	**0**	**x**	**x**	**1**	**x**	**x**	**0**

MFC Status byte 2

not used	not used	EJ_DATA_ PENDING	EJ_FULL	EJ_NEAR_ FULL	MFC_ CMD_ INTERRUPTED	MFC_ PRINTOUT_ INTERRUPTED	MFC_ FISCAL_ MODE
x	**x**	**x**	**x**	**x**	**x**	**0**	**x**

Only allowed in receipt types RETAIL_SALES_RECEIPT and RETAIL_SALES_RECEIPT_WITH_TAX_ID.

Computation:

Receipt.Tot[BRUTTO_SALES] += *<DiscountValue>*
Receipt.BruttoSale[SALES_Kn] += *<DiscountValue>*, n = *<VATCategory>*
Receipt.Tot[DISCOUNT] -= *<DiscountValue>*
Receipt.EventCnt[DISCOUNT] += 1

Note: Receipt.EventCnt[DISCOUNT] will represent the total number of discount actions including discount voids.
If the value voided is bigger than the actual sum in concerning totalizer the command is ignored and an error EM_RECEIPT_TOT_UNDERFLOW is issued.

Error Msg:

EM_CMD_WRONG_ORDER
EM_PARA_WRONG_FORMAT
EM_PARA_WRONG
EM_PARA_MISSING
EM_RECEIPT_TOT_UNDERFLOW
EM_RECEIPT_TOT_OVERFLOW
EM_EJ_FULL

Printout:

```
STORNARE
DISCOUNT                                   V.VVV.VVV,VV  C
```

7.7.5 ARTICLE_SURCHARGE

Function: This command specifies a surcharge amount to be applied, prints the sales line on the receipt and stores it inside the EJ.

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Code:

ESC MFB f e
ESC MFB1 a d <*SurchargeValue*>
ESC MFB1 c a <*VATCategory*>
ESC MFB1 d j <ArticleCategory>
[ESC MFB2 b d <PrintLinePre>] ESC MFE

Parameter	Value	Format	
<SurchargeValue> Price of the printed line.	30 H ≤ *n* ≤ 39 H	max. 9 bytes 1..9.999.999,99	**V**
<VATCategory> Index to VAT table	30 H ≤ *n* ≤ 34 H, 36H, 37H	1 byte	**C**
<ArticleCategory>	30 H ≤ *n* ≤ 39 H	Max. 7 bytes	
<PrintLinePre> Application specific information which is printed before the article line	00 H ≤ *n* ≤ FF H	max. 35 bytes	**I**

Condition:

MFC Status byte 1

MFC_ TKD_RESET	MFC_ RECEIPT_ TOTAL	MFC_ DOCU_OPEN	MFC_ REPORT_ OPEN	MFC_ RECEIPT_ OPEN	MFC_ DAY_OPEN	MFC_ TRAINING_ MODE	MFC_ BLOCKADE
x	**0**	**x**	**x**	**1**	**x**	**x**	**0**

MFC Status byte 2

not used	not used	EJ_DATA_ PENDING	EJ_FULL	EJ_NEAR_ FULL	MFC_ CMD_ INTERRUPTED	MFC_ PRINTOUT_ INTERRUPTED	MFC_ FISCAL_ MODE
x	**x**	**x**	**x**	**x**	**x**	**0**	**x**

Only allowed in receipt types RETAIL_SALES_RECEIPT and RETAIL_SALES_RECEIPT_WITH_TAX_ID.

Computation:

Receipt.Tot[BRUTTO_SALES] += <*SurchargeValue*>
Receipt.BruttoSale[SALES_Kn] += <*SurchargeValue*>, n = <*VATCategory*>
Receipt.Tot[SURCHARGE] += <*SurchargeValue*>
Receipt.EventCnt[SURCHARGE] += 1

Error Msg:

EM_CMD_WRONG_ORDER
EM_PARA_WRONG_FORMAT
EM_PARA_WRONG
EM_PARA_MISSING
EM_RECEIPT_TOT_OVERFLOW
EM_EJ_FULL

Printout:

The <*SurchargeValue*> (V in the “Appearance Masks”) can have a maximum length of 9 bytes. A user defined line is only printed if <*PrintLinePre*> is specified. The same print mask is used also in the SUBTOTAL_SURCHARGE command.

```
IIII
```

7.7.6 ARTICLE_SURCHARGE_VOID

Function: With this command the application cancels a previous ARTICLE_SURCHARGE, prints the line on the receipt and stores it inside the EJ.

Code:

ESC MFB f f
ESC MFB1 a d <*SurchargeValue*>
ESC MFB1 c a <*VATCategory*>
ESC MFB1 d j <ArticleCategory> ESC MFE

Parameter	Value	Format	
<SurchargeValue> Price of the printed line.	30 H ≤ *n* ≤ 39 H	max. 9 bytes 1..9.999.999,99	**V**
<VATCategory> Index to VAT table	30 H ≤ *n* ≤ 34 H, 36H, 37H	1 byte	**C**
<ArticleCategory>	30 H ≤ *n* ≤ 39 H	Max. 7 bytes	

Condition:

MFC Status byte 1

MFC_ TKD_RESET	MFC_ RECEIPT_ TOTAL	MFC_ DOCU_OPEN	MFC_ REPORT_ OPEN	MFC_ RECEIPT_ OPEN	MFC_ DAY_OPEN	MFC_ TRAINING_ MODE	MFC_ BLOCKADE
x	**0**	**x**	**x**	**1**	**x**	**x**	**0**

MFC Status byte 2

not used	not used	EJ_DATA_ PENDING	EJ_FULL	EJ_NEAR_ FULL	MFC_ CMD_ INTERRUPTED	MFC_ PRINTOUT_ INTERRUPTED	MFC_ FISCAL_ MODE
x	**x**	**x**	**x**	**x**	**x**	**0**	**x**

Only allowed in receipt types RETAIL_SALES_RECEIPT and RETAIL_SALES_RECEIPT_WITH_TAX_ID.

Computation:

Receipt.Tot[BRUTTO_SALES] -= <*SurchargeValue*>
Receipt.BruttoSale[SALES_Kn] -= <*SurchargeValue*>, n = <*VATCategory*>
Receipt.Tot[SURCHARGE] -= <*SurchargeValue*>
Receipt.EventCnt[SURCHARGE] += 1

Note: Receipt.EventCnt[SURCHARGE] will represent the total number of discount actions including discount voids.
If the value voided is bigger than the actual sum in concerning totalizer the command is ignored and an error EM_RECEIPT_TOT_UNDERFLOW is issued.

Error Msg:

EM_CMD_WRONG_ORDER
EM_PARA_WRONG_FORMAT
EM_PARA_WRONG
EM_PARA_MISSING
EM_RECEIPT_TOT_UNDERFLOW
EM_EJ_FULL

Printout: The same print mask is used also in the SUBTOTAL_SURCHARGE_VOID command.

```
STORNARE
MAJORARE                                  V.VVV.VVV,VV-C
```

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7.7.7 ARTICLE_RETURN

Function: With this command, the application prints returned items on the receipt and stores it inside the EJ.

Code:

ESC MFB f h
ESC MFB1 b a <*ArticleName*>
ESC MFB1 a a <*SinglePrice*>
ESC MFB1 c a <VATCategory>
ESC MFB1 d j <ArticleCategory> ESC MFE

Parameter	Value	Format	
<ArticleName>	00 H ≤ *n* ≤ FF H	max. 29 bytes	**D**
<SinglePrice> Price of the printed line.	30 H ≤ *n* ≤ 39 H	max. 9 bytes 1..9.999.999,99	**V**
<VATCategory> Index to VAT table	30 H ≤ *n* ≤ 34 H, 36H, 37H	1 byte	**C**
<ArticleCategory>	30 H ≤ *n* ≤ 39 H	Max. 7 bytes	

Condition:

MFC Status byte 1

MFC_ TKD_RESET	MFC_ RECEIPT_ TOTAL	MFC_ DOCU_OPEN	MFC_ REPORT_ OPEN	MFC_ RECEIPT_ OPEN	MFC_ DAY_OPEN	MFC_ TRAINING_ MODE	MFC_ BLOCKADE
x	**0**	**x**	**x**	**1**	**x**	**x**	**0**

MFC Status byte 2

not used	not used	EJ_DATA_ PENDING	EJ_FULL	EJ_NEAR_ FULL	MFC_ CMD_ INTERRUPTED	MFC_ PRINTOUT_ INTERRUPTED	MFC_ FISCAL_ MODE
x	**x**	**x**	**x**	**x**	**x**	**0**	**x**

Only allowed in receipt types RETAIL_SALES_RECEIPT and RETAIL_SALES_RECEIPT_WITH_TAX_ID.

Notes: Due to the limitation that the receipt can not go negative (<0) there are limitations to how ARTICLE_RETURN is applied. The maximum allowed price of the returned article is determined by the amount of that VAT category already booked in the open fiscal receipt. If the amount already booked for that VAT category is less than the return the system will return the error EM_RECEIPT_TOT_UNDERFLOW.

Computation:

Receipt.Tot[BRUTTO_SALES] -= <*SinglePrice*>
Receipt.BruttoSales[SALES_Kn] -= <*SinglePrice*> n = <*VATCategory*>
Receipt.Tot[RETURN] += <*SinglePrice*>
Receipt.EventCnt[RETURN] += 1

Error Msg:

EM_CMD_WRONG_ORDER
EM_PARA_WRONG_FORMAT
EM_PARA_WRONG
EM_PARA_MISSING
EM_RECEIPT_TOT_UNDERFLOW
EM_RECEIPT_TOT_OVERFLOW
EM_EJ_FULL

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Printout:

```
CORECTIE  ARTICOL
DDDD
```

EM_RECEIPT_TOT_UNDERFLOW
EM_RECEIPT_TOT_OVERFLOW
EM_EJ_FULL

Printout:

```
STORNARE
CORECTIE  ARTICOL
DDDD
```

Condition:

MFC Status byte 1

MFC_ TKD_RESET	MFC_ RECEIPT_ TOTAL	MFC_ DOCU_OPEN	MFC_ REPORT_ OPEN	MFC_ RECEIPT_ OPEN	MFC_ DAY_OPEN	MFC_ TRAINING_ MODE	MFC_ BLOCKADE
x	**0**	**x**	**x**	**1**	**x**	**x**	**0**

MFC Status byte 2

not used	not used	EJ_DATA_ PENDING	EJ_FULL	EJ_NEAR_ FULL	MFC_ CMD_ INTERRUPTED	MFC_ PRINTOUT_ INTERRUPTED	MFC_ FISCAL_ MODE
x	**x**	**x**	**x**	**x**	**x**	**0**	**x**

Only allowed in receipt types RETAIL_SALES_RECEIPT and RETAIL_SALES_RECEIPT_WITH_TAX_ID.

SUBTOTAL_ discount and surcharge commands are not allowed after ARTICLE_FUEL_SELL.

Computation: *<Quantity> * <SinglePrice> = TotalFuelPrice*

If the calculated value *TotalFuelPrice* is bigger than 9.999.999,99, the ARTICLE_FUEL_SELL command is rejected with the error message EM_PRICE_OVERFLOW.

Receipt.Tot[BRUTTO_SALES] += *<TotalFuelPrice>*
Receipt.BruttoSale[SALES_Kn] += *<TotalNettoPrice>*, n = *<VATCategory>*
Receipt.VAT[VAT_KF] += *<TotalRoadTax>*

Example: 1212,300 l fuel is sold for 333,62 Lei/l with a road tax amount of 123,27 Lei/l. This item is booked in VAT category A.

Within the ESC sequence the application sends the article name, tax name, single price, road tax, quantity and VAT category to the MFC:

<ArticleName> = Regular Fuel
<TaxName> = Road tax amount
<SinglePrice> = 33362 (= BruttoPrice per liter w/ road tax)
<RoadTax> = 12327
<Quantity> = 1212300
<VATCategory> = 0

Now the firmware calculates different necessary values:

TotalFuelPrice is the total price including road tax amount.

(1) $\boxed{\text{TotalFuelPrice} = Quantity * Single\ Price}$

Example: *TotalFuelPrice* = 1212300 l * 33362 LEI= 40444752600 LEI
The result is rounded in **2 steps**:

1.1 cut from 3 decimal by division by 1000
40444752600 / 1000 = 40444752

1.2 rounding by addition of 50 and division by 100
(40444752 + 50) / 100 = 404448

The *TotalFuelPrice* is added to the receipt totalizer for brutto sales (Receipt.Tot[BRUTTO_SALES]).

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(2) $TotalRoadTax = Quantity * RoadTax$

TotalRoadTax = 1212300 l * 12327 LEI= 14944022100 LEI.

2.1 after cut of the last 3 decimals, the result will be stored **with 2 additional** decimals (149.440,22) in the receipt totalizer for road tax (stReceipt.ulRoadtax).

2.2 rounding by addition of 50 and division by 100
(14944022 + 50) / 100 = 149440

(3) TotalNettoPrice = TotalFuelPrice - RoadTax

(3) $TotalNettoPrice = Quantity * (SinglePrice - RoadTax)$

TotalNettoPrice = 40444752600 LEI - 14944022100 LEI= 25500730500 LEI.

The *TotalNettoPrice* will be stored in the totalizer which belongs to the VAT category given as parameter (Receipt.BruttoSales[ucVatCategory]).

After these calculations the four print lines will be printed.

At this point ARTICLE_FUEL_SELL calculation is finished. It is possible to send further ARTICLE_FUEL_SELL commands. Each command starts the calculation new (at point 1).

After the TOTAL_LINE command the MFC calculates the VAT amount for each VAT category and rounds the cumulated sum of road tax.

(4) $$VATCategoryAmount = \frac{TotalNettoPrice * VATCategory}{1 + VATCategory}$$

VATCategoryAmount = 255.007 LEI * 0,19% / (1+0,19%) = 40.715 LEI.
This amount is printed after the total line.

Receipt.VAT[VAT_KF] += <*VATCategoryAmount*>

The cumulated sum of *TotalRoadTax* is rounded like described in point 2.2 and printed on the receipt.

```
*****   W I N C O R - N I X D O R F    *****

            receipt header line 2

            receipt header line 3

            receipt header line 4

     Cod Fiscal C.I.F.: RO1234567890123
```

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```
                                        Lei

Regular Fuel                        2.550,08 B

   12,12300 L   x     210,35

TAXA DRUM Regular Fuel              1.494,40 F

   12,12300 L   x     123,27

                                 -----------

T O T A L                      4.044,48


TVA B   9,00%                        210,56
TOTAL VÂNZĂRI ALTE TAXE F 0,00%    1.494,40


CASH                                5.000,00

REST CASH                            -955,52


EJTRZ: 00005

ID UNIC:12345678902017062612555500010001

BF. 00001    DATA:15/09/2007   ORA:15:12:21

               RL 1234567890

            B O N  F I S C A L
```

Error Msg: EM_CMD_WRONG_ORDER
EM_PARA_WRONG_FORMAT
EM_PARA_WRONG
EM_PARA_MISSING
EM_PRICE_OVERFLOW if *FuelAmount* + *RoadTaxAmount* is higher than 9.999.999,99 or if *RoadTaxAmount* >= *SinglePrice*
EM_RECEIPT_TOT_OVERFLOW
EM_EJ_FULL

Printout:

```
DDDDDDDDDDDDDDDDDDDDDDDDDDDDDDD  V.VVV.VVV,VV  C
Q.QQQ,QQQQQ  MMMMM  x   NN.NNN,NN
TTTTTTTTTTTTTTTTTTTTTTTTTTTTTTT  X.XXX.XXX,XX  F
Q.QQQ,QQQQQ  MMMMM  x   RR.RRR,RR
```

If the <*RoadTax*> = 0, then the third and fourth line are not printed.

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7.7.10 ARTICLE_FUEL_VOID

Function: With this command application cancels a previous ARTICLE_FUEL_SELL command, prints the line on the receipt and stores it inside the EJ.

Code: **ESC MFB f k**
ESC MFB1 d j <ArticleCategory> ESC MFE

Parameter	Value	Format	
<ArticleCategory>	30 H ≤ *n* ≤ 39 H	Max. 7 bytes	

Condition:

MFC Status byte 1

MFC_ TKD_RESET	MFC_ RECEIPT_ TOTAL	MFC_ DOCU_OPEN	MFC_ REPORT_ OPEN	MFC_ RECEIPT_ OPEN	MFC_ DAY_OPEN	MFC_ TRAINING_ MODE	MFC_ BLOCKADE
x	**0**	**x**	**x**	**1**	**x**	**x**	**0**

MFC Status byte 2

not used	not used	EJ_DATA_ PENDING	EJ_FULL	EJ_NEAR_ FULL	MFC_ CMD_ INTERRUPTED	MFC_ PRINTOUT_ INTERRUPTED	MFC_ FISCAL_ MODE
x	**x**	**x**	**x**	**x**	**x**	**0**	**x**

Only allowed in receipt types RETAIL_SALES_RECEIPT and RETAIL_SALES_RECEIPT_WITH_TAX_ID.

Computation: All computations are internal, so the application has not to transmit any parameters.

Receipt.Tot[BRUTTO_SALES] -= *<PreviousTotalFuelPrice>*
Receipt.BruttoSale [SALES_Kn] -= *<PreviousTotalNettoPrice>,*
n = *<VATCategory>*
Receipt.Tot[VOID] += *<PreviousTotalFuelPrice>*
Receipt.VAT[VAT_KF] -= *<PreviousTotalRoadTax>*
Receipt.EventCnt[VOID] += 1

Note: If the value voided is bigger than the actual sum in concerning totalizer the command is ignored and an error EM_RECEIPT_TOT_UNDERFLOW is issued.

Error Msg: EM_CMD_WRONG_ORDER
EM_RECEIPT_TOT_UNDERFLOW
EM_RECEIPT_TOT_OVERFLOW
EM_EJ_FULL

Printout:

```
STORNARE
DDDDDDDDDDDDDDDDDDDDDDDDDDDDDDD  V.VVV.VVV,VV-C
STORNARE
TTTTTTTTTTTTTTTTTTTTTTTTTTTTTTT  X.XXX.XXX,XX-F
```

7.7.11 ARTICLE_TAX

Function: This command is used to book as income a tax that is a not value added tax to an article or service. It might be used in some areas as a ‘green tax’ or ‘visitor’s tax’. The tax is printed on the receipt, and stored inside the EJ.

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Code: **ESC MFB f n**
ESC MFB1 b i <*TaxName*>
ESC MFB1 a m <*TaxValue*> ESC MFE

Parameter	Value	Format	
<TaxName>	00 H ≤ *n* ≤ FF H	max. 32 bytes	**T**
<TaxValue>	30 H ≤ *n* ≤ 39 H	max. 7 bytes 1..99.999,99	**R**

Condition:

MFC Status byte 1

MFC_ TKD_RESET	MFC_ RECEIPT_ TOTAL	MFC_ DOCU_OPEN	MFC_ REPORT_ OPEN	MFC_ RECEIPT_ OPEN	MFC_ DAY_OPEN	MFC_ TRAINING_ MODE	MFC_ BLOCKADE
x	**0**	**x**	**x**	**1**	**x**	**x**	**0**

MFC Status byte 2

not used	not used	EJ_DATA_ PENDING	EJ_FULL	EJ_NEAR_ FULL	MFC_ CMD_ INTERRUPTED	MFC_ PRINTOUT_ INTERRUPTED	MFC_ FISCAL_ MODE
x	**x**	**x**	**x**	**x**	**x**	**0**	**x**

Only allowed in receipt types RETAIL_SALES_RECEIPT and RETAIL_SALES_RECEIPT_WITH_TAX_ID.

Computation: If the calculated value *TaxValue* is bigger than 9.999.999,99, the command is rejected with the error message EM_RECEIPT_TOT_OVERFLOW.

Receipt.Tot[BRUTTO_SALES] += <TaxValue>
Receipt.BruttoSale [VAT_KF] += <*TaxValue*>
Receipt.VAT[VAT_KF] += <*TaxValue*>

Error Msg:
EM_CMD_WRONG_ORDER
EM_PARA_WRONG_FORMAT
EM_PARA_WRONG
EM_PARA_MISSING
EM_RECEIPT_TOT_OVERFLOW
EM_EJ_FULL

Printout:

```
TTTT
```

<TaxValue>	30 H ≤ *n* ≤ 39 H	max. 7 bytes 1..99.999,99	**R**

Condition:

MFC Status byte 1

MFC_ TKD_RESET	MFC_ RECEIPT_ TOTAL	MFC_ DOCU_OPEN	MFC_ REPORT_ OPEN	MFC_ RECEIPT_ OPEN	MFC_ DAY_OPEN	MFC_ TRAINING_ MODE	MFC_ BLOCKADE
x	**0**	**x**	**x**	**1**	**x**	**x**	**0**

MFC Status byte 2

not used	not used	EJ_DATA_ PENDING	EJ_FULL	EJ_NEAR_ FULL	MFC_ CMD_ INTERRUPTED	MFC_ PRINTOUT_ INTERRUPTED	MFC_ FISCAL_ MODE
x	**x**	**x**	**x**	**x**	**x**	**0**	**x**

Only allowed in receipt types RETAIL_SALES_RECEIPT and RETAIL_SALES_RECEIPT_WITH_TAX_ID.

Computation:

Receipt.Tot[BRUTTO_SALES]	-= <TaxValue>
Receipt.Tot[VOID]	+= *<TaxValue*
Receipt.BruttoSale [VAT_KF]	-= *<TaxValue>*
Receipt.VAT[VAT_KF]	-= *<TaxValue>*
Receipt.EventCnt[VOID]	+= 1

Note: If the value voided is bigger than the actual sum in concerning totalizer the command is ignored and an error EM_RECEIPT_TOT_UNDERFLOW is issued.

Error Msg:

EM_CMD_WRONG_ORDER
EM_PARA_WRONG_FORMAT
EM_PARA_WRONG
EM_PARA_MISSING
EM_RECEIPT_TOT_UNDERFLOW
EM_EJ_FULL

Printout:

```
STORNARE
TTTT
```

7.8 SUBTOTAL command

All SUBTOTAL_ commands may return error messages from the set of 8.4 Supplementary checks error codes.

7.8.1 SUBTOTAL_LINE

Function: With this command the application requests a subtotal line to be printed. The SUBTOTAL_LINE command is a “print-only” version of the TOTAL_LINE command.
No totalizers and counters are updated with this command. The line is stored inside the EJ.

Code: **ESC MFB c a**
[ESC MFB2 b e <*PrintLinePost* >] ESC MFE

Parameter	Value	Format	
<PrintLinePost> Application specific information which is printed after the subtotal line	00 H ≤ *n* ≤ FF H	max.35 bytes	**I**

Condition:

MFC Status byte 1

MFC_ TKD_RESET	MFC_ RECEIPT_ TOTAL	MFC_ DOCU_OPEN	MFC_ REPORT_ OPEN	MFC_ RECEIPT_ OPEN	MFC_ DAY_OPEN	MFC_ TRAINING_ MODE	MFC_ BLOCKADE
x	**0**	**x**	**x**	**1**	**x**	**x**	**x**

MFC Status byte 2

not used	not used	EJ_DATA_ PENDING	EJ_FULL	EJ_NEAR_ FULL	MFC_ CMD_ INTERRUPTED	MFC_ PRINTOUT_ INTERRUPTED	MFC_ FISCAL_ MODE
x	**x**	**x**	**x**	**x**	**x**	**0**	**x**

Only allowed in receipt types RETAIL_SALES_RECEIPT and RETAIL_SALES_RECEIPT_WITH_TAX_ID.

Computation: This command performs no computations as such, it simply outputs the contents of
Receipt.Tot[BRUTTO_SALES].

Note: For synchronizing, the running receipt totals are available at all times for the application, by using the commands GET_RECEIPT_BRUTTO_SALES(7.15.3), GET_RECEIPT_TOTALIZER(7.15.2), GET_RECEIPT_VAT_AMOUNT(7.15.4)

SUBTOTAL_LINE cannot be the first command inside a receipt.

Error Msg: EM_CMD_WRONG_ORDER
EM_PARA_WRONG_FORMAT
EM_PARA_WRONG

Printout:

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| | | |

$$\boxed{Sales\,[VATCategor\quad y] = Sales\,[VATCategor\quad y] + DicountVal\quad ue}$$

and also adds *DiscountValue*> to Receipt.Tot[DISCOUNT].

Example: Receipt with three items:
Item 1 = 1000, VAT category = A
Item 2 = 2000, VAT category = B
Item 3 = 3000, VAT category = C

The subtotal is 6000 and the subtotal discount should be 100.

Now the application has to calculate the discount value for all VAT categories as described in formula (5):

DiscountValue[A] = $100 * \frac{1000[A]}{6000} + 0{,}5 \rightarrow 17$

DiscountValue[B] = $100 * \frac{2000[B]}{6000} + 0{,}5 \rightarrow 33$

DiscountValue[C] = $100 * \frac{3000[C]}{6000} + 0{,}5 \rightarrow 50$

The sum of the discounts must be the total discount of 100.

The following sequences have to be submitted to the MFC:

ESC MFB “cb” **ESC MFB1** “ac” “17” **ESC MFB1** “ca” ‘0’
[ESC MFB2 “bd” “Discount VAT A”**] ESC MFE**

ESC MFB “cb” **ESC MFB1** “ac” “33” **ESC MFB1** “ca” ‘1’
[ESC MFB2 “bd” “Discount VAT B”**] ESC MFE**

ESC MFB “cb” **ESC MFB1** “ac” “50” **ESC MFB1** “ca” ‘2’
[ESC MFB2 “bd” “Discount VAT C”**] ESC MFE**

Texts in brackets [...] are optional.

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```
*****   W I N C O R - N I X D O R F    *****
           receipt header line 2
           receipt header line 3
           receipt header line 4
     Cod Fiscal C.I.F.: RO1234567890123
0123456789012345678901234567890123456789012 3
                                          Lei
     1,000 MMM x          1,00
Item 1                                1,00 A
     1,000 MMM x          2,00
Item 2                                2,00 B
     1,000 MMM x          3,00
Item 3                                3,00 C
                                  ----------
SUBTOTAL                              6,00

Discount VAT A
DISCOUNT                              0,17-A
Discount VAT B
DISCOUNT                              0,33-B
Discount VAT C
DISCOUNT                              0,50-C
                                  ----------
 T O T A L                         5 , 0 0

TVA A 00,00%                          0,00
TVA B 19,00%                          0,27
TVA C 00,00%                          0,00

CASH                                  5,00

EJTRZ: 00005
ID UNIC: 12345678902017062612555500010001
BF. 00001     DATA:15/091/2017  ORA:15-15-53
              RL   1234567890
            B O N   F I S C A L
```

Note: SUBTOTAL_DISCOUNT is accepted only after SUBTOTAL_LINE or SUBTOTAL_DISCOUNT.

This command is ignored if the receipt contains one or more ARTICLE_FUEL commands. (There is no exception if these commands have been cancelled afterwards before using SUBTOTAL_DISCOUNT!)

For a SUBTOTAL_DISCOUNT to more than one VAT category the application has to use this command multiple.

Error Msg: EM_CMD_WRONG_ORDER
EM_PARA_WRONG_FORMAT
EM_PARA_WRONG
EM_PARA_MISSING
EM_RECEIPT_TOT_UNDERFLOW
EM_RECEIPT_TOT_OVERFLOW

Printout: <*DiscountValue*> (V in the “Appearance Masks”) is printed right-aligned followed by minus and the VAT category. If no <*PrintLinePre*> field is supplied

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by the application, no additional line is printed. The same print mask is used also in the ARTICLE_DISCOUNT command.

I	I	I	I	I	I	I	I	I	I	I	I	I	I	I	I	I	I	I	I	I	I	I	I	I	I	I	I	I	I	I	I	I	I						
D	I	S	C	O	U	N	T																			V	.	V	V	V	.	V	V	V	,	V	V	-	C

7.8.3 SUBTOTAL_DISCOUNT_VOID

Function: With this command the application has the opportunity to cancel a part or all of the discount amount applied with a previous SUBTOTAL_DISCOUNT command.

Code:
ESC MFB c c
ESC MFB1 a c *<DiscountValue>*
ESC MFB1 c a *<VATCategory>* ESC MFE

Parameter	Value	Format	
<DiscountValue> Discount to previous article.	30 H ≤ *n* ≤ 39 H	max. 9 bytes 1..9.999.999,99	**V**
<VATCategory> Index to VAT table	30 H ≤ *n* ≤ 34 H, 36H, 37H	1 byte	**C**

Condition:

MFC Status byte 1

MFC_ TKD_RESET	MFC_ RECEIPT_ TOTAL	MFC_ DOCU_OPEN	MFC_ REPORT_ OPEN	MFC_ RECEIPT_ OPEN	MFC_ DAY_OPEN	MFC_ TRAINING_ MODE	MFC_ BLOCKADE
x	**0**	**x**	**x**	**1**	**x**	**x**	**0**

MFC Status byte 2

not used	not used	EJ_DATA_ PENDING	EJ_FULL	EJ_NEAR_ FULL	MFC_ CMD_ INTERRUPTED	MFC_ PRINTOUT_ INTERRUPTED	MFC_ FISCAL_ MODE
x	**x**	**x**	**x**	**x**	**x**	**0**	**x**

Only allowed in receipt types RETAIL_SALES_RECEIPT and RETAIL_SALES_RECEIPT_WITH_TAX_ID.

Computation:

Receipt.Tot[BRUTTO_SALES]	+= *<DiscountValue>*
Receipt.BruttoSale[SALES_Kn]	+= *<DiscountValue>*, n: *<VATCategory>*
Receipt.Tot[DISCOUNT]	-= *<DiscountValue>*
Receipt.EventCnt[DISCOUNT]	+= 1

Note: SUBTOTAL_DISCOUNT_VOID is accepted only after at least one SUBTOTAL_DISCOUNT command. Multiple SUBTOTAL_DISCOUNT_VOID commands are possible.

The event counter is increased by 1 for a discount void, because the counter is used to track discount events and not only discounts applied. A void is a type of discount event.

Error Msg: EM_CMD_WRONG_ORDER
EM_PARA_WRONG_FORMAT

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EM_PARA_WRONG
EM_PARA_MISSING
EM_RECEIPT_TOT_UNDERFLOW
EM_RECEIPT_TOT_OVERFLOW

Printout: The same print masks are used also in the ARTICLE_DISCOUNT_VOID command.

```
STORNARE
DISCOUNT                                   V.VVV.VVV,VV  C
```

7.8.4 SUBTOTAL_SURCHARGE

Function: With this command the application specifies a surcharge amount to be applied in the previous SUBTOTAL_LINE.

Code:
ESC MFB c d
ESC MFB1 a d <*SurchargeValue*>
ESC MFB1 c a <*VATCategory*>
[ESC MFB2 b d <*PrintLinePre*>] ESC MFE

Parameter	Value	Format	
<SurchargeValue> Discount to previous article.	30 H ≤ *n* ≤ 39 H	max. 9 bytes 9.999.999,99	**V**
<VATCategory> Index to VAT table	30 H ≤ *n* ≤ 34 H, 36H, 37H	1 byte	**C**
<PrintLinePre> Application specific information which is printed before the subtotal discount line	00 H ≤ *n* ≤ FF H	35 bytes	**I**

Condition:

MFC Status byte 1

MFC_ TKD_RESET	MFC_ RECEIPT_ TOTAL	MFC_ DOCU_OPEN	MFC_ REPORT_ OPEN	MFC_ RECEIPT_ OPEN	MFC_ DAY_OPEN	MFC_ TRAINING_ MODE	MFC_ BLOCKADE
x	**0**	**x**	**x**	**1**	**x**	**x**	**0**

MFC Status byte 2

not used	not used	EJ_DATA_ PENDING	EJ_FULL	EJ_NEAR_ FULL	MFC_ CMD_ INTERRUPTED	MFC_ PRINTOUT_ INTERRUPTED	MFC_ FISCAL_ MODE
x	**x**	**x**	**x**	**x**	**x**	**0**	**x**

Only allowed in receipt types RETAIL_SALES_RECEIPT and RETAIL_SALES_RECEIPT_WITH_TAX_ID.

Computation:
Receipt.Tot[BRUTTO_SALES] += <*SurchargeValue*>
Receipt.BruttoSale[SALES_Kn] += <*SurchargeValue*> n = <*VATCategory*>
Receipt.Tot[SURCHARGE] += <*SurchargeValue*>
Receipt.EventCnt[SURCHARGE] += 1

The application has to perform the calculation given in the formula below:

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$$Surch\ \arg eValue = Amount * \frac{Sales[VATCategor\ y]}{TicketTota\ l} + 0{,}5$$

The MFC handles the surcharge value as

$$Sales[VATCategor\ y] = Sales[VATCategor\ y] + Surch\ \arg eValue$$

and also adds <*SurchargeValue*> to Receipt.Tot[SURCHARGE].

Note: SUBTOTAL_SURCHARGE is accepted only after SUBTOTAL_LINE or SUBTOTAL_SURCHARGE.
For further information please refer to the opposite function in chapter in 7.8.2 SUBTOTAL_DISCOUNT

This command is ignored if the receipt contains one or more ARTICLE_FUEL commands.

Error Msg: EM_CMD_WRONG_ORDER
EM_PARA_WRONG_FORMAT
EM_PARA_WRONG
EM_PARA_MISSING
EM_RECEIPT_TOT_UNDERFLOW
EM_RECEIPT_TOT_OVERFLOW

Printout: <SurchargeValue> (V in the “Appearance Masks”) can have a maximum length of eight bytes. If no <PrintLinePre> field is supplied by application, no additional line is printed. The same print mask is used also in the ARTICLE_SURCHARGE command.

```
IIII
```

Condition:

MFC Status byte 1

MFC_ TKD_RESET	MFC_ RECEIPT_ TOTAL	MFC_ DOCU_OPEN	MFC_ REPORT_ OPEN	MFC_ RECEIPT_ OPEN	MFC_ DAY_OPEN	MFC_ TRAINING_ MODE	MFC_ BLOCKADE
x	**0**	**x**	**x**	**1**	**x**	**x**	**0**

MFC Status byte 2

not used	not used	EJ_DATA_ PENDING	EJ_FULL	EJ_NEAR_ FULL	MFC_ CMD_ INTERRUPTED	MFC_ PRINTOUT_ INTERRUPTED	MFC_ FISCAL_ MODE
x	**x**	**x**	**x**	**x**	**x**	**0**	**x**

Only allowed in receipt types RETAIL_SALES_RECEIPT and RETAIL_SALES_RECEIPT_WITH_TAX_ID.

Computation:

Receipt.Tot[BRUTTO_SALES] -= <*SurchargeValue*>
Receipt.BruttoSale[SALES_Kn] -= <*SurchargeValue*> n = <*VATCategory*>
Receipt.Tot[SURCHARGE] -= <*SurchargeValue*>
Receipt.EventCnt[SURCHARGE] += 1

Note:

SUBTOTAL_SURCHARGE_VOID is accepted only after a SUBTOTAL_SURCHARGE command. Multiple SUBTOTAL_SURCHARGE_VOID is possible.

The event counter is increased by 1 for a surcharge void, because the counter is used to track surcharge events and not only surcharges applied. A void is a type of surcharge event.

Error Msg:

EM_CMD_WRONG_ORDER
EM_PARA_WRONG_FORMAT
EM_PARA_WRONG
EM_PARA_MISSING
EM_RECEIPT_TOT_UNDERFLOW
EM_RECEIPT_TOT_OVERFLOW

Printout:

The same print mask is also used in the ARTICLE_SURCHARGE_VOID command.

```
STORNARE
MAJORARE  STORNARE                    V.VVV.VVV,VV-C
```

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7.9 TOTAL command

7.9.1 TOTAL_LINE

Function: With this command the application requests a total line to be printed and stored inside the EJ. The MFC sets MFC_RECEIPT_TOTAL. It has the effect of stopping the sales posting stage. If successful the command sets the MFC_RECEIPT_TOTAL bit.

Code: **ESC MFB d a ESC MFB1 a b <*TotalValue*> ESC MFE**

Parameter	Value	Format	
<TotalValue> Price of the printed line	30 H ≤ *n* ≤ 39 H	max. 9 bytes 9.999.999,99	**V**

Condition:

MFC Status byte 1

MFC_ TKD_RESET	MFC_ RECEIPT_ TOTAL	MFC_ DOCU_OPEN	MFC_ REPORT_ OPEN	MFC_ RECEIPT_ OPEN	MFC_ DAY_OPEN	MFC_ TRAINING_ MODE	MFC_ BLOCKADE
x	**0**	**x**	**x**	**1**	**x**	**x**	**0**

MFC Status byte 2

not used	not used	EJ_DATA_ PENDING	EJ_FULL	EJ_NEAR_ FULL	MFC_ CMD_ INTERRUPTED	MFC_ PRINTOUT_ INTERRUPTED	MFC_ FISCAL_ MODE
x	**x**	**x**	**x**	**x**	**x**	**0**	**x**

Only allowed in receipt types RETAIL_SALES_RECEIPT and RETAIL_SALES_RECEIPT_WITH_TAX_ID.

Computation: The totalizers Day.Vat[A...D, G, H] is handled as

$$VATAmount[n] = \frac{aulBruttoSales[n] * (100 * auiVATRate[n])}{(10000 + auiVATRate[n])}$$

Where *n* is the VAT rate A…D, G, H.

The total amount is rounded just for print-out purposes as follows:

$$VATAmount\ [n] \mid \text{Pr int}\ out = \frac{VATAmount\ [n] + 50}{100}$$

n=E is not possible because of “void” VAT rate category VAT_KE with 0%.
n=F is reserved for road tax

The above calculation is done with two implicit decimals.

Note: ***For derived values, the system always uses one floating point more than it actually prints. All internal calculations are done with three decimals. For print-out the last two digits is rounded up/down according to the value of the third digit (+0,5 and third digit is truncated). All three digits are***

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internally hold until the DAY_END command is issued. The application can obtain all values with the

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This command cannot be reversed or voided in any way, the only possibility of exiting MFC_RECEIPT_TOTAL mode is to issue a SPECIAL_ALL_VOID command.

If <*TotalValue*> is not equal to Receipt.Tot [BRUTTO_SALES], the system cancels the receipt automatically (same as 7.12.5 SPECIAL_ALL_VOID). For synchronising, the receipt totals are available for the application at all times inside a receipt (see commands GET_RECEIPT_BRUTTO_SALES(7.15.3), GET_RECEIPT_TOTALIZER(7.15.2), GET_RECEIPT_VAT_AMOUNT(7.15.4)).

TOTAL_LINE cannot be the first command inside a receipt.

<TotalValue> == 0 is allowed (Zero-Bon). The receipt will be fiscal receipt.

In legal receipts the application has the opportunity to issue FREEPRINT_NORMAL commands after TOTAL_LINE.

Error Msg:

EM_CMD_WRONG_ORDER
EM_PARA_WRONG
EM_PARA_MISSING
EM_RECEIPT_TOT_WRONG
EM_DAY_TOT_OVERFLOW
Error messages from the set of 8.3 EJ error codes

Error messages from the set of 8.4 Supplementary checks error codes

Printout:

The first line printed is a separator line with dashes. Receipt.Tot[BRUTTO_SALES] can have maximum length 9 bytes. The fixed text “TOTAL” and the value is printed in double width. After total line the VAT information part is printed by MFC automatically. The content of this block depends on the number of VAT classes used in the receipt. The VAT category for an ARTICLE_FUEL_SELL command is not printed out particularly.

C	Abbreviation for VAT category issued in article lines
R	VAT rate
X	VAT amount of VAT concerning category

Optional print lines are marked in *italic* (printout depends on previous article sell).

```
                                        ----------
T O T A L          V . V V V . V V V , V V
TOTAL  TVA                                X.XXX.XXX,XX
TVA  C   RR,RR%                           X.XXX.XXX,XX
TOTAL  VAN.  EXCEPTATE  TVA  E            X.XXX.XXX,XX
TOTAL  VANZARI  ALTE  TAXE  F             X.XXX.XXX,XX
```

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7.10 PAYMENT commands

All PAYMENT_ commands may return error messages from the set of 8.4 Supplementary checks error codes.

7.10.1 PAYMENT_IN

Function: This command specifies the amount and type of payment done by a customer. A corresponding payment line is printed on the receipt. All printed amounts update the appropriate CMOS totalizers.

Code:
ESC MFB o a
ESC MFB1 g a *<PaymentType>*
ESC MFB1 g b *<PaymentValue>* ESC MFE

Parameter	Value	Format	
<PaymentType> kind of payment	30 H ≤ *n* ≤ 39H 36 H not allowed	max. 2 bytes 0..19 (see 7.4.5 for media list)	**Y**
<PaymentValue>	30 H ≤ *n* ≤ 39 H	max.9 bytes 1..9.999.999,99	**V**
Payment Name is taken from the payment list (see 5.6 Payment list handling)			**D**

Condition:

MFC Status byte 1

MFC_ TKD_RESET	MFC_ RECEIPT_ TOTAL	MFC_ DOCU_OPEN	MFC_ REPORT_ OPEN	MFC_ RECEIPT_ OPEN	MFC_ DAY_OPEN	MFC_ TRAINING_ MODE	MFC_ BLOCKADE
x	**1**	**0**	**x**	**x**	**x**	**x**	**0**

MFC Status byte 2

not used	not used	EJ_DATA_ PENDING	EJ_FULL	EJ_NEAR_ FULL	MFC_ CMD_ INTERRUPTED	MFC_ PRINTOUT_ INTERRUPTED	MFC_ FISCAL_ MODE
x	**x**	**x**	**x**	**x**	**x**	**0**	**x**

Only allowed in receipt types RETAIL_SALES_RECEIPT, RETAIL_SALES_RECEIPT_WITH_TAX_ID and PAYMENT_CORRECTION.

The parameter *<PaymentType>* = “6” CASH_BACK is not allowed and refused with error message EM_PARA_WRONG.

Computation: Receipt.Payment[*n*] += *<PaymentValue>* *n* = *<PaymentType>*

If the given *<PaymentValue>* greater the sum of Receipt.Tot[BRUTTO_SALES] + Day.Tot[BRUTTO_SALES] then the command is refused with error messages EM_DAY_TOT_OVERFLOW.

If the given *<PaymentType>* is not defined (see chapter 7.4.5 SET_PAYMENT_LIST) the command will be refused with error 0x79 EM_PAYMENT_TYPE_NOT_PRESENT

Error Msg:
EM_CMD_WRONG_ORDER
EM_EJ_FULL
EM_PARA_WRONG
EM_PARA_WRONG_FORMAT
EM_PARA_MISSING

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EM_PAYMENT_TYPE_NOT_PRESENT
EM_RECEIPT_TOT_OVERFLOW

Printout:

```
DDDDDDDDDDDDDDDDDDDDDDDDDDDD  V.VVV.VVV,VV
```

7.10.2 PAYMENT_IN_VOID

Function: This command cancels previously applied payment amounts. A corresponding payment cancellation line is printed on the receipt and stored on the EJ. All printed amounts update the appropriate CMOS totalizers.

Code:
ESC MFB o c
ESC MFB1 g a *<PaymentType>*
ESC MFB1 g b *<PaymentValue>* ESC MFE

Parameter	Value	Format	
<PaymentType> kind of payment	30 H ≤ *n* ≤ 39H 36 H not allowed	max. 2 bytes 0..19 (see 7.4.5 for media list)	**Y**
<PaymentValue>	30 H ≤ *n* ≤ 39 H	max.9 bytes 1..9.999.999,99	**V**
Payment Name is taken from the payment list (see 5.6 Payment list handling)			**D**

Condition:

MFC Status byte 1

MFC_ TKD_RESET	MFC_ RECEIPT_ TOTAL	MFC_ DOCU_OPEN	MFC_ REPORT_ OPEN	MFC_ RECEIPT_ OPEN	MFC_ DAY_OPEN	MFC_ TRAINING_ MODE	MFC_ BLOCKADE
x	**1**	**0**	**x**	**x**	**x**	**x**	**0**

MFC Status byte 2

not used	not used	EJ_DATA_ PENDING	EJ_FULL	EJ_NEAR_ FULL	MFC_ CMD_ INTERRUPTED	MFC_ PRINTOUT_ INTERRUPTED	MFC_ FISCAL_ MODE
x	**x**	**x**	**x**	**x**	**x**	**0**	**x**

Only allowed in receipt types RETAIL_SALES_RECEIPT, RETAIL_SALES_RECEIPT_WITH_TAX_ID and PAYMENT_CORRECTION.

The parameter <*PaymentType*> = “6” CASH_BACK is not allowed and refused with error message EM_PARA_WRONG.

Computation: Receipt.Payment[*n*] -= <*PaymentValue*> *n* = <*PaymentType*>

Notes: PAYMENT_IN_VOID is accepted only after at least one PAYMENT_IN command.

If the <PaymentValue> is bigger as the referenced totalizer, the command is ignored with error EM_RECEIPT_TOT_UNDERFLOW.

If the given <PaymentType> not defined the command will be refused with error 0x79 EM_PAYMENT_TYPE_NOT_PRESENT

Error Msg:
EM_CMD_WRONG_ORDER
EM_EJ_FULL
EM_PARA_WRONG
EM_PARA_WRONG_FORMAT
EM_PARA_MISSING

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EM_RECEIPT_TOT_UNDERFLOW
EM_PAYMENT_TYPE_NOT_PRESENT

Printout: <*PaymentType*> = 0...19

```
STORNARE
DDDDDDDDDDDDDDDDDDDDDDDDDDDDDDD  V.VVV.VVV,VV-
```

7.10.3 PAYMENT_OUT

Function: This command specifies the amount of change due returned to the customer. A corresponding change due line is printed on the receipt and stored on the EJ. All printed amounts update the appropriate CMOS totalizers.

Code: **ESC MFB o b**
ESC MFB1 g a <*PaymentType*>
ESC MFB1 g b <*PaymentValue*> ESC MFE

Parameter	Value	Format	
<PaymentType> Kind of payment	30 H ≤ *n* ≤ 39 H	max. 2 bytes 0..19 (see 7.4.5 for media list)	
<PaymentValue>	30 H ≤ *n* ≤ 39 H	max.9 bytes 1..9.999.999,99	**V**
Payment Name is taken from the payment list (see 5.6 Payment list handling)			**D**

Condition:

MFC Status byte 1

MFC_ TKD_RESET	MFC_ RECEIPT_ TOTAL	MFC_ DOCU_OPEN	MFC_ REPORT_ OPEN	MFC_ RECEIPT_ OPEN	MFC_ DAY_OPEN	MFC_ TRAINING_ MODE	MFC_ BLOCKADE
x	**1**	**0**	**x**	**x**	**x**	**x**	**0**

MFC Status byte 2

not used	not used	EJ_DATA_ PENDING	EJ_FULL	EJ_NEAR_ FULL	MFC_ CMD_ INTERRUPTED	MFC_ PRINTOUT_ INTERRUPTED	MFC_ FISCAL_ MODE
x	**x**	**x**	**x**	**x**	**x**	**0**	**x**

Only allowed in receipt types RETAIL_SALES_RECEIPT, RETAIL_SALES_RECEIPT_WITH_TAX_ID and PAYMENT_CORRECTION.

Computation: If the receipt type is **PAYMENT_CORRECTION**:

The parameter <*PaymentType*> = “6” CASH_BACK is not allowed and refused with error message EM_PARA_WRONG.

Receipt.Payment[*n*] -= <*PaymentValue*> *n* = <*PaymentType*>

If the receipt type is **RETAIL_SALES_RECEIPT** or **RETAIL_SALES_RECEIPT_WITH_TAX_ID**:

The parameter <*PaymentType*> = “0” CASH and “6” CASH_BACK is allowed only. All other parameter is refused with error message EM_PARA_WRONG.

***<PaymentType> = 0* (CASH)**

Receipt.Payment[CASH] -= <*PaymentValue*>

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If the <*PaymentValue*> is bigger than the referenced day + receipt totalizer + + CashBalance* then the command is refused with error EM_DAY_TOT_UNDERFLOW.

***<PaymentType> = 6* (CASH BACK)**

Receipt.Payment[CASH_BACK] += <*PaymentValue*>

If the given <*PaymentValue*> greater the sum of Receipt.Payment[CASH] + Day.Payment[CASH] + CashBalance* then the command is refused with error messages EM_DAY_TOT_OVERFLOW.

If the receipt type is RETAIL_SALES_RECEIPT or RETAIL_SALES_RECEIPT_WITH_TAX_ID:
If the sum of all accumulated payments is greater than the total value of receipt (∑Receipt.Payment[<*all PaymentTypes*>] > Receipt.Tot[BRUTTO_SALES]), then
Receipt.Tot[REST] =
∑Receipt.Payment[<*all PaymentTypes*>] - Receipt.Tot[BRUTTO_SALES]

Error Msg:

EM_CMD_WRONG_ORDER
EM_EJ_FULL
EM_PARA_WRONG
EM_PARA_WRONG_FORMAT
EM_PARA_MISSING
EM_PAYMENT_TYPE_NOT_PRESENT
EM_DAY_TOT_UNDERFLOW
EM_DAY_TOT_OVERFLOW

Printout:

```
DDDDDDDDDDDDDDDDDDDDDDDDDDDD  V.VVV.VVV,VV-
```

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7.11 FREEPRINT commands

The application has the opportunity to print non-fiscal information lines within one legal or illegal receipt by using two different commands: FREEPRINT_NORMAL for printing without any firmware restrictions and FREEPRINT_RESTRICTED with the restriction that in the right part of the ticket are no numbers allowed. Please see the table below what command is allowed in what receipt type during article sales stage and before the total command:

Article sell stage	FREEPRINT	
	_NORMAL	_RESTRICTED
Within Legal receipt	-	✓
Within Illegal receipt	✓	✓
After TOTAL_LINE	✓	✓

7.11.1 FREEPRINT_RESTRICTED

Function: An optional text is printed on the receipt and stored inside the EJ.

Code:
ESC MFB j b
ESC MFB1 b c <*PrintLine*>
ESC MFB1 f a <*Station*> ESC MFE

Parameter	Value	Format	
<Station>	31 H ≤ *n* ≤ 33 H	„1“ – Journal „2“ – Receipt „3“ – Receipt & Journal	
<PrintLine>	00 H ≤ *n* ≤ FF H	44 bytes	**T**
		In positions 33 to 44 no numbers (ASCII 0x30 to 0x39 inclusive) are allowed!	**R**

Condition:

MFC Status byte 1

MFC_ TKD_RESET	MFC_ RECEIPT_ TOTAL	MFC_ DOCU_OPEN	MFC_ REPORT_ OPEN	MFC_ RECEIPT_ OPEN	MFC_ DAY_OPEN	MFC_ TRAINING_ MODE	MFC_ BLOCKADE
x	**0**	**x**	**x**	**1**	**x**	**x**	**x**

MFC Status byte 2

not used	not used	EJ_DATA_ PENDING	EJ_FULL	EJ_NEAR_ FULL	MFC_ CMD_ INTERRUPTED	MFC_ PRINTOUT_ INTERRUPTED	MFC_ FISCAL_ MODE
x	**x**	**x**	**x**	**x**	**x**	**0**	**x**

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Error Msg: EM_CMD_WRONG_ORDER
EM_CMD_TOO_LONG
EM_PARA_WRONG
EM_PARA_WRONG_FORMAT
EM_PARA_RANGE_ERROR
EM_PARA_MISSING

Note: PrintLine must not contain any printer commands. Each character 0x00-0x1F is replaced using " " (blank)character

Printout:

| | | |

The following printer commands are allowed in the <*PrintLine*> definition*:*

(Please refer to TH230+ Programmers Guide for further information about the printer commands)

Print Attributes	
ESC !	Select printer mode
ESC a	Select justification
ESC r	Set current color[1)]
ESC I	Turn italic mode on/off
GS B	white/black reverse printing
Graphic	
GS /	Print current logo[1) 2)]
GS (L fn=69	Print the specified graphic[1)]
Barcode	
GS H	Printing Position of HRI
GS f	Select Font for HRI
GS h	Set barcode height
GS w	Set barcode width
GS k	Print bar code (Only m>=65 accepted) [1)]
GS p	Select PDF417 parameters

1) see also printer command GS 0x81 (set paper type)
2) logo previously defined with printer command GS *

Graphic and barcode print commands are allowed inside this command at the beginning of the line (i. e. a previous print line has to be terminated by LF).

Only one graphic or barcode per line is allowed. Characters printed behind graphics or barcodes are cut.

Graphics cannot be stored on the electronic journal. The graphic will be replaced by the string “[GRAPHIC]”.

Barcodes cannot be stored on the electronic journal. The barcode will be replaced by the string “[BARCODE]”.

All printing parameters set by ESC z n, ESC c 0 n and GS h n are set to the fiscal default: Emphasized, double height, double width, underline (border) → OFF, receipt station selected, parallel printing → OFF, barcode height → 10 print lines, barcode width → scale factor 2

Error Msg:

EM_CMD_WRONG_ORDER
EM_CMD_TOO_LONG
EM_PARA_WRONG
EM_PARA_WRONG_FORMAT
EM_PARA_RANGE_ERROR
EM_PARA_MISSING

Printout:

| | | |

or optional graphic/barcode printout

7.12 SPECIAL commands

7.12.1 SPECIAL_CASH_IN

Function: This command specifies a cash-in amount to be added to the totalizer and prints a service receipt.

Code: **ESC MFB i a**
ESC MFB1 f p <*CashDrawBalanceType*>
ESC MFB1 a g <*CashValue*> ESC MFE

Parameter	Value	Format	
<CashDrawBalanceType>	30 H, 31 H, 33 H	0 – init float (SOLD INITIAL) 1 – extra float (ALIMENTARE EXTRA) 3 – paid in (INCASARI)	**D**
<CashValue>	30 H ≤ *n* ≤ 39 H	max. 9 bytes 9.999.999,99	**V**

Condition:

MFC Status byte 1

MFC_ TKD_RESET	MFC_ RECEIPT_ TOTAL	MFC_ DOCU_OPEN	MFC_ REPORT_ OPEN	MFC_ RECEIPT_ OPEN	MFC_ DAY_OPEN	MFC_ TRAINING_ MODE	MFC_ BLOCKADE
x	**x**	**x**	**x**	**1**	**x**	**x**	**0**

MFC Status byte 2

not used	not used	EJ_DATA_ PENDING	EJ_FULL	EJ_NEAR_ FULL	MFC_ CMD_ INTERRUPTED	MFC_ PRINTOUT_ INTERRUPTED	MFC_ FISCAL_ MODE
x	**x**	**x**	**x**	**x**	**x**	**0**	**x**

Only allowed in receipt type: CASH_IN_RECEIPT

Only one SPECIAL_CASH_IN command is allowed per CASH_IN receipt.

Computation: Day.CashDrawBalance[*n*] += <*CashValue*> *n* = <*CashDrawBalanceType*>

Day.CashDrawBalance[CASH_IN_DRAW] += <*CashValue*>

Note: Cash-in values cannot be voided or returned after they have been posted with the SPECIAL_CASH_IN command. While the CASH_IN_RECEIPT is still open, SPECIAL_ALL_VOID is possibility for cancelling this ticket.

Error Msg: EM_CMD_WRONG_ORDER
EM_PARA_WRONG
EM_PARA_WRONG_FORMAT
EM_PARA_MISSING
EM_PRICE_OVERFLOW
EM_DAY_TOT_OVERFLOW

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```
                INTRODUCERE ÎN CASĂ
        NUMERAR: DDDDDDDDDDDDDDDDD
                 V.VVV.VVV,VV LEI
```

7.12.2 SPECIAL_CASH_OUT

Function: This command specifies a cash-out amount to be added to the totalizer and prints a service receipt.

Code:
ESC MFB i b
ESC MFB1 f p <*CashDrawBalanceType*>
ESC MFB1 a g <*CashValue*> ESC MFE

Parameter	Value	Format	
<CashDrawBalanceType>	32 H, 34 H	2 – pick up (REMITERE) 4 – paid out (PLĂȚI)	**D**
<CashValue>	30 H ≤ *n* ≤ 39 H	max. 9 bytes 9.999.999,99	**V**

Condition:

MFC Status byte 1

MFC_ TKD_RESET	MFC_ RECEIPT_ TOTAL	MFC_ DOCU_OPEN	MFC_ REPORT_ OPEN	MFC_ RECEIPT_ OPEN	MFC_ DAY_OPEN	MFC_ TRAINING_ MODE	MFC_ BLOCKADE
x	**x**	**x**	**x**	**1**	**x**	**x**	**0**

MFC Status byte 2

not used	not used	EJ_DATA_ PENDING	EJ_FULL	EJ_NEAR_ FULL	MFC_ CMD_ INTERRUPTED	MFC_ PRINTOUT_ INTERRUPTED	MFC_ FISCAL_ MODE
x	**x**	**x**	**x**	**x**	**x**	**0**	**x**

Only allowed in receipt type: CASH_OUT_RECEIPT

Only one SPECIAL_CASH_OUT command is allowed per CASH_OUT receipt.

Computation: Day.CashDrawBalance[*n*] += <*CashValue*> *n* = <*CashDrawBalanceType*>

Day.CashDrawBalance[CASH_IN_DRAW] -= <*CashValue*>

Note: Cash-out values cannot be voided or returned after they have been posted with the SPECIAL_CASH_OUT command. While the CASH_OUT_RECEIPT is still open, SPECIAL_ALL_VOID is possibility for cancelling this ticket.

Error Msg:
EM_CMD_WRONG_ORDER
EM_PARA_WRONG
EM_PARA_WRONG_FORMAT
EM_PARA_MISSING
EM_PRICE_OVERFLOW
EM_DAY_TOT_OVERFLOW
EM_DAY_TOT_UNDERFLOW

Printout:

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```
                    EXTRAGERE DIN CASĂ
          NUMERAR: DDDDDDDDDDDDDDDDD
                   -V.VVV.VVV,VV LEI
```

7.12.3 SPECIAL_CUT

Function: The ticket full cut is executed. It is to be used to cut the paper after a RECEIPT_END_MFLINE was used to close the receipt but allow for additional information to be printed on the ticket.

Code: **ESC MFB i l**
[ESC MFB2 f f <*CutType*>] ESC MFE

Parameter	Value	Format
<CutType>	30 H ≤ *n* ≤ 31 H	0 – no cut 1 – full cut

Condition:

MFC Status byte 1

MFC_ TKD_RESET	MFC_ RECEIPT_ TOTAL	MFC_ DOCU_OPEN	MFC_ REPORT_ OPEN	MFC_ RECEIPT_ OPEN	MFC_ DAY_OPEN	MFC_ TRAINING_ MODE	MFC_ BLOCKADE
x	**x**	**x**	**0**	**0**	**1**	**x**	**0**

MFC Status byte 2

not used	not used	EJ_DATA_ PENDING	EJ_FULL	EJ_NEAR_ FULL	MFC_ CMD_ INTERRUPTED	MFC_ PRINTOUT_ INTERRUPTED	MFC_ FISCAL_ MODE
x	**x**	**x**	**x**	**x**	**0**	**x**	**x**

Note: Only available outside of a fiscal ticket.

The parameter <*CutType*> = 0 (no cut) has no effect.

Error Msg: EM_CMD_WRONG_ORDER
EM_PARA_WRONG
EM_PARA_RANGE_ERROR

7.12.4 SPECIAL_PRINT_HEADER_CUT

Function: The command has the same effect as SPECIAL_CUT. It is available only due to historical reasons.

Code: *ESC MFB i g ESC MFE*

Error Msg: EM_CMD_WRONG_ORDER
EM_PARA_WRONG
EM_PARA_RANGE_ERROR

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7.12.5 SPECIAL_ALL_VOID

Function: This command cancels a receipt or a fiscal report. The “cancel information” lines are printed on the receipt and stored inside the EJ.

Code: **ESC MFB i i ESC MFE**

Condition:

MFC Status byte 1

MFC_ TKD_RESET	MFC_ RECEIPT_ TOTAL	MFC_ DOCU_OPEN	MFC_ REPORT_ OPEN	MFC_ RECEIPT_ OPEN	MFC_ DAY_OPEN	MFC_ TRAINING_ MODE	MFC_ BLOCKADE
x	x		1 OR	1	x	x	x

MFC Status byte 2

not used	not used	EJ_DATA_ PENDING	EJ_FULL	EJ_NEAR_ FULL	MFC_ CMD_ INTERRUPTED	MFC_ PRINTOUT_ INTERRUPTED	MFC_ FISCAL_ MODE
x	x	x	x	x	0	x	x

Computation: If the receipt to be closed is <*ReceiptType*> = RETAIL_SALES_RECEIPT|RETAIL_SALES_RECEIPT_WITH_TAX_ID:

Day.Tot[VOID] += Receipt.Tot[VOID]
Day.Tot[VOID] += Receipt.Tot[BRUTTO_SALES]

Day.EventCnt[VOID] += 1
Day.ReceiptTypeCnt[RETAIL_SALES_RECEIPT] -= 1
Day.ReceiptGroupCnt[LEGAL_RECEIPT] -= 1
stGrand.ulGrandLegalReceiptCnt -= 1

Day.ReceiptTypeCnt[GENERIC_RECEIPT] += 1
Day.ReceiptGroupCnt[ILLEGAL_RECEIPT] += 1

If <*ReceiptType*> = GENERIC_RECEIPT or
CASH_IN_RECEIPT or
CASH_OUT_RECEIPT

Day.EventCnt[VOID] += 1

MFC_RECEIPT_TOTAL, MFC_RECEIPT_OPEN, MFC_REPORT_OPEN, MFC_DOCU_OPEN, MFC_PRINTOUT_INTERRUPTED will be reset as needed.

Note: The ticket is always ended as an illegal receipt.

The MFC_PRINTOUT_INTERRUPTED bit can only be reset by sending a SPECIAL_ALL_VOID command.

In case of an interruption during printing, the MFC_CMD_INTERRUPTED bit will be set and the application can restart this interrupted command by send a SPECIAL_CMD_RESTART command (see chapter 7.12.6 SPECIAL_CMD_RESTART).

Error Msg: EM_CMD_WRONG_ORDER
Error messages from the set of 8.3 EJ error codes
Error messages from the set of 8.4 Supplementary checks error codes

Printout:

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```
*******  BON  ANULAT  *******  BON  ANULAT  ******
                                 ----------
TOTAL                                     0,00

EJTRZ:  00005
BNF. 12345    DATA:02/12/2007    ORA:10:20:25
                                BU4710123456 7
*****  BON  NEFISCAL  ******  BON  NEFISCAL  *****
```

7.12.6 SPECIAL_CMD_RESTART

Function: With this command, the application can restart a previously interrupted command by a power failure, printer or EJ error. This command applies to DAY_END (7.5.2), RECEIPT_END (7.6.2) and SPECIAL_ALL_VOID (7.12.5).

Code: **ESC MFB i j ESC MFE**

Condition:

MFC Status byte 1

MFC_ TKD_RESET	MFC_ RECEIPT_ TOTAL	MFC_ DOCU_OPEN	MFC_ REPORT_ OPEN	MFC_ RECEIPT_ OPEN	MFC_ DAY_OPEN	MFC_ TRAINING_ MODE	MFC_ BLOCKADE
x	x	x	x	x	x	x	x

MFC Status byte 2

not used	not used	EJMEM_DATA_ PENDING	EJ_FULL	EJ_NEARLY_ FULL	MFC_ CMD_ INTERRUPTED	MFC_ PRINTOUT_ INTERRUPTED	MFC_ FISCAL_ MODE
x	x	x	x	x	1	x	x

Note: The MFC_CMD_INTERRUPTED bit can only be reset by sending a SPECIAL_CMD_RESTART command.

Error Msg:
EM_CMD_WRONG_ORDER
EM_SC_DEFECT
EM_TAX_TERMINAL_WRONG
EM_PARAM_WRONG
Error messages from the set of 8.2 FM error codes.
Error messages from the set of 8.3 EJ error codes
Error messages from the set of 8.4 Supplementary checks error codes

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7.13 REPORT commands

7.13.1 General Note

After issuing the REPORT_ command, the MFC_REPORT_OPEN flag is set. This is only reset if the entire report is printed or if a SPECIAL_ALL_VOID command is applied.

Fiscal reports can only be printed while the FISCAL_MODE flag or BLOCKADE flag is set.

Fiscal reports are printed on the receipt station only and not stored in the EJ.

If possible, the relevant SNO, VAT rates and TH block of the given report period are printed before.

Paper near-end status is ignored in all cases.

The optional <Station> parameter is reserved for future development and only accepts the value “2” at this time.

All REPORT_ commands may return error messages from the set of 8.2 FM error codes. Reports without any data within the defined range will not return an error, but print the header and footer.

All REPORT_ commands return an error message, if the Tax Terminal is not enabled (paired), any Tax Terminal component is not available or the customer display is not connected.

7.13.2 REPORT_BLOCK2BLOCK

Function: With this command the application prints a fiscal report (of every block type) from logical block number to logical block number. This report is either printed on the receipt station or is held ready as soft copy to be fetched by the application.

Code:

ESC MFB g a
ESC MFB1 d a <*FromValue*>
ESC MFB1 d b <*ToValue*>
[ESC MFB2 f a <*Station*>]
[ESC MFB2 f d < *ReportType*>] ESC MFE

Parameter	Value	Format	
<FromValue>	1 – max. block no. (5.7.10)	ASCII, 30 H ≤ *n* ≤ 39 H, max. 5 bytes	F
<ToValue>	1 – max. block no. (5.7.10)	ASCII, 30 H ≤ *n* ≤ 39 H, max. 5 bytes	T
<Station>	30 H ≤ *n* ≤ 39 H	“2“ - Receipt (Default) “16” - Host – For EJ soft copy	

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		"32" - External Storage	
<ReportType>	30 H ≤ n ≤ 39 H	see 5.7.1 for valid types	R

Condition:

MFC Status byte 1

MFC_ TKD_RESET	MFC_ RECEIPT_ TOTAL	MFC_ DOCU_OPEN	MFC_ REPORT_ OPEN	MFC_ RECEIPT_ OPEN	MFC_ DAY_OPEN	MFC_ TRAINING_ MODE	MFC_ BLOCKADE
x	**x**	**x**	**0**	**0**	**x**	**x**	**x**

MFC Status byte 2

not used	not used	EJ_DATA_ PENDING	EJ_FULL	EJ_NEAR_ FULL	MFC_ CMD_ INTERRUPTED	MFC_ PRINTOUT_ INTERRUPTED	MFC_ FISCAL_ MODE
x	**x**	**x**	**x**	**x**	**x**	**0**	**1**

Note:

The input parameters are logical block numbers. See chapter 3.4 Block handling (logical and physical blocks).

If the parameter <ReportType> is given, then only this block type will be printed, otherwise every block type. To print EOD blocks only, it is recommended to use command REPORT_EOD2EOD.

Report commands which may include EOD blocks (<ReportType> empty or EOD) will be rejected with an error message 76 EM_CURRENCY_CHANGE_IN_RANGE, if the range includes a currency change block.

The command 7.15.22 GET_MFMEM_LAST_LOGICAL_BL_NUM returns the last logical block numbers.

Error Msg:

EM_CMD_WRONG_ORDER
EM_CMD_WRONG_FORMAT
EM_PARA_WRONG
EM_PARA_WRONG_FORMAT
EM_PARA_MISSING
EM_PARAM_RANGE_ERROR
EM_CHANGE_CURRENCY_IN_RANGE

Printout:

The report consists of a standard header, a (fixed) report receipt title, the required data and the illegal receipt footer.

```
0123456789012345678901234567890123456789012 3
*****   W I N C O R - N I X D O R F    *****   Header Line 1
            receipt header line 2               Header Line 2
            receipt header line 3               Header Line 3
            receipt header line 4               Header Line 4
     Cod Fiscal C.I.F.: 12345678901234          Header Line 5
          R A P O R T  F I S C A L              Report receipt title
                          …
BLOC: RRRRRRRRRRRRRRRRRRRRRRRRRRRRRRRRRRRRRRR    REPORT_BLOCK2BLOCK Report range
        DE LA FFFFF  LA TTTTT                    <FromValue> <ToValue>
                          or
```

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```
  BLOC: RRRRRRRRRRRRRRRRRRRRRRRRRRRRRRRRRRRRR
        DE LA  FF:FF:FFFF
           LA  TT:TT:TTTT
```

REPORT_DATE2DATE
<FromDate>
<ToDate>

or

```
  BLOC: RRRRRRRRRRRRRRRRRRRRRRRRRRRRRRRRRRRRR
        DE LA  FF:FF:FFFF FF:FF:FF
           LA  TT:TT:TTTT TT:TT:TT
```

REPORT_DATE2DATE
<FromDateTime>
<ToDateTime>

or

```
      RAPORT-Z: DE LA FFFFF  LA TTTTT
```

REPORT_EOD2EOD

or

```
    SUMA DE LA  FF:FF:FFFF FF:FF:FF
             LA  TT:TT:TTTT TT:TT:TT
```

REPORT_TOTAL_DATE2DATE

…

```
   PRINTER              : FFC-TH230+
   LOGOTIP              : RL
   SERIA NR.            : BU4710123456
   DATA INITIALZÂRII    : 06/01/2008
   TIMPUL INITIALZÂRII  : 09:48:35
   VERSIUNEA FIRMWARE   : 00/14/99

      1 20/11/2015 12:23:00

BLOC SNO     1
---------------------------------------------
      2  20/11/2015 12:25:00

BLOC TVA 1
BLOC DATE TVA
VECHI   0,00%  NOU    9,00%  (A)
VECHI   0,00%  NOU   19,00%  (B)
VECHI   0,00%  NOU    7,50%  (C)
VECHI   0,00%  NOU   32,25%  (D)
VECHI   0,00%  NOU    0,00%  (E)
VECHI   0,00%  NOU    0,00%  (F)
VECHI   0,00%  NOU   10,00%  (G)
VECHI   0,00%  NOU   20,00%  (H)
---------------------------------------------
      3  20/11/2015  12:26:00

BLOC ANTET     1

*****   W I N C O R - N I X D O R F    *****

            receipt header line 2

            receipt header line 3

            receipt header line 4

     Cod Fiscal C.I.F.: RO1234567890123



---------------------------------------------
      4  20/11/2015 12:27:00

BLOC MODIFICĂRI JURNAL ELECTRONIC 1
  EJ Code 001
  EJ UMID SXXXXXXXXX03201C43E4
---------------------------------------------
      6  21/11/2015 08:05:14

BLOC TKD   1
20.11.2015 21:08:49 00-14-99 Error: 84
---------------------------------------------
      7  21/11/2015 10:22:05

BLOC FIRMWARE CHANGE   1

OLD FIRMWARE VERSION 00-14-99
NEW FIRMWARE VERSION 01-14-01
```

Printer
Logo type
Serial number
Date of initialisation (DD/MM/YY)
Time of initialisation
Firmware release

SNO block (always printed if *ReportType* is not defined)

Valid VAT rate (always printed if *ReportType* is not defined)

Fixed 0
Road Tax
Optional
Optional

Valid receipt header (always printed if *ReportType* is not defined)

EJ change block [optional when block is in range]

TKD block with information about date/time and error [optional when block is in range]

FW block with information about when an update was done from version to version [optional when block is in range]

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----------	
8 21/11/2015 20:12:32	**1st EOD block**
RAPORT Z 1	
BONURI FISCALE 2	**number of fiscal receipts**
TOTAL VÂNZĂRI BRUTE (A) 0,99	
TOTALIZATOR TVA 9,00% 0,08	
TOTAL VÂNZĂRI BRUTE (B 5,00	
TOTALIZATOR TVA 19,00% 0,80	
TOTAL VÂNZĂRI BRUTE (C) 9,00	
TOTALIZATOR TVA 7,50% 0,63	
TOTAL VÂNZĂRI BRUTE (D) 20,00	
TOTALIZATOR TVA 32,25% 4,88	
TOTAL VÂN. EXCEPT. TVA 100,00	
TOTAL VÂNZĂRI ALTE TAXE 2,00	
TOTAL VÂNZĂRI BRUTE (G) 9,00	**Only if VAT G or H are active**
TOTALIZATOR TVA 10,00% 0,63	
TOTAL VÂNZĂRI BRUTE (H) 20,00	
TOTALIZATOR TVA 20,00% 4,88	
TOTAL GENERAL	**see 7.5.4 for details**
VÂNZĂRI BRUTE 65,99	
VÂNZĂRI TAXABILE 63,99	
VALOARE TVA 11,90	
NR. BONURI CU CIF 1	**Number of tickets with CIF**
TOTAL VÂNZĂRI CU CIF 2,34	**Total value of sales on tickets with CIF**
VALORARE TVA CIF 0,10	**Total value of VAT on tickets with CIF(VAT A,B,C,D,G,H) w/o E and F**
ID JURNAL ELECTRONIC SXXXX	**Id of Electronic Journal**
NR. SOLDURI INIŢIALE 0	**No. of Initial Float operations**
SOLD INIŢIAL 100,00	**Value of Initial Float**
NR. ALIMENTĂRI EXTRA 0	**No. of Extra Float operations**
ALIMENTARE EXTRA 500,00	**Value of Extra Float**
NR. REMITERI 0	**No. of Pick-Up operations**
REMITERE 300,00	**Value of Pick-Ups**
NR. ÎNCASĂRI 0	**No. of Paid In operations**
ÎNCASĂRI 14.189,56	**Value of Paid In**
NR. PLAŢI 0	**No. of Paid Out operations**
PLĂŢI 10.113,04	**Value of Paid Out**
CASH 2.820,70	**Value of cash payment**
CASH ÎN SERTAR 2.010,00	**Value of cash payment - REST + CASH_IN - CASH_OUT**
CARD 20.000,00	**Value of card payment**
TICHETE 25,00	
VOUCHER 100,00	**Value of voucher payment**
ALTE ÎNCASĂRI 23,00	**Value of other payment**
MIJLOC de PLATĂ 5 95,82	**Value of type 5 payment**
MIJLOC de PLATĂ 6 997,00	**Value of type 6 payment**
...	...
MIJLOC de PLATĂ 19 215,22	**Value of type 19 payment**
FB383AFE C99ACDCC 768BD99F 1C40A3AD C0974C43 DB493AFF D39B5DCF 99FF	**Message digest of this Ej Session**

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9 22/11/2015 20:14:03	**2nd EOD block**
RAPORT Z 2	
•	
•	
•	
---	
29 07/12/2015 07:24:13	
BLOC TKD 1 05.12.2007 07:06:19 00-14-99 Error: 84	**TKD block with information about date/time and error**
---	
30 07/12/2007 14:10:40	**17th EOD block**
RAPORT Z 17	
•	
•	
•	
FB383AFE C99ACDCC 768BD99F 1C40A3AD C0974C43	

SUMA PE PERIOADA RAPORTATA:	**Summary of all EOD blocks (printed only, if EOD block was printed)**
BONURI FISCALE 27	
TOTAL VÂN. BRUTE (A) 3,99	
TOTALIZATOR TVA 0,08	
TOTAL VÂN. BRUTE (B) 12.000.175,47	
TOTALIZATOR TVA 1.189.206,89	
TOTAL VÂN. BRUTE (C) 14,00	
TOTALIZATOR TVA 2,25	
TOTAL VÂN. BRUTE (D) 20,00	
TOTALIZATOR TVA 4,88	
TOTAL VÂN. EXCEPT. TVA 100,00	
TOTAL VÂN. ALTE TAXE 2,00	
TOTAL VÂN. BRUTE (G) 300,00	**Only if VAT G or H are active**
TOTALIZATOR TVA 28,50	
TOTAL VÂN. BRUTE (H) 1.500,00	
TOTALIZATOR TVA 136,25	
TOTAL VÂNZĂRI BRUTE 12.002.015,46	**see section** TOTAL GENERAL on Raport-Z ReportLayout
TOTAL VÂNZĂRI TAXABILE 12.002.013,46	
TOTAL TVA 1.189.378,85	
TOTAL NR. DE BONURI CU CIF 234	Sum of "Nr. BONURI CU CIF"
TOTALIZATOR VÂNZĂRI CU CIF 1.879,89	Sum of "TOTAL VÂNZĂRI CU CIF"
TOTALIZATOR TVA CU CIF 289,33	Sum of "VALOARE TVA CIF"
TOTAL NR. SOLDURI INIŢIALE 30	Sum of "Nr. SOLDURI INIŢIALE"
TOTALIZATOR SOLD INIŢIAL 100,00	Sum of "SOLD INIŢIAL"
TOTAL NR. ALIMENTĂRI EXTRA 3	Sum of "Nr. ALIMENTĂRI EXTRA"
TOTALIZATOR ALIMENTARE EXTRA 500,00	Sum of "ALIMENTARE EXTRA"
TOTAL NR. REMITERI 2	Sum of "NR. REMITERI"
TOTALIZATOR REMITERE 300,00	Sum of "REMITERI"
TOTAL NR. ÎNCASĂRI 5	Sum of "NR. ÎNCASĂRI"
TOTALIZATOR ÎNCASĂRI 14.189,56	Sum of "ÎNCASĂRI"
TOTAL NR. PLĂŢI 3	Sum of "NR. PLĂŢI"
TOTALIZATOR PLĂŢI 10.113,04	Sum of "PLĂŢI"
TOTALIZATOR CASH 2.820,70	Sum of "CASH"
TOTALIZATOR CASH ÎN SERTAR 2.010,00	Sum of "CASH ÎN SERTAR"
TOTALIZATOR CARD 995,82	Sum of "CARD"
TOTALIZATOR TICHETE 5,00	Sum of "TICHETE"
TOTALIZATOR VOUCHER 0,00	Sum of "VOUCHER"
TOTALIZATOR ALTE ÎNCASĂRI 0,00	Sum of "ALTE ÎNCASĂRI"
TOTALIZATOR MIJLOC de PLATĂ 5 5,82	Sum of "MIJLOC de PLATĂ 5"
…	

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TOTALIZATOR MIJLOC de PLATĂ 19 5,00	Sum of "MIJLOC de PLATĂ 19"

7 21/11/2015 10:22:05 BLOC MODIFICĂRI MONEDĂ 1 New Currency: EURO ----------	**CRY-BlockInformation (printed only if *ReportType* is defined)**

DATA:14/12/2007 ORA:13:26:42 *RL* 47101234567 R A P O R T F I S C A L	**Legal footer**

The report subtitle is:

| | | |

ESC MFB1 (e b <*ToDate*> | e f <*ToDateTime*>)
[ESC MFB2 f a <*Station*>]
[ESC MFB2 f d <*ReportType*>] ESC MFE

Parameter	Value	Format	
<FromDate>	DD/MM/YY ASCII, 30 H ≤ *n* ≤ 39 H	DD – day MM – month YY – year	F
<ToDate>	DD/MM/YY ASCII, 30 H ≤ *n* ≤ 39 H	DD – day MM – month YY – year	T
<FromDateTime>	DD.MM.YYYY hh:mm:ss	DD – day hh - hour MM – month mm - month YYYY – year ss - second	F
<ToDateTime>	DD.MM.YYYY hh:mm:ss	DD – day hh - hour MM – month mm - month YYYY – year ss - second	T
<Station>	30 H ≤ n ≤ 39 H	"2" - Receipt (Default) "16" - Host – For EJ soft copy "32" - External Storage	
<ReportType>	30 H ≤ n ≤ 39 H	see 5.7.1 for valid types	R

Condition:

MFC Status byte 1

MFC_ TKD_RESET	MFC_ RECEIPT_ TOTAL_	MFC_ DOCU_OPEN	MFC_ REPORT_ OPEN_	MFC_ RECEIPT_ OPEN_	MFC_ DAY_OPEN	MFC_ TRAINING_ MODE_	MFC_ BLOCKADE
x	**x**	**x**	**0**	**0**	**x**	**x**	**x**

MFC Status byte 2

not used	not used	EJ_DATA_ PENDING	EJ_FULL	EJ_NEAR_ FULL	MFC_ CMD_ INTERRUPTED	MFC_ PRINTOUT_ INTERRUPTED	MFC_ FISCAL_ MODE
x	**x**	**x**	**x**	**x**	**x**	**0**	**1**

Note:

If the parameter <FromDate> and <ToDate > is given then they was translated in <FromDateTime> and <ToDateTime>. The "FromTime" is 00:00:00 and the "ToTime" is 23:59:59.

If the parameter <ReportType> is given, then only this block type will be printed, otherwise every block type.

Report commands which may include EOD blocks (<ReportType> empty or EOD) will be rejected with an error message 76 EM_CURRENCY_CHANGE_IN_RANGE, if the range includes a currency change block.

Error Msg:

EM_CMD_WRONG_ORDER
EM_PARA_WRONG
EM_PARA_DATE_WRONG_FORMAT
EM_PARA_MISSING

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EM_CHANGE_CURRENCY_IN_RANGE

Printout: See REPORT_BLOCK2BLOCK (7.13.2).

7.13.4 REPORT_EOD2EOD

Function: With this command the application prints a fiscal report from DAY_END to DAY_END. Only EOD blocks are evaluated. This report is either printed on the receipt station or is held ready as soft copy to be fetched by the application.

Code:

ESC MFB g c
ESC MFB1 d a <*FromValue*>
ESC MFB1 d b <*ToValue*>
[ESC MFB2 f a <*Station*>] ESC MFE

Parameter	Value	Format	
<FromValue>	1 – max. EOD no. (5.7.10)	ASCII, 30 H ≤ *n* ≤ 39 H, max. 5 bytes	F
<ToValue>	1 – max. EOD no. (5.7.10)	ASCII, 30 H ≤ *n* ≤ 39 H, max. 5 bytes	T
<Station>	30 H ≤ n ≤ 39 H	"2" - Receipt (Default) "16" - Host – For EJ soft copy "32" - External Storage	

Condition:

MFC Status byte 1

MFC_ TKD_RESET	MFC_ RECEIPT_ TOTAL	MFC_ DOCU_OPEN	MFC_ REPORT_ OPEN	MFC_ RECEIPT_ OPEN	MFC_ DAY_OPEN	MFC_ TRAINING_ MODE	MFC_ BLOCKADE
x	**x**	**x**	**0**	**0**	**x**	**x**	**x**

MFC Status byte 2

not used	not used	EJ_DATA_ PENDING	EJ_FULL	EJ_NEAR_ FULL	MFC_ CMD_ INTERRUPTED	MFC_ PRINTOUT_ INTERRUPTED	MFC_ FISCAL_ MODE
x	**x**	**x**	**x**	**x**	**x**	**0**	**1**

Note: Input values for this command are EOD counters. (e.g. print from the 1st EOD to the 20th EOD) not logical block numbers.

With this command only the EOD, the VAT and the TH blocks will be printed.

Report commands with a range over two currencies will be rejected with an error message 76 EM_CURRENCY_CHANGE_IN_RANGE.

Error Msg:

EM_CMD_WRONG_ORDER
EM_CMD_WRONG_FORMAT
EM_PARA_WRONG
EM_PARA_WRONG_FORMAT
EM_PARA_MISSING
EM_PARAM_RANGE_ERROR
EM_CHANGE_CURRENCY_IN_RANGE

Printout: See REPORT_BLOCK2BLOCK (7.13.2).

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7.13.5 REPORT_TOTAL_DATE2DATE

Function: With this command the application prints a fiscal report with only the totals for the specified period. This report is either printed on the receipt station or is held ready as soft copy to be fetched by the application.

Code:

ESC MFB g d
ESC MFB1 (e a <*FromDate*> | e e <*FromDateTime*>)
ESC MFB1 (e b <*ToDate*> | e f <*ToDateTime*>)
[ESC MFB2 f a <*Station*>] ESC MFE

Parameter	Value	Format	
<FromDate>	DD/MM/YY ASCII, 30 H ≤ *n* ≤ 39 H	DD – day MM – month YY – year	F
<ToDate>	DD/MM/YY ASCII, 30 H ≤ *n* ≤ 39 H	DD – day MM – month YY – year	T
<FromDateTime>	DD.MM.YYYY hh:mm:ss	DD – day hh - hour MM – month mm - month YYYY – year ss - second	F
<ToDateTime>	DD.MM.YYYY hh:mm:ss	DD – day hh - hour MM – month mm - month YYYY – year ss - second	T
<Station>	30 H ≤ n ≤ 39 H	"2" - Receipt (Default) "16" - Host – For EJ soft copy "32" - External Storage	

Condition:

MFC Status byte 1

MFC_ TKD_RESET	MFC_ RECEIPT_ TOTAL	MFC_ DOCU_OPEN	MFC_ REPORT_ OPEN	MFC_ RECEIPT_ OPEN	MFC_ DAY_OPEN	MFC_ TRAINING_ MODE	MFC_ BLOCKADE
x	**x**	**x**	**0**	**0**	**x**	**x**	**x**

MFC Status byte 2

not used	not used	EJ_DATA_ PENDING	EJ_FULL	EJ_NEAR_ FULL	MFC_ CMD_ INTERRUPTED	MFC_ PRINTOUT_ INTERRUPTED	MFC_ FISCAL_ MODE
x	**x**	**x**	**x**	**x**	**x**	**0**	**1**

Note: If the parameter <FromDate> and <ToDate > is given then they was translated in <FromDateTime> and <ToDateTime> with the "FromTime" set to 00:00:00 and the "ToTime" set to 23:59:59.

Report commands with a range over two currencies will be rejected with an error message 76 EM_CURRENCY_CHANGE_IN_RANGE.

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Error Msg: EM_CMD_WRONG_ORDER
EM_PARA_WRONG
EM_PARA_DATE_WRONG_FORMAT
EM_PARA_MISSING
EM_CURRENCY_CHANGE_IN_RANGE

Printout: See REPORT_BLOCK2BLOCK (7.13.2).

Example for REPORT_TOTAL_DATE2DATE:

```
0123456789012345678901234567890123456789012 3

*****   W I N C O R - N I X D O R F    *****

            receipt header line 2

            receipt header line 3

            receipt header line 4

     Cod Fiscal C.I.F.: 12345678901234

          R A P O R T   F I S C A L

    SUMA DE LA  01.12.2015 08:00:00
            LA  01.12.2015 22:30:00
   PRINTER             : FFC-TH230+
   LOGOTIP             : RL
   SERIA NR.           : BU4710123456
   DATA INITIALZÂRII   : 06/01/2008
   TIMPUL INITIALZÂRII : 09:48:35
   VERSIUNEA FIRMWARE  : 00/14/99

      1  20.11.2015 12:23:00

BLOC SNO     1
---------------------------------------------
      3  20.11.2015 12:25:00

BLOC TVA     1
BLOC DATE TVA
VECHI  0,00%  NOU  9,00% (A)
VECHI  0,00%  NOU 19,00% (B)
VECHI  0,00%  NOU  7,50% (C)
VECHI  0,00%  NOU 32,25% (D)
VECHI  0,00%  NOU  0,00% (E)
VECHI  0,00%  NOU  0,00% (F)
VECHI  0,00%  NOU 10,00% (G)
VECHI  0,00%  NOU 20,00% (H)
---------------------------------------------
      4  20.11.2015  12:26:00

BLOC ANTET     1

*****   W I N C O R - N I X D O R F    *****

           receipt header line 2

     Cod Fiscal C.I.F.: 12345678901234


---------------------------------------------

SUMA PE PERIOADA RAPORTATA:

BONURI FISCALE                            27
```

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```
TOTAL VÂN. BRUTE (A)                    3,99
TOTALIZATOR TVA                         0,08
TOTAL VÂN. BRUTE (B)           12.000.175,47
TOTALIZATOR TVA                 1.189.206,89
TOTAL VÂN. BRUTE (C)                   14,00
TOTALIZATOR TVA                         2,25
TOTAL VÂN. BRUTE (D)                   20,00
TOTALIZATOR TVA                         4,88
TOTAL VÂN. EXCEPT. TVA                100,00
TOTAL VÂN. ALTE TAXE                    0,00
TOTAL VÂN. BRUTE (G)                  300,00
TOTALIZATOR TVA                        28,50
TOTAL VÂN. BRUTE (H)                1.500,00
TOTALIZATOR TVA                       136,25

TOTAL VÂNZĂRI BRUTE            12.000.213,46
TOTAL TVA                       1.189.214,10

       DATA:14/12/2007    ORA:13-26-42
             RL      47101234567

          R A P O R T  F I S C A L
```

Note: This report contains a printout of all kinds of blocks **except EOD blocks** and a summary of all turnovers in the given period.

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7.14 TRAINING commands

7.14.1 General Note

Training mode is allowed only before the printer has started a fiscal day. Once the printer has been used for fiscal transactions it is no longer allowed to be switched into training mode.

The receipt and day totalizers will be kept correctly as if fiscal tickets were generated, but not stored in the MF controller or EJ. This means that when DAY_END is issued all counters and totalizers will be reset to 0.

7.14.2 TRAINING_BEGIN

Function: This command informs the MFC that the application requests a change to the training mode. The status changes to MFC_TRAINING_MODE. The training mode serves to train the POS operators under realistic conditions without the entered totals being stored. The receipts created in this mode are identified as non-fiscal and therefore are not saved into the EJ.

Code: **ESC MFB t a ESC MFE**

Code old: ESC MFB I a ESC MFE (expired - not available for new implementations)

Condition:

MFC Status byte 1

MFC_ TKD_RESET	MFC_ RECEIPT_ TOTAL	MFC_ DOCU_OPEN	MFC_ REPORT_ OPEN	MFC_ RECEIPT_ OPEN	MFC_ DAY_OPEN	MFC_ TRAINING_ MODE	MFC_ BLOCKADE
x	x	x	x	x	0	0	0

MFC Status byte 2

not used	not used	EJ_DATA_ PENDING	EJ_FULL	EJ_NEAR_ FULL	MFC_ CMD_ INTERRUPTED	MFC_ PRINTOUT_ INTERRUPTED	MFC_ FISCAL_ MODE
x	x	x	x	x	0	0	x

Note: This command is only available before the first EOD block is stored.

In training mode a special information line is printed after the ticket header and after the ticket footer. The MF logo and the SNO number are not printed in training mode.

After the command is successfully executed the status bit MFC_TRAINING_MODE is set.

Error Msg: EM_CMD_WRONG_ORDER
EM_DATE_NOT_PRESENT
EM_MFMEM_NOT_FORMATTED

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Printout: Administrative receipt with optional header lines:

```
******  W I N C O R - N I X D O R F      *******
              receipt header line 2
              receipt header line 3
              receipt header line 4
        Cod Fiscal C.I.F.:1234567890123 4

****   BON NEFISCAL   ****   BON NEFISCAL    ****

          T R A I N I N G _ B E G I N

        DATA: 16/05/2015      ORA:17-57-56
****   BON NEFISCAL   ****   BON NEFISCAL    ****
```

After all receipt titles the following line must be printed.

```
****   BON NEFISCAL   ****   BON NEFISCAL    ****
```

7.14.3 TRAINING_END

Function: This command informs the MFC that the application requests a change to the normal operation mode.

Code: **ESC MFB t b ESC MFE**

Code old: ESC MFB I b ESC MFE (expired - not available for new implementations)

Condition:

MFC Status byte 1

MFC_ TKD_RESET	MFC_ RECEIPT_ TOTAL	MFC_ DOCU_OPEN	MFC_ REPORT_ OPEN	MFC_ RECEIPT_ OPEN	MFC_ DAY_OPEN	MFC_ TRAINING_ MODE	MFC_ BLOCKADE
x	**x**	**x**	**x**	**x**	**0**	**1**	**0**

MFC Status byte 2

not used	not used	EJ_DATA_ PENDING	EJ_FULL	EJ_NEAR_ FULL	MFC_ CMD_ INTERRUPTED	MFC_ PRINTOUT_ INTERRUPTED	MFC_ FISCAL_ MODE
x	**x**	**x**	**x**	**x**	**x**	**x**	**x**

Computation: After the command is successfully executed the status bit MFC_TRAINING_MODE is cleared.

Error Msg: EM_CMD_WRONG_ORDER

Printout: Administrative receipt with optional header lines:

```
******  W I N C O R - N I X D O R F      *******
              receipt header line 2
              receipt header line 3
              receipt header line 4
        Cod Fiscal C.I.F.:1234567890123 4

****   BON NEFISCAL   ****   BON NEFISCAL    ****
```

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```
TRAINING_END

DATA: 16/05/2015   ORA:17-57-56
**** BON NEFISCAL **** BON NEFISCAL ****
```

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7.15 GET commands

7.15.1 General Note

- GET commands output ASCII data strings.
- GET commands never print anything.
- MFC status or other CMOS data are never affected by GET commands.
- All numbers are sent without leading zero.
- Decimals are always implied for amount and percentage values (e. g.: "123" means "1,23")
- GET commands are not executable if an error is pending
- GET commands are accepted from display port or application port and sends the answer comes back on the same port.

7.15.2 GET_RECEIPT_TOTALIZER

Function: This command requests the receipt totalizers.

Code: **ESC MFB k a ESC MFE ESC + 09H**

Condition: The totalizers are valid while the receipt is open (MFC_RECEIPT_OPEN = 1). The totalizers will be cleared after either RECEIPT_END or SPECIAL_ALL_VOID is executed.

Answer:

```
ESC s DLH DLL 09H
ESC MFB  <Receipt.Tot[BRUTTO_SALES]>
ESC MFB1 <Receipt.Tot[VOID]>
ESC MFB1 <Receipt.Tot[DISCOUNT]>
ESC MFB1 <Receipt.Tot[SURCHARGE]>
ESC MFB1 <Receipt.Tot[RETURN]>
ESC MFB1 <Receipt.Tot[REST]>
ESC MFB1 <Receipt.Tot[SPECIAL VOID]>ESC MFE
```

Update: After every fiscal command within the receipt. Set to zero after RECEIPT_END command.

7.15.3 GET_RECEIPT_BRUTTO_SALES

Function: This command requests the receipts brutto sales amounts per VAT category.

Code: **ESC MFB k b ESC MFE ESC + 09H**

Condition: The totalizers are available while the receipt is open (MFC_RECEIPT_OPEN = 1).
The totalizers will be cleared after either RECEIPT_END or SPECIAL_ALL_VOID is executed.

Answer:

```
ESC s DLH DLL 09H
ESC MFB  <Receipt.BruttoSale[SALES_KA]>
ESC MFB1 <Receipt.BruttoSale[SALES_KB]>
ESC MFB1 <Receipt.BruttoSale[SALES_KC]>
ESC MFB1 <Receipt.BruttoSale[SALES_KD]>
ESC MFB1 <Receipt.BruttoSale[SALES_KE]>
ESC MFB1 <Receipt.BruttoSale[SALES_KF]>
```

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```
ESC MFB1 <Receipt.BruttoSale[SALES_KG]>
ESC MFB1 <Receipt.BruttoSale[SALES_KH]> ESC MFE
```

Note: SALES_KF (RoadTax) = fix 0

Update: After every fiscal command within the receipt. Set to zero after RECEIPT_END command.

7.15.4 GET_RECEIPT_VAT_AMOUNT

Function: This command requests the amounts of all VAT totalizers.

Code: **ESC MFB k u ESC MFE ESC + 09H**

Answer:

```
ESC s DLH DLL 09H
ESC MFB  <Receipt.Vat[SALES_KA]>
ESC MFB1 <Receipt.Vat[SALES_KB]>
ESC MFB1 <Receipt.Vat[SALES_KC]>
ESC MFB1 <Receipt.Vat[SALES_KD]>
ESC MFB1 <Receipt.Vat[SALES_KE]>
ESC MFB1 <Receipt.Vat[SALES_KF]>
ESC MFB1 <Receipt.Vat[SALES_KG]>
ESC MFB1 <Receipt.Vat[SALES_KH]> ESC MFE
```

Note: This command do not make sense before the TOTAL_LINE command has been processed by the MFC. Received before the receipt has been totalled the MFC will send an answer will all totalizers set to zero.

SALES_KE (VAT_E fix 0%) = 0, SALES_KF = Road tax

Update: After receipt is totalled. Set to zero after RECEIPT_END command.

7.15.5 GET_RECEIPT_PAYMENT

Function: This command requests the payment amounts for the current receipt.

Code: **ESC MFB k z ESC MFE ESC + 09H**

Condition: The totalizers are available while the receipt is open (MFC_RECEIPT_OPEN = 1).
The totalizers will be cleared after either RECEIPT_END or SPECIAL_ALL_VOID is executed.

Answer:

```
ESC s DLH DLL 09H
ESC MFB  <Receipt.Payment[PAYMENT_TYPE_0]>
ESC MFB1 <Receipt.Payment[PAYMENT_TYPE_1]>
ESC MFB1 <Receipt.Payment[PAYMENT_TYPE_2]>
.
.
ESC MFB1 <Receipt.Payment[PAYMENT_TYPE_19]>
ESC MFE
```

Update: After every fiscal command within the receipt. Set to zero after RECEIPT_END command.

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7.15.6 GET_RECEIPT_STATISTIC (in preparation)

Function: This command requests the statistic amounts e.g. CIF amounts.

Code: **ESC MFB w t ESC MFE ESC + 09H**

Condition: The totalizers are available while the receipt is open (MFC_RECEIPT_OPEN = 1).
The totalizers will be cleared after either RECEIPT_END or SPECIAL_ALL_VOID is executed.

Answer:
```
ESC s DLH DLL 09H
ESC MFB  <Receipt.Statistic[CIF]>
ESC MFB1 <Receipt.Statistic[CIF_VAT]>
ESC MFE
```

Update: After every fiscal command within the receipt. Set to zero after RECEIPT_END command.

7.15.7 GET_DAY_TOTALIZER

Function: This command requests the day totalizers.

Code: **ESC MFB k c ESC MFE ESC + 09H**

Condition: The totalizers are available while the day is open (MFC_DAY_OPEN = 1).
The totalizers are updated after every RECEIPT_END command and cleared after executing DAY_END.

Answer:
```
ESC s DLH DLL 09H
ESC MFB  <Day.Tot[BRUTTO_SALES]>
ESC MFB1 <Day.Tot[VOID]>
ESC MFB1 <Day.Tot[DISCOUNT]>
ESC MFB1 <Day.Tot[SURCHARGE]>
ESC MFB1 <Day.Tot[RETURN]>
ESC MFB1 <Day.Tot[REST]>
ESC MFB1 <Day.Tot[SPECIAL VOID]> ESC MFE
```

Update: After every receipt end command or after synchronisation error caused by wrong ticket total. Set to zero after DAY_END command.

7.15.8 GET_DAY_BRUTTO_SALES

Function: This command requests all brutto totalizers of the actual day.

Code: **ESC MFB k p ESC MFE ESC + 09H**

Answer:
```
ESC s DLH DLL 09H
ESC MFB  <Day.BruttoSale[SALES_KA]>
ESC MFB1 <Day.BruttoSale[SALES_KB]>
ESC MFB1 <Day.BruttoSale[SALES_KC]>
ESC MFB1 <Day.BruttoSale[SALES_KD]>
ESC MFB1 <Day.BruttoSale[SALES_KE]>
ESC MFB1 <Day.BruttoSale[SALES_KF]>
ESC MFB1 <Day.BruttoSale[SALES_KG]>
ESC MFB1 <Day.BruttoSale[SALES_KH]> ESC MFE
```

Note: All totalizers are updated after every RECEIPT_END command.

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Update: After every receipt end command or after synchronisation error caused by wrong ticket total. Set to zero after DAY_END command.

7.15.9 GET_DAY_VAT_AMOUNT

Function: This command requests the day VAT amount per VAT category

Code: **ESC MFB k d ESC MFE ESC + 09H**

Condition: The totalizers are available while the day is open (MFC_DAY_OPEN = 1). The totalizers are updated after every RECEIPT_END command and cleared after executing DAY_END.

Answer:

```
ESC s DLH DLL 09H
ESC MFB  <Day.Vat[SALES_KA]>
ESC MFB1 <Day.Vat[SALES_KB]>
ESC MFB1 <Day.Vat[SALES_KC]>
ESC MFB1 <Day.Vat[SALES_KD]>
ESC MFB1 <Day.Vat[SALES_KE]>
ESC MFB1 <Day.Vat[SALES_KF]>
ESC MFB1 <Day.Vat[SALES_KG]>
ESC MFB1 <Day.Vat[SALES_KH]> ESC MFE
```

Note: SALES_KE (VAT_E fix 0%) = 0, SALES_KF = Road tax

Update: After every receipt end command or after synchronisation error caused by wrong ticket total. Set to zero after DAY_END command.

7.15.10 GET_DAY_PAYMENT

Function: This command requests the payment amounts per day.

Code: **ESC MFB k y ESC MFE ESC + 09H**

Condition: The totalizers are available while the day is open (MFC_DAY_OPEN = 1). The totalizers are updated after every RECEIPT_END command and cleared after executing DAY_END.

Answer:

```
ESC s DLH DLL 09H
ESC MFB  <CASH>
ESC MFB1 <Day.Payment[PAYMENT_TYPE_0]>
ESC MFB1 <Day.Payment[PAYMENT_TYPE_2]>
ESC MFB1 <Day.Payment[PAYMENT_TYPE_3]>
.
.
ESC MFB1 <Day.Payment[PAYMENT_TYPE_19]>
ESC MFE
```

Update: After every receipt end command or after synchronisation error caused by wrong ticket total. Set to zero after DAY_END command.

7.15.11 GET_DAY_STATISTIC

Function: This command requests the statistic amounts e.g. CIF amounts per day.

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Code: **ESC MFB w u ESC MFE ESC + 09H**

Condition: The totalizers are available while the day is open (MFC_DAY_OPEN = 1). The totalizers are updated after every RECEIPT_END command and cleared after executing DAY_END.

Answer:
```
ESC s DLH DLL 09H
ESC MFB  <Day.Statistic[CIF]>
ESC MFB1 <Day.Statistic[CIF_VAT]>
ESC MFE
```

Update: After every receipt end command or after synchronisation error caused by wrong ticket total. Set to zero after DAY_END command.

7.15.12 GET_DAY_DRAWER_CONTENT

Function: This command requests the amount of cash drawer balance types per day.

Code: **ESC MFB r m ESC MFE ESC + 09H**

Condition: The totalizers are available while the day is open (MFC_DAY_OPEN = 1). The totalizers are updated after every RECEIPT_END (CASH_IN/_OUT_RECEIPT) command and cleared after executing DAY_END.

Answer:
```
ESC s DLH DLL 09H
ESC MFB  <Day.CashDrawBalance[CASH_IN_DRAW]>
ESC MFB1 <Day.CashDrawBalance[INIT_FLOAT]>
ESC MFB1 <Day.CashDrawBalance[EXTRA_FLOAT]>
ESC MFB1 <Day.CashDrawBalance[PICK_UP]>
ESC MFB1 <Day.CashDrawBalance[PAID_IN]>
ESC MFB1 <Day.CashDrawBalance[PAID_OUT]> ESC MFE
```

Update: After every receipt end command or after synchronisation error caused by wrong ticket total. Set to zero after DAY_END command.

7.15.13 GET_DAY_RECEIPT_TYPE_CNT

Function: This command requests all receipt type counters.

Code: **ESC MFB w j ESC MFE ESC + 09H**

Condition: The counters are available while the day is open (MFC_DAY_OPEN = 1). The totalizers are updated after every RECEIPT_END command and cleared after executing DAY_END.

Answer:
```
ESC s DLH DLL 09H
ESC MFB  <Day.ReceiptTypeCnt[RETAIL_SALES_RECEIPT]>
ESC MFB1 <Day.ReceiptTypeCnt[GENERIC_RECEIPT]>
ESC MFB1 <Day.ReceiptTypeCnt[CASH_IN_RECEIPT]>
ESC MFB1 <Day.ReceiptTypeCnt[CASH_OUT_RECEIPT]>
ESC MFB1 <Day.ReceiptTypeCnt[FISCAL_REPORT]>
ESC MFB1 <Day.ReceiptTypeCnt[DAY_END_REPORT]>
ESC MFB1 <Day.ReceiptTypeCnt[DAY_X_REPORT]>
ESC MFB1 <Day.ReceiptTypeCnt[INFO_TICKET]>
ESC MFB1 <Day.ReceiptTypeCnt[REPRINT]>
ESC MFB1
<Day.ReceiptTypeCnt[RETAIL_SALES_RECEIPT_WITH_TAX_ID]>
ESC MFB1 <Day.ReceiptTypeCnt[PAYMENT_CORRECTION]>
```

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```
ESC MFB1 <Day.ReceiptTypeCnt[TAX_TERMINAL]>
ESC MFE
```

7.15.14 GET_DAY_RECEIPT_GROUP_CNT

Function: This command requests all receipt group counters.

Code: **ESC MFB w k ESC MFE ESC + 09H**

Condition: The counters are available while the day is open (MFC_DAY_OPEN = 1). The totalizers are updated after every RECEIPT_END command and cleared after executing DAY_END.

Answer:

```
ESC s DLH DLL 09H
ESC MFB <Day.ReceiptGroupCnt[LEGAL_RECEIPTS]>
ESC MFB1<Day.ReceiptGroupCnt[ILLEGAL_RECEIPTS]>
ESC MFB1<Day.ReceiptGroupCnt[EJ_TICKETS]>
ESC MFB1<Day.ReceiptGroupCnt[INFO_RECEIPTS]>

ESC MFE
```

7.15.15 GET_DAY_EVENT_CNT

Function: This command requests all fiscal event counters.

Code: **ESC MFB w l ESC MFE ESC + 09H**

Condition: The counters are available while the day is open (MFC_DAY_OPEN = 1). The totalizers are updated after every RECEIPT_END command and cleared after executing DAY_END.

Answer:

```
ESC s DLH DLL 09H
ESC MFB  <DISCOUNT>
ESC MFB1 <SURCHARGE>
ESC MFB1 <RETURN>
ESC MFB1 <VOID>
ESC MFB1 <SPECIAL_VOID>
ESC MFB1 <CIF>
ESC MFB1 <CASH_IN_INIT_FLOAT>
ESC MFB1 <CASH_IN_EXTRA_FLOAT>
ESC MFB1 <CASH_IN_PAID_IN>
ESC MFB1 <CASH_OUT_PICK_UP>
ESC MFB1 <CASH_OUT_PAID_OUT> ESC MFE
```

7.15.16 GET_GRAND_BRUTTO_SALES

Function: This command requests the grand totalizers per VAT category.

Code: **ESC MFB k v ESC MFE ESC + 09H**

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Answer:

```
ESC s DLH DLL 09H
ESC MFB  <stGrand.aullBruttoSale[SALES_KA]>
ESC MFB1 <stGrand.aullBruttoSale[SALES_KB]>
ESC MFB1 <stGrand.aullBruttoSale[SALES_KC]>
ESC MFB1 <stGrand.aullBruttoSale[SALES_KD]>
ESC MFB1 <stGrand.aullBruttoSale[SALES_KE]>
ESC MFB1 <stGrand.aullBruttoSale[SALES_KF]>
ESC MFB1 <stGrand.aullBruttoSale[SALES_KG]>
ESC MFB1 <stGrand.aullBruttoSale[SALES_KH]> ESC MFE
```

Note: This command is available at all time but is only updated after DAY_END command.

Update: After DAY_END command.

7.15.17 GET_GRAND_VAT_AMOUNT

Function: This command requests the VAT grand totals per VAT category.

Code: **ESC MFB k g ESC MFE ESC + 09H**

Answer:

```
ESC s DLH DLL 09H
ESC MFB  <stGrand.aullVat[VAT_KA]>
ESC MFB1 <stGrand.aullVat[VAT_KB]>
ESC MFB1 <stGrand.aullVat[VAT_KC]>
ESC MFB1 <stGrand.aullVat[VAT_KD]>
ESC MFB1 <stGrand.aullVat[VAT_KE]>
ESC MFB1 <stGrand.aullVat[VAT_KF]>
ESC MFB1 <stGrand.aullVat[VAT_KG]>
ESC MFB1 <stGrand.aullVat[VAT_KH]> ESC MFE
```

Update: After DAY_END command.

Note: SALES_KE (VAT_E fix 0%) = 0, SALES_KF = Road tax+ARTICLE_TAX sales

7.15.18 GET_GRAND_FISCAL_CNT

Function: This command request the grand fiscal counters.

Code: **ESC MFB k i ESC MFE ESC + 09H**

Answer:

```
ESC s DLH DLL 09H
ESC MFB  <stGrand.ulGrandFiscalReceiptCnt> ESC MFE
```

Update: At the beginning of each RECEIPT_BEGIN with type=RETAIL_SALES_RECEIPT| RETAIL_SALES_RECEIPT_WITH_TAX_ID. If the receipt is subsequently cancelled, the counter is reduced again.

7.15.19 GET_MFMEM_CNT

Function: This command requests the MF memory statistic counters.

Code: **ESC MFB k j ESC MFE ESC + 09H**

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Answer:

```
ESC s DLH DLL 09H
ESC MFB  <SNO block counter>
ESC MFB1 <EOD block counter>
ESC MFB1 <TKD block counter>
ESC MFB1 <EJ CHANGE block counter>
ESC MFB1 <TH block counter>
ESC MFB1 <VAT block counter>
ESC MFB1 <CRY block counter>
ESC MFB1 <TT block counter>
ESC MFB1 <FW block counter>
ESC MFE
```

Update: After burning one block into MFMEM.

7.15.20 GET_MFMEM_LAST_BL_DATE

Function: This command requests the date and time for each block type last written to the fiscal memory.

Code: **ESC MFB r z ESC MFE ESC + 09H**

Answer:

```
ESC s DLH DLL 09H
ESC MFB  <Date of last SNO block>
ESC MFB1 <Date of last EOD block>
ESC MFB1 <Date of last TKD block>
ESC MFB1 <Date of last EJ_CHANGE block>
ESC MFB1 <Date of last TH block>
ESC MFB1 <Date of last VAT block>
ESC MFB1 <Date of last CRY block>
ESC MFB1 <Date of last TT block>
ESC MFB1 <Date of last FW block>
ESC MFE
```

Code old: ESC MFB r a ESC MFE ESC +09H

old date format is DD/MM/YY.

Note: Date format is DD.MM.YY hh:mm:ss.

If a certain block type is not present in the MF memory the system returns blanks but keeps the separators (“ . . : : “).

It is not recommended to use the old command syntax for new implementations. This syntax is obsolete and may be not supported in the future.

Update: After burning one block into MFMEM.

7.15.21 GET_MFMEM_LAST_BL_NUM

Function: This command requests the first **physical block number** for each logical block type last written to the MF memory.

Code: **ESC MFB k m ESC MFE ESC + 09H**

Answer:

```
ESC s DLH DLL 09H
ESC MFB  <Last SNO block number>
ESC MFB1 <Last EOD block number>
```

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```
ESC MFB1 <Last TKD  block number>
ESC MFB1 <Last EJ_CHANGE block number>
ESC MFB1 <Last TH block number>
ESC MFB1 <Last VAT block number>
ESC MFB1 <Last CRY block number>
ESC MFB1 <Last TT block number>
ESC MFB1 <Last FW block number>
ESC MFE
```

Note: If a certain block type is not present in the MF memory, its address will be given as 65535.

Update: After burning one block into MFMEM.

7.15.22 GET_MFMEM_LAST_LOGICAL_BL_NUM

old name: **GET_MFMEM_LAST_BL_ADDRESS**

Function: This command requests the **logical** block number for each block type last written to the fiscal memory.

Code: **ESC MFB r d ESC MFE ESC + 09H**

Answer:

```
ESC s DLH DLL 09H
ESC MFB  <Last SNO block number>
ESC MFB1 <Last EOD block number>
ESC MFB1 <Last TKD block number>
ESC MFB1 <Last EJ_CHANGE block number>
ESC MFB1 <Last TH block number>
ESC MFB1 <Last VAT block number>
ESC MFB1 <Last CRY block number>
ESC MFB1 <Last TT block number>
ESC MFB1 <Last FW block number>
ESC MFE
```

Update: After burning one block into MFMEM.

Note: If one of these blocks is not present inside the MFMEM, 65535 will be returned instead.

7.15.23 GET_MFMEM_BL_NUM

Function: This command translates the number of a block type to the logical and physical BlockNumber within the fiscal memory as described in 3.4 Block handling (logical and physical blocks).

Code:
ESC MFB w v
ESC MFB1 d a *<BlockTypeNo>*
ESC MFB1 f d *<BlockType >*
ESC MFE

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Parameter	Value	Format
<BlockTypeNo>	1 – max. block no. (5.7.10)	ASCII, 30 H ≤ *n* ≤ 39 H, max. 5 bytes
<BlockType>	30 H ≤ n ≤ 39 H	see 5.7.1 for valid types

Note: This command is mainly needed for testing.

Answer:

```
ESC s DLH DLL 09H
ESC MFB  <logical block number>
ESC MFB1 <physical block number>
ESC MFE
```

7.15.24 GET_MFMEM_BL_HEADER

Function: This command reads general information of a logical block in the fiscal memory as described in 3.4 Block handling (logical and physical blocks).

Code: **ESC MFB w w**
ESC MFB1 d a *<BlockNo>*
ESC MFE

Parameter	Value	Format
<BlockNo>	1 – max. block no. (5.7.10)	ASCII, 30 H ≤ *n* ≤ 39 H, max. 5 bytes

Note: This command is mainly needed for testing.

Answer:

```
ESC s DLH DLL 09H
ESC MFB  <BlockStatus>
ESC MFB1 <DateAndTime>
ESC MFB1 <logicalBlockNumber>
ESC MFB1 <BlockType>
ESC MFB1 <BlockTypeNumber>
ESC MFB1 <PreviousBlockFromType>
ESC MFE
```

Parameter	Meaning	Format
`<BlockStatus>`		1 bytes in ASCII
`<DateAndTime>`		DD.MM.YYYY hh:mm:ss, 19 bytes in ASCII including separators
`<logicalBlock Number>`		
`<BlockType>`		
`<BlockTypeNum ber>`		

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<PreviousBlockFromType>		

7.15.25 GET_DATE_TIME

Function: With this command the application receives the date and time from MFC real-time clock.

Code: **ESC MFB w b ESC MFE ESC +09H**

Answer: `ESC s DLH DLL 09H ESC MFB <DateTime> ESC MFE`
format <DateTime> : “DD.MM.YYYY hh:mm:ss”

Code old: ESC MFB k k ESC MFE ESC +09H

Answer: `ESC s DLH DLL 09H`

`ESC MFB <hh:mm:ss> ; <DD/MM/YY> ESC MFE`

Note: DD = Day, MM = Month, YYYY = Year, hh = hours, mm = minutes, ss = seconds

If a hardware error has occurred (RTC error, battery error), the system returns blanks but keeps its separators (“ . . : : “).

It is not recommended to use the old command syntax for new implementations. This syntax is obsolete and may be not supported in the future.

7.15.26 GET_VAT_RATES

Function: This command requests the VAT rates from the system.

Code: **ESC MFB k l ESC MFE ESC + 09H**

Answer:
```
ESC s DLH DLL 09H
ESC MFB  <VAT Rate A>
ESC MFB1 <VAT Rate B>
ESC MFB1 <VAT Rate C>
ESC MFB1 <VAT Rate D>
ESC MFB1 <VAT Rate E>
ESC MFB1 <VAT Rate F>
ESC MFB1 <VAT Rate G>
ESC MFB1 <VAT Rate H> ESC MFE
```

Note: VAT Rate E and F fix 0,00%,

VAT rates sent by the MFC have 2 implicit decimals (range 0...99,99%).

Example: VAT Rate "A" = 0%, VAT Rate "B" = 20%, VAT Rate "C" = 20.00%, VAT Rate "D" = 07,00%, VAT rate E..H unused:

ESC s 00h 17h 09h ESC MFB 0 ESC MFB1 2000 ESC MFB1 2000 ESC MFB1 0700 ESC MFB1 0000 ESC MFB1 0000 ESC MFB1 9999 ESC MFB1 9999 ESC MFE

Update: After burning one VAT block into MFMEM.

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7.15.27 GET_CURRENCY_COUNTRY

Function: This command requests the currency label of the MFC controller.

Code: **ESC MFB k o ESC MFE ESC + 09H**

Answer:
```
ESC s DLH DLL 09H
ESC MFB <CurrencyLabel> ESC MFE
```

Note: `<CurrencyLabel>` = label given by SET_CURRENCY_COUNTRY (7.4.4)

7.15.28 GET_PAYMENT_LIST

Function: This command requests the content of one entry in the payment list.

Code: **ESC MFB r w**
ESC MFB1 g a *<PaymentType>*
ESC MFE
ESC + 09H

Parameter	Value	Format
<PaymentType> Index of the entry in the payment list	30 H ≤ *n* ≤ 39 H	max. 2 bytes 0…19

Answer:
```
ESC s DLH DLL 09H
ESC MFB  <EntryValid>
ESC MFB1 <PaymentName>
ESC MFB1 <PaymentCategory> ESC MFE
```

Parameter	Value	Format
<EntryValid>	30 H, 31 H	1 byte 0 – not valid 1 – valid
<PaymentName>	see 7.4.5	see 7.4.5
<PaymentCategory>	see 7.4.5	see 7.4.5

Update: The commands SET_PAYMENT_LIST and SET_PAYMENT_LIST_VOID change all parameters of one payment list entry after successful command execution.

7.15.29 GET_ERROR_LIST

Function: This command requests the error list with the last errors. At most 10 error entries including date and time information are returned. If date/time information is not available the particular entry fields are filled up with blanks (20h).

Code: **ESC MFB k x ESC MFE ESC + 09H**

Answer:
```
ESC s DLH DLL 09H
ESC MFB  <Error Message 1><DateTime of Entry 1>
```

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```
ESC MFB1 <Error Message 2><DateTime of Entry 2>
ESC MFB1 …
ESC MFE
```

Parameter	Meaning	Format
<ErrorMessage>	Fiscal error (ref.8.1 Error messages)	2 bytes in ASCII
<DateTimeOfEntry>	Error happened at specific date	DD.MM.YYYY hh:mm:ss, 19 bytes in ASCII including separators

Update: The internal list of the fiscal controller is updated with every new error entry in the list. That means that always the latest list is available when this command is applied.

7.15.30 GET_FIRMWARE_VERSION

Function: Sends back the firmware versions.

Code: **ESC MFB w c ESC MFE ESC + 09h**

Old Code: *ESC MFB r f ESC MFE ESC + 09h*

Answer:

```
ESC s DLH DLL 09H
ESC MFB <FFCFirmwareVersion>
ESC MFB1 <TTFirmwareVersion>
ESC MFE
```

Parameter	Meaning	Format
<FFCFirmwareVersion>	FFC Firmware version	ASCII: “MM-CC-SS” MM = main version CC = country SS = subversion
<TTFirmwareVersion>	Tax Terminal Firmware version	max. 10 characters

7.15.31 GET_TKD_INFO

Function: This command reads out the blockade error time and date.

Code: **ESC MFB r g ESC MFE ESC + 09h**

Answer:

```
ESC s DLH DLL 09H
ESC MFB <BlockadeError>
ESC MFB1 <BlockadeDateTime>
ESC MFE
```

Parameter	Meaning	Format

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<BlockadeError>	Fiscal error (see chapter 8.1 Error messages)	2 bytes in ASCII
<BlockadeDateTime>	Date and time occurs to blockade	ASCII "DD.MM.YYYY hh:mm:ss" DD = Day, MM = Month, YYYY = Year, hh = hours, mm = minutes, ss = seconds

Note: If no blockade pending, the system returns blanks but keeps its separators
("ESC MFB 00 ESC MFB1 . . : : ESC MFE").

7.15.32 GET_HEADER

Function: This command requests the current receipt header.

Code: **ESC MFB r y ESC MFE ESC + 09H**

Answer:

```
ESC s DLH DLL 09H
ESC MFB  <ReceiptHeaderLine1>
ESC MFB1 <ReceiptHeaderLine2>
ESC MFB1 <ReceiptHeaderLine3>
ESC MFB1 <ReceiptHeaderLine4>
ESC MFB1 <ReceiptHeaderLine5>
ESC MFB1 <CIF_ID> ESC MFE
```

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Parameter	Value	Format	
<ReceiptHeaderLine1> □ □ □ *<ReceiptHeaderLine4>*	20h* ≤ *n* ≤ FFh	Maximal numbers of printable characters per line: 44 Maximal numbers of printable characters and printer commands per line: 79 * See Application hint: for further details	
<ReceiptHeaderLine5>	20h* ≤ *n* ≤ FFh	Maximal numbers of printable characters per line: 25 No printer commands allowed	**D**
<CIF_ID>	20 H ≤ n ≤ FF H	2 bytes (alphanumeric characters) allowed range: a-z, A-Z, 0-9, Space(0x20)	**I**

7.15.33 GET_SNO_NUMBER

Function: Sends back the serial number defined by INIT_WRITE_SNO.

Code: **ESC MFB w a ESC MFE ESC + 09h**

Answer:

```
ESC s DLH DLL 09H
ESC MFB  <FiscalId>
ESC MFB1 <CIF>
ESC MFB1 <PrinterSerialNumber> ESC MFE
```

Parameter	Value	Format
<FiscalId>	30 H ≤ *n* ≤ 39 H	10 bytes (digits)
<CIF>	30 H ≤ *n* ≤ 39 H	13 bytes (digits)
<PrinterSerialNumber>	20 H ≤ *n* ≤ FF H	Up to 20 bytes (alphanumeric characters)

7.15.34 GET_DATE_FIRST_RECEIPT

Function: With this command the application retrieves the date and time of the first RECEIPT_BEGIN command after DAY_OPEN from the MF controller.

Code: **ESC MFB r b ESC MFE ESC + 09H**

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Answer: `ESC s DLH DLL 09H ESC MFB <DateTime> ESC MFE`
format <DateTime> : “DD.MM.YYYY hh:mm:ss”

Note: DD = Day, MM = Month, YYYY = Year, hh = hours, mm = minutes, ss = seconds
The <*DateTime*> was set by receipt type RETAIL_SALES_RECEIPT or RETAIL_SALES_RECEIPT_WITH_TAX_ID only.

If the fiscal day is not open and before the first RECEIPT_BEGIN the return value is "DD.MM.YYYY hh:mm:ss“. In all other cases the date and time of the first RECEIPT_BEGIN command is returned.

Update: After the very first RECEIPT_BEGIN command (store date and time) and after a successful DAY_END command (delete date and time).

7.15.35 GET_DEVICE_CONNECTED

Function: This command checks whether a device is connected on the internal interfaces.

Code: **ESC MFB w e ESC MFB1 d c <*GetIndex*> ESC MFE**

Parameter	Value	Format
<***GetIndex***>	30 H ≤ *n* ≤ 32 H	„0“ – printer (in general TH230) „1“ – customer display (e.g. BA63) „2“ – tax modul

Answer: `ESC s DLH DLL 09H ESC MFB <DeviceStatus> ESC MFE`

Parameter	Meaning	Format
<***DeviceStatus***>	Is the device connected 0 – device is not connected 1 – device is connected	1 Bytes, “0”/”1” (ASCII)

7.15.36 GET_WORKING_MODE

Function: With this command the application retrieves the Working Mode.

Code: **ESC MFB w r ESC MFE ESC + 09H**

Answer: `ESC s DLH DLL 09H ESC MFB <Mode> ESC MFE`

7.15.37 GET_FA_STATUS

Function: With this command the application retrieves the fiscal authority status.

Code: **ESC MFB w g ESC MFE ESC + 09H**

Answer:
```
ESC s DLH DLL 09H
ESC MFB  <ConnectionState>
```

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```
ESC MFB1 <DaysUntilBlock>
ESC MFB1 <ServiceIntervention>
ESC MFE
```

Parameter	Value	Format
<ConnectionState>	30 H ≤ *n* ≤ 39 H	Tax terminal connection status: “0” – online regular (connection OK) “1” – online eclipse mode (connection problems) “2” – offline regular (no connection, but not blocked) “3” – offline blocked (no connection and blocked, no sales operation possible)
<DaysUntilBlock>	30 H ≤ *n* ≤ 39 H	Number of days left before device gets blocked due tax terminal is not connected to authority server, max. 4 bytes (digits)
<ServiceIntervention>	20 H ≤ *n* ≤ FF H	“0” – no action required “1” – service intervention is required

7.15.38 GET_FA_DAILY_STATUS

Function: With this command the application retrieves the daily fiscal authority status.

Code: **ESC MFB w x ESC MFE ESC + 09H**

Answer:

```
ESC s DLH DLL 09H
ESC MFB  <ActiveProfile>
ESC MFB1 <ActiveProvileDate>
ESC MFB1 <ZreportSchedule>
ESC MFB1 <MaxDaysUntilBlock>
ESC MFB1 <PingPeriodMinutes>
ESC MFB1 <ZreportPeriodMinutes>
ESC MFB1 <CertChange>
ESC MFB1 <Profile1ActivationDate>
ESC MFB1 <Profile0SwitchDateTime>
ESC MFB1 <NumUnsentFiles>
ESC MFE
```

Parameter	Value	Format
<ActiveProfile>	30 H ≤ *n* ≤ 39 H	Profile number (defined by authority) currently active, 1 byte (digit)
<ActiveProfileDate>	30 H ≤ *n* ≤ 39 H	Current profile activation date in format "DD.MM.YYYY"
<ZreportSchedule>	20 H ≤ *n* ≤ FF H	“0” – send immediately after printout “1” – send periodically as defined in <ZreportPeriodMinutes>

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<MaxDaysUntilBlock>	30 H ≤ *n* ≤ 39 H	Number of days when tax terminal gets blocked after having become offline, max. 4 bytes (digits)
<PingPeriodMinutes>	30 H ≤ *n* ≤ 39 H	Time between connection attempts to authority server in minutes, max. 6 bytes (digits)
<ZreportPeriodMinutes>	30 H ≤ *n* ≤ 39 H	Period in minutes between sending Z report data, max. 6 bytes (digits)
<CertChange>	30 H ≤ *n* ≤ 39 H	“0” – none action necessary “1” – certificate change required
<Profile1ActivationDate>	30 H ≤ *n* ≤ 39 H	Profile 1 activation date in format "DD.MM.YYYY"
<Profile0SwitchDateTime>	30 H ≤ *n* ≤ 39 H	Upcoming profile 0 switch date in format "DD.MM.YYYY HH:MM:SS"
<NumUnsentFiles>	30 H ≤ *n* ≤ 39 H	Number of files not sent yet to the authority server due to an offline situation, max. 3 bytes (digits)

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7.16 EJ_ commands

All EJ_ commands may return error messages from the set of 8.3 EJ error codes. Reprints without any ticket data within the defined range will not return an error, but print the header and footer.

All EJ_ commands return an error message, if the Tax Terminal is not enabled (paired), any Tax Terminal component is not available or the customer display is not connected.

7.16.1 EJ_INIT

Function: This command initialises a new electronic journal medium

Code: **ESC MFB q d**
[ESC MFB2 h a *<FreeSpace>*]
ESC MFE

Parameter	Value	Format
<FreeSpace> in byte	<= 8 * 1024 * 1024 * 1024	ASCII

Example: ESC MFB q d ESC MFB2 h a 515000 ESC MFE

Initializes an electronic journal medium with a free space of 515.000 byte

Condition:

MFC Status byte 1

MFC_ TKD_RESET	MFC_ RECEIPT_ TOTAL	MFC_ DOCU_OPEN	MFC_ REPORT_ OPEN	MFC_ RECEIPT_ OPEN	MFC_ DAY_OPEN	MFC_ TRAINING_ MODE	MFC_ BLOCKADE
x	**x**	**x**	**x**	**x**	**0**	**x**	**0**

MFC Status byte 2

not used	not used	EJ_DATA_ PENDING	EJ_FULL	EJ_NEAR_ FULL	MFC_ CMD_ INTERRUPTED	MFC_ PRINTOUT_ INTERRUPTED	MFC_ FISCAL_ MODE
x	**x**	**x**	**x**	**x**	**x**	**x**	**x**

Note: To test the ‘EJ nearly full’ and ‘EJ full’ conditions, the remaining EJ space can be specified.

One EJ Change Block (EJ change) is created and stored into the MFMEM. Please refer to chapter 5.7.5 EJ change block for the EJ change block format

Error Msg: EM_PARA_WRONG
EM_PARA_RANGE_ERROR
EM_MFMEM_BL_LIMIT_OVERRUN
EM_EJ_UNSUPPORTED_MEDIUM
EM_EJ_NOT_EMPTY

Error messages from the set of 8.2 FM error codes

Error messages from the set of 8.4 Supplementary checks error codes

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Printout: Administrative receipt with optional header lines:

```
******  W I N C O R - N I X D O R F      *******
        receipt header line 2
        receipt header line 3
        receipt header line 4
    Cod Fiscal C.I.F.:1234567890 1234

****   BON NEFISCAL   ****   BON NEFISCAL    ****

                E J _ I N I T

    EJ Code:    001
    EJ UMID:    SXXXXXXXXX0320C43E4

**  NUMĂR DE MODIFICARE JURNAL ELECTRONIC  **
**  CARE SUNT INCA PERMISE: 199            **

        DATA: 16/05/2015     ORA:17-57-56
              RL 1234567890
****   BON NEFISCAL   ****   BON NEFISCAL    ****
```

7.16.2 EJ_REPRINT_T2T_BY_SESSION

Function: The application prints a non-fiscal report (illegal receipt) over a range of EJ data specified by session and ticket info. A session is an EOD block number. This report is either printed on the receipt station or is held ready as soft copy to be fetched by the application. This command is not allowed in training mode as there is no data stored in the EJ during training mode.

Code:

ESC MFB q b
ESC MFB1 d a *<FromValue>*
ESC MFB1 d g *<FromTicket>*
ESC MFB1 d b *<ToValue>*
ESC MFB1 d h *<ToTicket>*
[ESC MFB2 f e *<TicketType>*]
[ESC MFB2 f q *<SessionType>*]
[ESC MFB2 f a *<Station>*]
ESC MFE

Parameter	Value	Format
<FromValue>	1 – max. EOD no. (5.7.10)	ASCII, 30 H ≤ *n* ≤ 39 H, max. 5 bytes
<FromTicket>	1 – 9999	ASCII, 30 H ≤ *n* ≤ 39 H, max. 4 bytes
<ToValue>	1 – max. EOD no. (5.7.10)	ASCII, 30 H ≤ *n* ≤ 39 H, max. 5 bytes
<ToTicket>	1 – 9999	ASCII, 30 H ≤ *n* ≤ 39 H, max. 4 bytes

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<TicketType>	**Bit**	**Hexadecimal ASCII, bit coded (MSB.LSB), max. 4 byte**
	15-12	Reserved
	11	0800h: PAYMENT_CORRECTION
	10	0400h: FISCAL_SALES_RECEIPT_WITH_TAX_ID
	9	0200h: reserved
	8	0100h: reserved
	7	0080h: reserved
	6	0040h: CASH_OUT_RECEIPT
	5	0020h: CASH_IN_RECEIPT
	4	0010h: reserved
	3	0008h: DAY_END_REPORT
	2	0004h: reserved
	1	0002h: GENERIC_RECEIPT
	0	0001h: FISCAL_SALES_RECEIPT
<SessionType>	30 H ≤ *n* ≤ 33 H	max. 1 byte 0..3 0 DEFAULT Mode 1 TEST Mode 2 PROBA Mode 3 INITIERE Mode
<Station>	30 H ≤ *n* ≤ 39 H	“2“ - Receipt (Default) “16” - Host – For EJ soft copy. “32” - External Storage

Example: ESC MFB q b ESC MFB1 d a 3 ESC MFB1 d g 1 ESC MFB1 d b 11 ESC MFB1 d h 4 ESC MFB1 f e 60 MFB2 f q 0 1 ESC MFE

Prints a report from session 3 ticket 1 to session 11 ticket 4 containing only the ticket types CASH_OUT_RECEIPT and CASH_IN_RECEIPT.

Condition:

MFC Status byte 1

MFC_ TKD_RESET	MFC_ RECEIPT_ TOTAL	MFC_ DOCU_OPEN	MFC_ REPORT_ OPEN	MFC_ RECEIPT_ OPEN	MFC_ DAY_OPEN	MFC_ TRAINING_ MODE	MFC_ BLOCKADE
x	**x**	**x**	**0**	**0**	**x**	**0**	**x**

MFC Status byte 2

not used	not used	EJ_DATA_ PENDING	EJ_FULL	EJ_NEAR_ FULL	MFC_ CMD_ INTERRUPTED	MFC_ PRINTOUT_ INTERRUPTED	MFC_ FISCAL_ MODE
x	**x**	**x**	**x**	**x**	**0**	**0**	**x**

Note: This non-fiscal report (illegal receipt) is not stored inside the EJ.

If the parameter <TicketType> is not given then all ticket types are taken into account.

The command rejects with EM_PARA_RANGE_ERROR, if the defined range <FromValue>.<FromTicket>..<ToValue>.<ToTicket> filtered with <TicketType> does not contain at least one ticket.

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If the parameter <SessionType> is not given then only Normal working fiscal mode Sessions are considered.

Error Msg: EM_CMD_WRONG_ORDER
EM_PARA_MISSING
EM_PARA_WRONG_FORMAT
EM_PARA_WRONG
EM_PARA_RANGE_ERROR

Printout:

```
0123456789012345678901234567890123456789012345678901234567890123
*****   W I N C O R - N I X D O R F    *****   Header Line 1
            receipt header line 2               Header Line 2
            receipt header line 3               Header Line 3
            receipt header line 4               Header Line 4
     Cod Fiscal C.I.F.: 12345678901234          Header Line 5

     RETIPĂRIRE DIN JURNALUL ELECTRONIC
          BON LA BON DUPĂ SESIUNE               Report title

De la sesiunea: xxxxx Bonul: yyyyy
   la sesiunea: xxxxx Bonul: yyyyy
                                                "TOATE" if no ticket
    Tipul bonurilor: <TicketType>               type is defined

    •
    •
    •

            DATA:03/04/2008    ORA:15-42-23
                RL 1234567890
***  BON NEFISCAL  ******  BON NEFISCAL  ***
```

7.16.3 EJ_REPRINT_T2T_BY_DATE_TIME

Function: The application prints a non-fiscal report (illegal receipt) over a range of EJ data specified by date. This report is either printed on the receipt station or is held ready as soft copy to be fetched by the application. This command is not

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allowed in training mode as there is no data stored in the EJ during training mode.

Code:

ESC MFB q j
ESC MFB1 e e<*FromDateTime*>
ESC MFB1 e f <ToDateTime>
[ESC MFB2 f e<*TicketType*>]
[ESC MFB2 f a<*Station*>]
ESC MFE

Parameter	Value	Format
<FromDateTime>	DD.MM.YYYY hh:mm:ss	ASCII
<ToDateTime>	DD.MM.YYYY hh:mm:ss	ASCII
<TicketType>	see 7.16.2	see 7.16.2
<Station>	30 H ≤ *n* ≤ 39 H	"2" - Receipt (Default) "16" - Host – For EJ soft copy "32" - External Storage

Condition:

MFC Status byte 1

MFC_ TKD_RESET	MFC_ RECEIPT_ TOTAL	MFC_ DOCU_OPEN	MFC_ REPORT_ OPEN	MFC_ RECEIPT_ OPEN	MFC_ DAY_OPEN	MFC_ TRAINING_ MODE	MFC_ BLOCKADE
x	**x**	**x**	**0**	**0**	**x**	**0**	**x**

MFC Status byte 2

not used	not used	EJ_DATA_ PENDING	EJ_FULL	EJ_NEAR_ FULL	MFC_ CMD_ INTERRUPTED	MFC_ PRINTOUT_ INTERRUPTED	MFC_ FISCAL_ MODE
x	**x**	**x**	**x**	**x**	**0**	**0**	**x**

Note:

This non-fiscal report (illegal receipt) is not stored inside the EJ.

If the parameter <TicketType> is not given then the default value FFFFh is used.

The command rejects with EM_PARA_RANGE_ERROR, if the defined range <FromDateTime>..<ToDateTime> filtered with <TicketType> does not contain at least one ticket.

Error Msg:

EM_CMD_WRONG_ORDER
EM_PARA_MISSING
EM_PARA_WRONG_FORMAT
EM_PARA_WRONG
EM_PARA_DATE_WRONG_FORMAT
EM_PARA_RANGE_ERROR

Printout:

```
012345678901234567890123456789012345678901 23
*****   W I N C O R - N I X D O R F    *****   Header Line 1
            receipt header line 2               Header Line 2
            receipt header line 3               Header Line 3
```

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receipt header line 4	*Header Line 4*
Cod Fiscal C.I.F.: 12345678901234	*Header Line 5*
RETIPĂRIRE DIN JURNALUL ELECTRONIC	
DUPĂ DATĂ ŞI ORĂ	Report title
De la: DD.MM.YYYY hh:mm:ss	
la: DD.MM.YYYY hh:mm:ss	
Tipul bonurilor: <TicketType>	*"TOATE" if no ticket type is defined*
•	
•	
•	
DATA:03/04/2008 ORA:15-42-23	
RL 1234567890	
*** BON NEFISCAL ****** BON NEFISCAL ***	

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7.17 EJ_GET commands

All EJ_GET_ commands may return error messages from the set of 8.3 EJ error codes.

7.17.1 EJ_GET_STATUS

Function: This command transmits current EJ Status

Code: **ESC MFB p s ESC MFE ESC + 09h**

Answer: `ESC s DLH DLL 09H ESC MFB <EJStatus High><EJStatus Low> ESC MFE`

*with (DLH * 255)+DLL= 7 [DLH = 0 , DLL = 7]*

Parameter	Meaning	Format
<EJStatus High>	Highbyte (B15-B8) of EJ Status	Binary
<EJStatus Low>	Lowbyte (B7-B0) of EJ Status	Binary

Please refer to chapter 4.4 Electronic journal status for details about the electronic journal status.

7.17.2 EJ_GET_SIZE

Function: This command transmits the size of the installed EJ memory card.

Code: **ESC MFB p a ESC MFE ESC + 09h**

Answer: `ESC s DLH DLL 09H ESC MFB <Size> ESC MFE`

Parameter	Meaning	Format
<Size>	Size of the EJ MMC in Byte	ASCII

7.17.3 EJ_GET_FREE_SPACE

Function: This command transmits the free memory space of the EJ.

Code: **ESC MFB p b ESC MFE ESC + 09h**

Answer: `ESC s DLH DLL 09H ESC MFB <FreeSpace> ESC MFE`

Parameter	Meaning	Format
<FreeSpace>	Remaining EJ space in Byte	ASCII

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Update: The value is updated after every write cycle into the EJ.

7.17.4 EJ_GET_INIT_INFO

Function: This command transmits the initialisation date and time, the EJ_UMID and the EJMediumNumber of the installed EJ memory card.

Code: **ESC MFB p w ESC MFE ESC + 09h**

Answer:

```
ESC s DLH DLL 09H
ESC MFB <InitDateTime>
ESC MFB1 <EJ_UMID>
ESC MFB1 <EJMediumNumber> ESC MFE
```

Parameter	Meaning	Format
<InitDateTime>	Returns the initialisation date of the EJ in the format “DD.MM.YYYY hh.mm.ss”.	ASCII
<EJ_UMID>	Returns the unique medium identifier. see chapter 4.5 „Unique medium identification (UMID)“	ASCII
<EJMediumNumber>	Ascending number of the EJ MMC of that fiscal printer.	ASCII

Note: If *<EJMediumNumber>* is an empty string, the MMC can be treated as not initialised. Also the *<InitDateTime>* and *<EJ_UMID>* fields are not filled.

7.17.5 EJ_GET_MEDIUM_INFO

Function: This command transmits the EJ_UMID of the existing EJ memory card.

Code: **ESC MFB p d ESC MFE ESC + 09h**

Answer:

```
ESC s DLH DLL 09H
ESC MFB < EJ_UMID >
ESC MFB1 <EJMediumNumber> ESC MFE
```

Parameter	Meaning	Format
<EJ_UMID>	Returns the unique medium identifier. see chapter 4.5 „Unique medium identification (UMID)“	ASCII
<EJMediumNumber>	Ascending number of MMC	ASCII

Note: If *<EJMediumNumber>* is an empty string, the EJ Medium is not initialised by the fiscal firmware.

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7.17.6 EJ_GET_LAST_MEDIUM_INFO

Function: This command transmits the EJ_UMID and the EJMediumNumber of the last installed EJ memory card. This information is retrieved out of the fiscal memory.

Code: **ESC MFB p f ESC MFE ESC + 09h**

Answer:

```
ESC s DLH DLL 09H
ESC MFB < EJ_UMID >
ESC MFB1 <EJMediumNumber> ESC MFE
```

Parameter	Meaning	Format
<EJ_UMID>	Returns the unique medium identifier. see chapter 4.5 „Unique medium identification (UMID)“	ASCII
<EJMediumNumber>	Ascending number of MMC	ASCII

Note: If *<EJ_UMID>* and/or *<EJMediumNumber>* is an empty string, no EJ change block (see chapter 5.7.5 EJ change block) is found in the fiscal memory.

7.17.7 EJ_GET_EOD_ENTRY

Function: This command transmits EJ information about one selected session / EOD block.

Code: **ESC MFB p e ESC MFB1 d a <*FromValue*> ESC MFE ESC + 09h**

Parameter	Value	Format
<FromValue>, i. e. EOD block number	1 – EOD BlockCount (5.7.10 Block restriction)	ASCII, 30 H ≤ *n* ≤ 39 H, max. 5 bytes

Answer:

```
ESC s DLH DLL 09H
ESC MFB <SerialNumber>
ESC MFB1 <EODOrderNumber>
ESC MFB1 <CreationDateTime>
ESC MFB1 <FileAttribute>
ESC MFB1 <StartingClusterNumber>
ESC MFB1 <FileSize>
ESC MFB1 <FileExtension>
ESC MFE
```

Parameter	Meaning	Format
<SerialNumber>	*<DistrictID><AutorizationID><Fiscal SerialNumber>* see chapter 7.3.4 INIT_WRITE_SNO	ASCII
<EODOrderNumber>	EOD order number corresponding to the EJ file	ASCII

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<CreationDateTime>	EJ file creation date “DD.MM.YYYY hh:mm:ss”)	ASCII
<FileAttribute>	EJ file attribute (0-255)	ASCII
<StartingClusterNumber>	EJ file first cluster number	ASCII
<FileSize>	EJ file size in bytes	ASCII
<FileExtension>	EJ file name extension ‘EJ ‘	ASCII

Error Msg: EM_PARA_MISSING
EM_PARA_WRONG
EM_PARA_WRONG_FORMAT

7.17.8 EJ_GET_NUMBER_OF_EOD

Function: This command transmits the number of stored sessions (including open session) on the installed EJ memory card.

Code: **ESC MFB p g ESC MFE ESC + 09h**

Answer: `ESC s DLH DLL 09H ESC MFB <NumOfSessions> ESC MFE`

Parameter	Meaning	Format
<NumOfSessions>	Number of sessions of the EJ MMC	ASCII

7.17.9 EJ_GET_EOD_RANGE

Function: This command is used to get the first *<FirstSession>* and the last *<CurrentSession>* EOD session number found in the EJ file names in the SD card directory. The last EOD session number corresponds to the current EJ. After the EJMEM formatting, *<FirstSession>* and the *<CurrentSession>* are equal.

Code: **ESC MFB p h ESC MFE ESC + 09h**

Answer:
```
ESC s DLH DLL 09H
ESC MFB <FirstSession>
ESC MFB1 <LastSession> ESC MFE
```

Parameter	Meaning	Format
<FirstSession>	Number of the first EOD stored on the EJ	ASCII
<LastSession>	Number of the last opened session on the EJ	ASCII

7.17.10 EJ_GET_DATE_RANGE

Function: This command transmits the first EOD session end date and the last EOD session end date stored on the installed EJ memory card.

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Code: **ESC MFB p i ESC MFE ESC + 09h**

Answer:

```
ESC s DLH DLL 09H
ESC MFB <FirstSessionDate>
ESC MFB1 <LastSessionDate>
ESC MFE
```

Parameter	Meaning	Format
<FirstSessionDate>	First EOD date stored on the EJ, 0 in case of no sessions stored	ASCII DD.MM.YYYY hh:mm:ss
<LastSessionDate>	Last EOD date stored on the EJ, 0 in case of no sessions stored	ASCII DD.MM.YYYY hh:mm:ss

7.17.11 EJ_GET_DATA

Function: This command reads a specified EJMEM file data.

Code: **ESC MFB p k ESC MFB1 d a <*FromValue*> ESC MFB1 d i <*SectorNumber*> ESC MFE ESC + 09H**

Parameter	Meaning	Value	Format
<FromValue>	EOD order number as ASCII decimal number starting with 1.	1 – EOD BlockCount (5.7.10 Block restriction)	ASCII, 30 H ≤ *n* ≤ 39 H, max. 5 bytes
<SectorNumber>	File read sector offset as ASCII decimal number starting with 0.	30 H ≤ *n* ≤ 39 H	Max 10 bytes

Answer:

```
ESC s DLH DLL 09H
ESC MFB <SectorNumber>
ESC MFB1 <FileData>
ESC MFE
```

Parameter	Meaning	Format
< SectorNumber>	Sector number starting with 0	ASCII
< FileData >	*d1...dn*: EJ file data (max 512 char)	ASCII

Note: The Command needs to be repeated as long as the <FileData>string length is 512 Bytes.
If the <*FileData*>string length is less than 512, then this string is the end of the file.
If <*FileData*>is an empty string, then the previous sector content is the last string in the file.

Example: **ESC MFB p k ESC MFB1 d a 4 4 ESC MFB1 d i 0 ESC MFE ESC + 09H**
This command returns the sector #0 of the EJ file 'NNNN0044.EJ' where NNNN is the last 4 digit of the MFMEM serial number.
ESC MFB p k ESC MFB1 d a 4 4 ESC MFB1 d i 4 3 ESC MFE ESC + 09H

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This command returns the sector #43 of the EJ file 'NNNN0044.EJ' where NNNN is the last 4 digit of the MFMEM serial number.

7.17.12 EJ_GET_EOD_NUMBER_BY_DATE

Function: This command returns the number of the first and last closed EJ session of the given date.

Code: **ESC MFB p l ESC MFB1 e a <*FromDate*> ESC MFE ESC + 09h**

Parameter	Value	Format
<FromDate>	"DD.MM.YYYY"	ASCII, 30h ≤ *n* ≤ 39h DD = Day MM = Month YYYY = Year

Answer:

```
ESC s DLH DLL 09H
ESC MFB <FirstSession>
ESC MFB1 <LastSession> ESC MFE
```

Parameter	Meaning	Format
<FirstSession>	first session number	ASCII
<LastSession>	last session number	ASCII

The command rejects with EM_PARA_RANGE_ERROR, if the defined day <FromDate> does not contain at least one closed session.

Error Msg:
EM_PARA_MISSING
EM_PARA_WRONG
EM_PARA_WRONG_FORMAT
EM_PARA_RANGE_ERROR

7.17.13 EJ_GET_INFO_PER_SESSION

Function: This command transmits EJ information about one specified EOD block (selected session). Furthermore the checksum of the selected session is verified.

Code: **ESC MFB p y ESC MFB1 d a <*FromBlock*> ESC MFE ESC + 09h**

Parameter	Value	Format
<***FromBlock***>, EOD block number	1 – EOD Block count (5.7.10 Block restriction)	ASCII, 30 H ≤ *n* ≤ 39 H, max. 5 bytes

Answer:

```
ESC s DLH DLL 09H
ESC MFB  <NumberOfTickets>
ESC MFB1 <Session checksum>
ESC MFB1 <StartSessionDateTime>
ESC MFB1 <EndSessionDateTime> ESC MFE
```

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Parameter	Meaning	Format
<NumberOfTickets>	Number of tickets of the slected session	ASCII
<SessionChecksum>	Checksum (20 Bytes)	ASCII
<StartSessionDateTime>	Start date/time of session,	ASCII DD.MM.YYYY hh:mm:ss
<EndSessionDateTime>	End date/time of session	ASCII DD.MM.YYYY hh:mm:ss

Note: Data fields that are still not available do not keep any data.

Error Msg: EM_EJ_NOT_PRESENT
EM_EJ_FILE_NOT_FOUND
EM_EJ_CHECKSUM_WRONG
EM_PARA_RANGE_ERROR
EM_PARA_WRONG_FORMAT
EM_CMD_WRONG_FORMAT

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7.18 TEST commands - general

7.18.1 TEST_READ_MFMEM_STATUS

Function: This command sends back the MFMEM status byte.

Code: **ESC MFB l b ESC MFE ESC + 09h**

Answer: `ESC s 00H 05H 09H` *`d0 d1 d2 d3`*

The content of *d0* is described in chapter 3.3 MFMEM status.

7.18.2 TEST_MFMEM_FORMAT

Function: This command reads the number of different MF blocks.

Code: **ESC MFB l c ESC MFE ESC + 09h**

Code old: ESC MFB L C ESC MFE ESC + 09H

Answer: `ESC s 00H 1BH 09H` *`d0 d1 ... d16 d17`*

d0, *d1*: Number of formatted blocks in HEX
d2, *d3*: Number of unformatted blocks in HEX
d4, *d5*: Number of used blocks in HEX
d6, *d7*: Number of free blocks in HEX
d8, *d9*: Number of faulty blocks in HEX
d10, *d11*: Number of SNO blocks in HEX
d12, *d13*: Number of EOD blocks in HEX
d14, *d15*: Number of TKD blocks in HEX
d16, *d17*: Number of EJ_CHANGE blocks in HEX
d18, *d19*: Number of TH blocks in HEX
d20, *d21*: Number of VAT blocks in HEX
d22, *d23*: Number of CRY blocks in HEX
d24, *d25*: Number of TT blocks in HEX

7.18.3 TEST_MFMEM

Function: All blocks of MF memory are read. The check sums of used blocks are verified.

Code: **ESC MFB l d ESC MFE ESC + 09h**

Code old: ESC MFB L D ESC MFE ESC + 09H

Answer: `ESC s 00H 09H 09H` *`d0 d1 ... d6 d7`*

d0, *d1*: Number of free blocks which are okay in HEX
d2, *d3*: Number of free blocks which are faulty in HEX
d4, *d5*: Number of used blocks which are okay in HEX
d6, *d7*: Number of used blocks which are faulty in HEX

Note: **The execution time is around 1 minute.** During that time the red led is blinking. (See chapter 12.1 Printer and MF LED's for exact blink codes)

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7.18.4 TEST_READ_RTC

Function: This command reads the actual date and time from the internal RTC.

Code: **ESC MFB I g ESC MFE ESC + 09h**

Code old: ESC MFB L G ESC MFE ESC + 09H

Answer: `ESC s 00H 08H 09H d0 ... d6`

Address Location	Function	BCD Data Mode Range
d0	Day	01 - 31
d1	**Month**	01 - 12
d2	Year	00 – 99
d3	**Century**	20
d4	Hours	00 - 23
d5	**Minutes**	00 - 59
d6	**Seconds**	00 - 59

7.18.5 TEST_READ_MFMEM_BL

Function: This command transfers the contents of one physical MFMEM block.

Code: **ESC MFB I h ESC MFB1 d a *<BlockNo>* ESC MFE ESC + 09h**

Code old: ESC MFB L H *<BlockNo>* ESC MFE ESC + 09H

Parameter	Value	Format
<BlockNo>	1 – max. phys. block no. - 1 (5.7.10)	ASCII, 30 H ≤ *n* ≤ 39 H, max. 5 bytes

Note: The *<BlockNo>* counting starts by zero, but the first physical block (0) can not be read with this command. Use command TEST_READ_MFMEM instead of.

Answer: `ESC s DLH DLL 09H <Data>`

Parameter	Meaning	Format
<Data>	*d1...dn*: MFMEM Block data (max 256 byte)	HEX

7.18.6 TEST_READ_MFMEM

Function: This command sends back one page of 32 bytes data from the MFMEM in HEX format.

Code: **ESC MFB I j ESC MFB1 d c *<GetIndex>* ESC MFE ESC + 09h**

Code old: ESC MFB L J *<MFMEMPage>* ESC MFE ESC + 09H

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Parameter	Value	Format
<GetIndex>	0 – 131071 (4 Mb / 32 byte)	ASCII, 30 H ≤ *n* ≤ 39 H, max. 6 bytes

Answer: `ESC s 00H 21H 09H` *`d0 ... d31`*
d0..d31: One 32 byte page contents of the MFMEM in HEX.

Example: ESC MFB L J 0 ESC MFE ESC + 09H
This command read out the data of the MFMEM from address 0000H – 001FH.

7.18.7 TEST_PRINT_SELFTEST

Function: This command prints a self-test including information about the printer, the fiscal status, the EJ status and tax terminal. The output is either printed on the receipt station or is held ready as soft copy to be fetched by the application.

Code: **ESC MFB I o**
[ESC MFB2 f a *<Station>*]
ESC MFE ESC + 09h

Code old: ESC MFB L O ESC MFE ESC + 09H

Parameter	Value	Format
<Station>	30 H ≤ *n* ≤ 39 H	"2" – Receipt (Default) "16" – Host – For EJ soft copy.

Answer: `ESC s 00H 01H 09H`

7.18.8 TEST_PRODUCTION

Function: This command performs a specified production test and returns the result.

Condition: This command is only possible if the system is not yet fiscalized
(No SNO block burned).

Code: **ESC MFB I u**
ESC MFB1 d o *<Number>*
ESC MFE ESC + 09h

Parameter	Value	Format
<Number>	1 – 255	ASCII (30 H ≤ n ≤ 39 H)

Answer: `ESC s 00H 03H 09H` *`<Testnumber> <Result>`*

Parameter	Meaning	Format
<Testnumber>	Number of test performed	HEX (0x00 -0xFF)
<Result>	Result of test	HEX (0x00 -0xFF)

Standard results:

```
0x00 = Test passed
0xff = Test failed
```

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```
0xfe = Test not implemented
```

Test Number	Description	Result
1	Test NV_RAM	0x00 = OK 0xff = FAILED
2	Stimulate MF LED's	0x00 = OK
3	Check Battery	0x00 = OK 0xff = FAILED
10	SD Slot / SD Card Test	0x00 = OK 0x38 = No SD card detected 0x6E = SD card is write protected 0x6A = SD communication error
13	Display	0x00 = OK 0xff = FAILED (test not possible, because Tax Terminal is already paired)
255	Temporarily enable transparent mode	Not enabled in Romania !! This command immediately enables the transparent mode. There is no answer. 0xfe = Test not implemented

7.19 TAX TERMINAL commands

7.19.1 General Note

These commands are not intended for application programmers. This commands group handles the interface between the tax terminal connected on the display port and the fiscal firmware.

7.19.2 TAX_TERMINAL_PAIR_REQUEST

Function: This command starts the pairing process between FFC and a tax terminal.

Code: **ESC MFB v f ESC MFE**

Condition:

MFC Status byte 1

MFC_ TKD_RESET	MFC_ RECEIPT_ TOTAL	MFC_ DOCU_OPEN	MFC_ REPORT_ OPEN	MFC_ RECEIPT_ OPEN	MFC_ DAY_OPEN	MFC_ TRAINING_ MODE	MFC_ BLOCKADE
x	**x**	**x**	**x**	**x**	**x**	**x**	**x**

MFC Status byte 2

not used	not used	EJ_DATA_ PENDING	EJ_FULL	EJ_NEAR_ FULL	MFC_ CMD_ INTERRUPTED	MFC_ PRINTOUT_ INTERRUPTED	MFC_ FISCAL_ MODE
x	**x**	**x**	**x**	**x**	**x**	**x**	**1**

Note: A new pairing is necessary after production or a replacement of either the FFC controller or the tax terminal controller.

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If the tax terminal contains files, which are still not sent to the fiscal authority, then the error EM_TAX_TERMINAL_NOT_EMPTY is replied. In this case the files must be transferred to the fiscal authority manually (USB stick and other PC).

Error Msg: EM_CMD_WRONG_ORDER
EM_TAX_TERMINAL_NOT_EMPTY
EM_TAX_TERMINAL_WRONG

Printout: Administrative receipt with optional header lines:

```
******  W I N C O R - N I X D O R F   ********
        receipt header line 2
        receipt header line 3
        receipt header line 4
   Cod Fiscal C.I.F.:1234567890123 4

****   BON NEFISCAL   ****   BON NEFISCAL   ****

       TT_PAIR_REQUEST

   TT serial no.:  XXXXXXXXXXXXXX

       DATA:16/05/2015     ORA:17-57-56
****   BON NEFISCAL   ****   BON NEFISCAL   ****
```

7.19.3 TAX_TERMINAL_FIRMWARE_UPDATE

Function: With this command, the application can start a Tax Terminal firmware update.

Code: **ESC MFB v g ESC MFE**

Condition:

MFC Status byte 1

MFC_ TKD_RESET	MFC_ RECEIPT_ TOTAL_	MFC_ DOCU_OPEN	MFC_ REPORT_ OPEN_	MFC_ RECEIPT_ OPEN_	MFC_ DAY_OPEN	MFC_ TRAINING_ MODE	MFC_ BLOCKADE
x	x	x	x	x	0	x	x

MFC Status byte 2

not used	not used	EJMEM_DATA_ PENDING	EJ_FULL	EJ_NEARLY_ FULL	MFC_ CMD_ INTERRUPTED	MFC_ PRINTOUT_ INTERRUPTED	MFC_ FISCAL_ MODE
x	x	x	x	x	x	x	x

Note: If none firmware update is available, then the error EM_CMD_WRONG_ORDER is replied.

Error Msg: EM_CMD_WRONG_ORDER
EM_TAX_TERMINAL_WRONG

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7.19.4 TAX_TERMINAL_TEST

Function: With this command, the application can start a Tax Terminal test as alternative to the display service menu (11.2.5).

Code: **ESC MFB v h ESC MFE**

Condition:

MFC Status byte 1

MFC_ TKD_RESET	MFC_ RECEIPT_ TOTAL	MFC_ DOCU_OPEN	MFC_ REPORT_ OPEN	MFC_ RECEIPT_ OPEN	MFC_ DAY_OPEN	MFC_ TRAINING_ MODE	MFC_ BLOCKADE
x	x	x	x	x	0	x	x

MFC Status byte 2

not used	not used	EJMEM_DATA_ PENDING	EJ_FULL	EJ_NEARLY_ FULL	MFC_ CMD_ INTERRUPTED	MFC_ PRINTOUT_ INTERRUPTED	MFC_ FISCAL_ MODE
x	x	x	x	x	x	x	x

Error Msg: EM_CMD_WRONG_ORDER
EM_TAX_TERMINAL_NOT_PAIRED
EM_TAX_TERMINAL_WRONG

7.19.5 TAX_TERMINAL_IMPORT_XML_PROFILE

Function: With this command, the application starts the import of a XML profile from an external storage connected to the Tax Terminal.

Code: **ESC MFB v i ESC MFE**

Condition:

MFC Status byte 1

MFC_ TKD_RESET	MFC_ RECEIPT_ TOTAL	MFC_ DOCU_OPEN	MFC_ REPORT_ OPEN	MFC_ RECEIPT_ OPEN	MFC_ DAY_OPEN	MFC_ TRAINING_ MODE	MFC_ BLOCKADE
x	x	x	x	x	0	x	x

MFC Status byte 2

not used	not used	EJMEM_DATA_ PENDING	EJ_FULL	EJ_NEARLY_ FULL	MFC_ CMD_ INTERRUPTED	MFC_ PRINTOUT_ INTERRUPTED	MFC_ FISCAL_ MODE
x	x	x	x	x	x	x	x

Error Msg: EM_CMD_WRONG_ORDER
EM_TAX_TERMINAL_NOT_PAIRED
EM_TAX_TERMINAL_WRONG

7.19.6 TAX_TERMINAL_EXPORT_XML_BY_SESSION

Function: With this command, the application starts the export of all daily XML files of given range to an external storage connected to the Tax Terminal. Additionally a summary XML file will be exported.

Code: **ESC MFB v j**
ESC MFB1 d a <*FromSession*>

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ESC MFB1 d b *<ToSession>*
ESC MFE

Parameter	Value	Format	
<FromSession>	1 – max. EOD no. (5.7.10)	ASCII, 30 H ≤ *n* ≤ 39 H, max. 5 bytes	
<ToSession>	1 – max. EOD no. (5.7.10)	ASCII, 30 H ≤ *n* ≤ 39 H, max. 5 bytes	

Condition:

MFC Status byte 1

MFC_ TKD_RESET	MFC_ RECEIPT_ TOTAL	MFC_ DOCU_OPEN	MFC_ REPORT_ OPEN	MFC_ RECEIPT_ OPEN	MFC_ DAY_OPEN	MFC_ TRAINING_ MODE	MFC_ BLOCKADE
x	**x**	**x**	**0**	**0**	**x**	**x**	**x**

MFC Status byte 2

not used	not used	EJMEM_DATA_ PENDING	EJ_FULL	EJ_NEARLY_ FULL	MFC_ CMD_ INTERRUPTED	MFC_ PRINTOUT_ INTERRUPTED	MFC_ FISCAL_ MODE
x	**x**	**x**	**x**	**x**	**x**	**0**	**1**

Note: The command will be rejected with an error, if the Tax Terminal is not enabled (paired), any Tax Terminal component is not available or the customer display is not connected.

Error Msg:
EM_CMD_WRONG_ORDER
EM_CMD_WRONG_FORMAT
EM_PARA_MISSING
EM_PARA_WRONG
EM_TAX_TERMINAL_NOT_PRESENT
EM_TAX_TERMINAL_NOT_PAIRED
EM_TAX_TERMINAL_WRONG
EM_SC_DEFECT

7.20 MFC commands

7.20.1 MFC_STATUS

Function: With this command the application obtains the MFC status

Code: **ESC + 07H**

Answer: `ESC s DLH DLL 07H d0 d1`
d0....d1: <MFC_STATUSBYTE_1><MFC_STATUSBYTE_2> in HEX

7.20.2 MFC_SYNC

Function: With this command the application obtains the printer paper status and the MFC error.

Code: **ESC + 08H**

Note: The MFC error will be cleared after the command is executed.

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Answer: `ESC s DLH DLL 08H <PrinterStatus> <MFC error>`

with

<PrinterStatus> : printer paper status (result of the sequence ESC "v")

<MFC error> : Current MFC error

7.20.3 MFC_TEST

Function: With this command the application obtains the data which are requested with the last TEST or GET command.

Code: **ESC + 09H**

Note: If none data are available (e.g. previous command failed or was not sent) a response with no data is sent.

Answer: `ESC s DLH DLL 09H <data>`

(DLH * 256) + DLL = length of following data

<data> = data previously requested by a TEST_ or GET_ command.

7.20.4 MFC_INFO

This command is not recommended for new implementations. It will not be supported in the future.

Function: With this command the application obtains the MFC information.

Code: **ESC + 0AH**

Answer: `ESC s DLH DLL 0AH d0 ..... dn`

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DataByte	Content	Meaning
d0-d1	<MFC_STATUSBYTE_1> <MFC_STATUSBYTE_2>	Hex
d2	;	Separator
d3-d10	<RTC_SECOND> ,<RTC_MINUTE> ,<RTC_HOUR>	ASCII
d11	;	Separator
d12-d19	<RTC_DAY> ,<RTC_MONTH> ,<RTC_YEAR>	ASCII
d20	;	Separator
d21-d26	<LegalReceiptNumber>	in ASCII, 6 Digits, leading Blanks
d27	;	Separator
d28-d35	<LAST_EOD_DAY> ,<LAST_EOD_MONTH> ,<LAST_EOD_YEAR>	ASCII
d36	;	Separator
D37-d46	<Fiscal_ID>	10
D47	;	Separator
D48-d49	<CIF_ID>	2
d50	;	Separator
d51-64	<CIF>	3-13 digits
d65	;	Separator

Note: <*LegalReceiptNumber*> is the number of legal receipts within the currently opened fiscal day. After DAY_END as well as after DAY_BEGIN the <*LegalReceiptNumber*> is zero.

7.20.5 MFC_OPEN_D

This command is not recommended for new implementations. It will not be supported in the future.

Function: The MFC_OPEN specifies the display configuration of the printer

Code: **ESC + 0DH**
<DISPType>
<DISPInterfaceType>
<DISPInterfaceNo>
<DISPConnection>
<PRTInterfaceType>

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<PRTInterfaceNo>

DataByte	Content	Meaning
<DISPType>	“1”	MFC open for customer display
<DISPInterfaceType>	“C”	Display connected to COM port
<DISPInterfaceNo>	“1”-“9”	number of display COM port
<DISPConnection>	“0” “1”	=customer display is connected to the CPU =customer display is connected to a subdevice (printer or MFC)
<PRTInterfaceType>	"C"	Printer connected to COM port
<PRTInterfaceNo	“1”-“9”	number of printer COM port

Answer: `ESC s DLH DLL 0DH <MFC error>`
with
<MFC error> : Current MFC error

7.20.6 MFC_READ_LAST_INTERRUPTED_CMD

Function: With this command the application obtains the last command code, which was interrupted by an error.

Code: **ESC + 0FH**

Answer: `ESC s DLH DLL 0FH <uiLastInterruptedCmd>`
with *<uiLastInterruptedCmd>* = 2 byte command code (e.g. "B" for DAY_END).

Note: This command is required, if the status bit MFC_CMD_INTERRUPTED is set and the application needs the information which command was interrupted by an error. To precede it is necessary to send again the same command, which was interrupted before.

7.20.7 MFC_READ_HOST_DATA

Function: With this command, the application obtains the relatively high volume data generated by different commands with host destination.

Code: **ESC “+” 10H**

Answer: `ESC “s” DLH DLL 10H <uiCmdCode><HostData>`

Parameter	Value	Format

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<uiCmdCode> CommandCode	20 H ≤ *n* ≤ 7F H	max. 2 bytes
<HostData> the data generated by the command with host destination		Maximum length of this data is 512 bytes.

<uiCmdCode> = 2 byte command code (e.g. "ga" for REPORT_BLOCK2BLOCK).
<HostData> = the data generated by the command with host destination

Note: This MFC command should be executed just after the command generating the data with the host destination. If an empty answer (**ESC** “s” 00H 01L 10H) is returned, no data has been generated by the previous command; otherwise the MFC_READ_HOST_DATA should be repeated until an empty answer (**ESC** “s” 00H 01L 10H) is returned.
The **ESC** “+” 10H sequence should follow immediately the command generating the data.

8 ERROR HANDLING

8.1 Error messages

Error: **0x10 EM_POWER_FAIL**

Meaning: This error is transmitted with the first ASB message, which is send from the MFC to the host after power on.

Reaction: The application must initialize the printer and the MFC if necessary.

Error: **0x11 EM_CPU_DEFECT (BLOCKADE)**.

Meaning: During the power-on test a CPU error was detected

Reaction: Replace the microprocessor on the MFC board or change the whole MFC board before executing a TKD reset.

Error: **0x12 EM_RAM_DEFECT (BLOCKADE)**

Meaning: Not used in FFC_TH230+

Error: **0x13 EM_EPROM_CHECKSUM_ERROR (BLOCKADE)**

Meaning: Not used in FFC_TH230+

Error: **0x14 EM_RTC_BATTERY_EMPTY (BLOCKADE)**

Meaning: The battery, which supplies the NV-RAM and the RTC during power-off, is empty.

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Reaction: Replace the battery on the MF board and set the current time before executing a TKD reset.

Error: **0x15 EM_RAM_CHECKSUM_ERROR (BLOCKADE)**

Meaning: The checksum of the fiscal data within the NV-RAM is wrong. The totalizers of the current day are lost.

Reaction: Execute a TKD reset.

Meaning: Not used in TH230-MF

Error: **0x16 EM_RTC_RAM_ERROR (BLOCKADE)**

Meaning: Not used in FFC_TH230+

Error: **0x17 EM_FW_CHANGE_WHILE_DAY_OPEN (BLOCKADE)**

Meaning: The FW was changed while the Day was open (MFC Status Bit MFC_DAY_OPEN = 1). The totalizers of the current day are lost.

Reaction: Execute a TKD reset.

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Error: **0x18 EM_NVRAM_ERROR (BLOCKADE)**

Meaning: Read/Write error on non-volatile memory.

Reaction: Execute a TKD reset.

Error: **0x19 EM_EVENT_ERROR**

Meaning: Not used in FFC_TH230+

Error: **0x20 EM_MFMEM_BL_DEFECT**

Meaning: Not used in FFC_TH230+

Error: **0x21 EM_MFMEM_NOT_CONNECTED (BLOCKADE)**

Meaning: The MFMEM isn't connected to the MFC (this is detected by write/read to the shift register of the MFMEM board) or the memory chip isn't assembled (this is detected by reading only 0xff in ReadHwData ()) on the MFMEM print.

Reaction: Connect a fully assembled MFMEM board to the MFC before executing a TKD reset.

Error: **0x22 EM_MFMEM_BYTE_NOT_BURNED (BLOCKADE)**

Meaning: It wasn't possible to burn a data byte into the MFMEM.

Reaction: Check the MFMEM board, the MFC (especially the program voltage) and the data connection between the two boards. If the problem is solved, execute a TKD reset.

Error: **0x23 EM_TKD_RESET_LIMIT_OVERRUN (BLOCKADE)**.

Meaning: The maximum amount of TKD resets has been reached.

Reaction: Replace the MFMEM board by a new one and start the initialisation procedure.

Error: **0x24 EM_MFMEM_DEFECT (BLOCKADE)**

Meaning: The MFMEM is defect.

Reaction: Replace the MFMEM board by a new one and start the initialisation procedure.

Error: **0x25 EM_MFMEM_BL_CHECKSUM_ERROR (BLOCKADE)**

Meaning: The checksum of a MFMEM block is wrong.

Reaction: Replace the MFMEM board by a new one and start the initialisation procedure.

Error: **0x26 EM_MFMEM_FULL (BLOCKADE)**

Meaning: The maximum number of MFMEM blocks, which can be stored in the MFMEM, has been reached.

Reaction: Replace the MFMEM board by a new one and start the initialisation procedure.

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Error: **0x27 EM_MFMEM_PTR_ERROR (BLOCKADE)**

Meaning: Not used in FFC_TH230+

Error: **0x28 EM_SNO_COMPARE_ERROR (BLOCKADE)**

Meaning: The serial number in the battery buffered RAM of the MFC is not the same as the serial number which is stored in the SNO block of the connected MFMEM board.

Reaction: Execute a TKD reset.

Error: **0x29 EM_HWDATA_CHECKSUM_ERROR (BLOCKADE)**

Meaning: 1. The checksum of the HWData, which has been transmitted within the INIT_HWDATA command, is wrong.
2. After power-on this error will be issued if the checksum of the HWData, which are stored in the MFMEM, is wrong.

Reaction: 1. Send an INIT_HWDATA command with a correct checksum for the HWData.
2. Replace the MFMEM board by a new one and start the initialisation procedure.

Error: **0x2A EM_MFMEM_BL_NOT_BURNED (BLOCKADE)**

Meaning: After 3 attempts it wasn't possible to store a MFMEM block.

Reaction: Check the MFMEM board, the MFC (especially the program voltage) and the data connection between the two boards. Use the test command TEST_READ_MFMEM_BL to analyse the problem. If the problem is solved, execute a TKD reset.

Not used in FFC_TH230+

Error: **0x2B EM_MFMEM_BL_LIMIT_OVERRUN**

Meaning: The maximal number of blocks, which can be stored in the MFMEM for a special MFMEM block type, has been reached (e.g. in Romania changes of TH and VAT rates).

Reaction: If the storage of the block is absolutely necessary (e.g. VAT change) the MFMEM must be replaced by a new one. If it's not absolutely necessary to store the block (e.g. TH change) it is possible to go on working.

Error: **0x2C EM_MFMEM_BL_READ_ERROR (BLOCKADE)**

Meaning: After 3 attempts it wasn't possible to read a MFMEM block.

Reaction: Check the MFMEM board, the MFC and the data connection between the two boards. Use the test command TEST_READ_MFMEM_BL to analyse the problem. If the problem is solved, execute a TKD reset.

Error: **0x2D EM_EJ_DEFECT (BLOCKADE)**

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Meaning: The EJ medium or the interface has a hardware defect.

Reaction: Replace the EJ medium or the MFC main board and execute a TKD reset.

Error: **0x2E EM_SC_DEFECT (BLOCKADE)**

Meaning: Smart Card doesn't work as expected. This is a Hardware defect that can't be solved by the application.

Error: **0x2F EM_EJMEM_DATA_CORRUPTED (BLOCKADE)**

Meaning: Not used in FFC_TH230+

Error: **0x30 EM_MFMEM_BL_NOT_EXIST**

Meaning: The requested block does not exist in the MFMEM

Error: **0x31 EM_PRT_TIMEOUT**

Meaning: There is no response from the printer.
In case this error happens while EJ_REPRINT or REPORT this happens because of Thermal Printer getting to hot. Please use smaller Ranges for Printout.

Error: **0x33 EM_PRT_ERROR**

Meaning: -

Error: **0x35 EM_EJ_ERROR**

Meaning: Not used in TH230-MF

Error: **0x36 EM_PRT_PAPER**

Meaning: 1. In a RETAIL_SALES_RECEIPT, RETAIL_SALES_RECEIPT_WITH_TAX_ID or a SERVICE_SALES_RECEIPT a paper end situation has been detected.
2. Before executing a BEGIN_RECEIPT, DAY_END or a REPORT command a paper near end situation has been detected.

Reaction: Insert a new paper roll.

Error: **0x38 EM_EJ_NOT_PRESENT**

Meaning: There is no EJ medium present.

Reaction: Insert an EJ medium

Error: **0x39 EM_EJ_NOT_THE_SAME**

Meaning: The inserted EJ medium is not the authorized EJ medium. Write access to this medium can not be granted.

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Reaction: Insert the authorized medium.

Error: **0x3A EM_EJ_NOT_EMPTY**

Meaning: 1. The EJ medium cannot be formatted because it is not empty
2. A write access to the EJ medium has failed, because this part of the medium is already used (not empty)

Reaction: Insert a new EJ medium and perform EJ_INIT

Error: **0x3C EM_EJ_SESSION_STILL_OPEN**

Meaning: A session is still open on the EJ

Error: **0x3D EM_EJ_SESSION_NOT_OPEN**

Meaning: There is no session open on the EJ.

Error: **0x3E EM_EJ_TRACK_STILL_OPEN**

Meaning: There is still a track open on the EJ.

Error: **0x3F EM_EJ_TRACK_NOT_OPEN**

Meaning: There is no track open on the EJ.

Error: **0x40 EM_DISP_CONFIG_UNDEFINED**

Meaning: Not used in TH230-MF

Error: **0x41 EM_INT_OPERATOR_DISP_NOT_PRESENT**

Meaning: Not used in TH230-MF

Error: **0x42 EM_INT_CUSTOMER_DISP_NOT_PRESENT**

Meaning: Not used in TH230-MF

Error: **0x43 EM_EXT_CUSTOMER_DISP_NOT_PRESENT**

Meaning: Not used in TH230-MF

Error: **0x44 EM_EXT_OPERATOR_DISP_NOT_PRESENT**

Meaning: Not used in TH230-MF

Error: **0x45 EM_MFC_CUSTOMER_DISP_NOT_PRESENT**

Meaning: There is no customer display connected to the printer.

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Error: **0x46 EM_PRT_CUSTOMER_DISP_NOT_PRESENT**

Meaning: Not used in TH230-MF

Error: **0x47 EM_TAX_TERMINAL_NOT_PAIRED**

Meaning: The tax terminal is not paired with FFC. Use command TAX_TERMINAL_PAIR_REQUEST.

Error: **0x48 EM_TAX_TERMINAL_NOT_PRESENT**

Meaning: The tax terminal is not connected or ready for operation.

Error: **0x49 EM_TAX_TERMINAL_WRONG**

Meaning: The tax terminal is wrong.

Error: **0x4A EM_RECEIPT_ALREADY_PRINTED**

Meaning: Not used in TH230-MF

Error: **0x4B EM_LAST_RECEIPT_NOT_FISCAL**

Meaning: Not used in TH230-MF

Error: **0x4D EM_TAX_TERMINAL_NOT_EMPTY**

Meaning: Pairing with tax terminal not allowed because it contains unsent XML files.

Error: **0x50 EM_SNO_NOT_PRESENT**

Meaning: The SNO is not stored in MFMEM. Fiscal commands are only allowed after SNO is present.

Reaction: Set SNO for fiscal command apply.

Error: **0x51 EM_CMD_WRONG_FORMAT**

Meaning: A fiscal command (ESC MFB ....) was interrupted by a MFC command (ESC "+" .....).

Reaction: Send a corrected command.

Error: **0x52 EM_CMD_WRONG_ORDER**

Meaning: The command is at the current MFC status not available.

Reaction: The MFC status depends on the two MFC_STATUSBYTES (e.g. a DAY_BEGIN command is only available if the MFC_OPEN bit is cleared). Send a command which is allowed at the current MFC status.

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Error: **0x53 EM_CMD_TOO_LONG**

Meaning: The fiscal command (ESC MFB .... ESC MFE) is too long.

Reaction: Send a corrected command.

Error: **0x54 EM_PARA_MFB1_MISSING**

Meaning: Not used in TH230-MF

Error: **0x55 EM_VAT_NOT_PRESENT**

Meaning: Any other fiscal relevant command will be rejected with this error message if no VAT rates are stored in the MF memory.

Reaction: Execute SET_VATRATES command.

Error: **0x56 EM_TH_NOT_PRESENT**

Meaning: DAY_BEGIN command will be rejected with this error message if no header is stored in the MFC or MF memory.

Reaction: Execute command SET_HEADER

Error: **0x57 EM_PARA_DATE_WRONG_FORMAT**

Meaning: One of the transmitted parameters <Date>, <Time> or <TimeDate> has a wrong format.

Reaction: Send a corrected command.

Error: **0x58 EM_HWDATA_NOT_PRESENT**

Meaning: No hardware data is stored in the connected MF memory

Reaction: Send an INIT_HWDATA command.

Error: **0x59 EM_MFMEM_NOT_FORMATTED**

Meaning: The connected MFMEM isn't fully formatted.

Reaction: Send an INIT_FORMAT_MFMEM command for fully formatting.

Error: **0x5A EM_CMD_UNKNOWN**

Meaning: The sent fiscal command is unknown or not available.

Reaction: Send a corrected command.

Error: **0x5B EM_PARA_RANGE_ERROR**

Meaning: The parameter <FromDate> is greater than the parameter <ToDate> or the parameter <FromValue> is greater than the parameter <ToValue>.

Reaction: Send a corrected command.

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Error: **0x5C EM_PARA_WRONG**

Meaning: One of the parameters within a fiscal command is wrong or not allowed

Reaction: Send a corrected command.

Error: **0x5D EM_PARA_WRONG_FORMAT**

Meaning: One of the parameters has the wrong content, the wrong length.

Reaction: Send a corrected command.

Error: **0x5E EM_PARA_MISSING**

Meaning: A required parameter for a fiscal command is missing.

Reaction: Send a corrected command.

Error: **0x5F EM_PARA_UNKNOWN**

Meaning: One of the parameters sent within a fiscal command is unknown or not expected

Reaction: Send a corrected command.

Error: **0x60 EM_CMD_CHECKSUM_ERROR**

Meaning: The LRC computed by the MFC does not match with the LRC received. This error can be generated by a transmission error or a wrong LRC computation.

Reaction: Resend the command with a correct LRC.

Error: **0x61 EM_DATE_NOT_PRESENT**

Meaning: The date and time of the RTC isn't set. This will be checked during power-on, TKD Reset and before storing the SNO block in the MFMEM.

Reaction: Set the date and time of the RTC with the command SET_TIME.

Error: **0x62 EM_DATE_NOT_ALLOWED**

Meaning: The command SET_TIME is rejected, because the time is older than the last block date.

Reaction: Set the date and time of the RTC with the command SET_TIME, which is equal or greater than the date of the last block.

Error: **0x63 EM_CRY_CHANGE_DATE_NOT_PRESENT**

Meaning: Not used in TH230-MF

Error: **0x64 EM_EJ_FULL**

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Meaning: The EJMEM free space is less than 6kB:

Reaction: Close fiscal day and replace EJ medium.

Error: **0x65 EM_EJ_FILE_NOT_FOUND**

Meaning: The requested data does not exists on the EJ medium

Reaction: Send a corrected command.

Error: **0x66 EM_EJ_FILE_OFFSET_WRONG**

Meaning: In the EJ_GET_DATA command the '<*Offset*>' parameter is higher than the EJ file size or is not a multiple of 512.

Reaction: Send a corrected command.

Error: **0x67 EM_EJ_NOT_INITIALIZED**

Meaning: The EJ medium is not initialized for use as a WN EJ medium.

Reaction: Insert and initialize a WN EJ card..

Error: **0x68 EM_SC_NOT_INITIALIZED**

Meaning: The Smart Card installed in the MFC isn't prepared for using in the field. (The Smart Card PERSO bit is not set).

Reaction: Prepare the Smart Card with the necessary tool for field usage.

Error: **0x69 EM_SC_READER_ERROR**

Meaning: Not used in TH230-MF

Error: **0x6A EM_EJ_INTERFACE_ERROR**

Meaning: A hardware error has been detected while accessing the EJ medium reader.

Reaction: Check EJ medium reader hardware.

Error: **0x6B EM_EJ_FILE_CHECKSUM_WRONG**

Meaning: The checksum of the read EJ data is wrong.

Error: **0x6C EM_EJ_NEAR_FULL**

Meaning: The EJ is nearly full. Please change this EJ medium.

Error: **0x6D EM_EJ_REINSERTED**

Meaning: The EJ medium has been reinserted**.**

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Error: **0x6E EM_EJ_WRITE_PROTECTED**

Meaning: The EJ medium is write protected.

Error: **0x6F EM_EJ_WRITE_ERROR**

Meaning: The EJ medium is not writeable.

Error: **0x70 EM_EJ_UNSUPPORTED_MEDIUM**

Meaning: This EJ medium is not supported.
The reason could be:
-wrong vendor
- wrong size,
- missing password protection.

Reaction: Insert a WN EJ medium

Error: **0x71 EM_RECEIPT_TOT_OVERFLOW**

Meaning: The last command is rejected because it will generate an overflow of one totalizer in the structure sReceipt.aulTot [].

Reaction: Close the current receipt and open a new one.

Error: **0x72 EM_PRICE_OVERFLOW**

Meaning: The last command is rejected because the given single price is too high.

Error: **0x73 EM_RECEIPT_TOT_UNDERFLOW**

Meaning: The last command is rejected because it will generate an underflow for one totalizer in the structure sReceipt.aulTot [].

Reaction: Send a corrected command.

Error: **0x74 EM_DAY_TOT_OVERFLOW**

Meaning: The last command is rejected because it will generate an overflow for one totalizer in the structure sDay.aulTot [].

Reaction: Close the current day and open a new one.

Error: **0x75 EM_GRAND_TOT_OVERFLOW (BLOCKADE)**

Meaning: During the execution of the TOTAL_LINE command it has been detected that the current receipt totalizers will generate an overflow for one totalizer in the structure sGrand.aulVAT [] or sGrand.aelNetSale []. The receipt will be cancelled and the day will be closed by MFC.

Reaction: Replace the MFMEM board by a new one and start the initialisation procedure.

Error: **0x76 EM_CURRENCY_CHANGE_IN_RANGE**

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Meaning: Report commands with a range over two currencies will be rejected with this error message.

Reaction: Send new correct range parameter.

Error: **0x77 EM_PAYMENT_TYPE_NOT_PRESENT**

Meaning: The payment type is not defined.

Reaction: Send a valid type parameter or defined a type with SET_PAYMENT_LIST.

Error: **0x79 EM_DAY_CNT_OVERFLOW**

Meaning: Day Counter would overflow if the operation was done.

Reaction: Check the size of DayCounters.

Error: **0x7A EM_PAYMENT_TOTAL_WRONG**

Meaning: The amount paid is smaller than the receipt Total.

Reaction: Either issue another PAYMENT_IN or cancel the current receipt using SPECIAL_ALL_VOID.

Error: **0x7B EM_DAY_TOT_UNDERFLOW**

Meaning: The last command is rejected because it will generate an underflow for one totalizer in the structure Day.Tot [].

Reaction: Send a corrected command.

Error: **0x81 EM_RECEIPT_TOTAL_WRONG**

Meaning: The parameter <TotalValue>, which is transmitted with the command TOTAL_LINE, isn't equal to the totalizer sReceipt.aulTot [BRUTTO_SALES]. The receipt will automatically cancelled by the MFC.

Reaction: Open a new receipt and transmit the corrected parameter <TotalValue> with the command TOTAL_LINE.

Error: **0x82 EM_BUTTON1_PRESSED**

Meaning: Not used in TH230-MF

Error: **0x83 EM_BUTTON2_PRESSED**

Meaning: Not used in TH230-MF

Error: **0x84 EM_TKD_JUMPER_IN_RESET**

Meaning: The TKD jumper is in reset position during power on.

Error: **0x85 EM_PASSWORD_REQUIRED**

Meaning: Not used in TH230-MF

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Error: **0x86 EM_DAY_END_REQUIRED**

Meaning: At the RECEIPT_BEGIN command execution, the elapsed time from the first RECEIPT_OPEN command in this fiscal day is greater than 24 hours.

Reaction: Execute the DAY_END and DAY_BEGIN command sequence.

Error: **0x87 EM_PASSWORD_WRONG**

Meaning: Not used in TH230-MF

Error: **0x88 EM_TKD_RESET_REQUIRED**

Meaning: Not used in TH230-MF

Error: **0x89 EM_INVOICE_RANGE**

Meaning: Not used in TH230-MF

Error: **0x8F EM_PROGRAM_ERROR**

Only for debugging purpose

Error: **0x91 EM_TEXT_NOT_ALLOWED**

Meaning: Not used in Romania

8.2 FM error codes

The following list is a logical grouping of error codes that may be returned by some fiscal commands. They are grouped here to keep a clean overview of return codes listed in the commands.

EM_MFMEM_NOT_CONNECTED
EM_MFMEM_FULL
EM_MFMEM_NOT_FORMATTED
EM_MFMEM_BYTE_NOT_BURNED
EM_MFMEM_BL_NOT_EXIST
EM_MFMEM_BL_CHECKSUM_ERROR
EM_MFMEM_BL_READ_ERROR

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8.3 EJ error codes

The following list is a logical grouping of error codes that may be returned by some fiscal commands. They are grouped here to keep a clean overview of return codes listed in the commands.

EM_EJ_INTERFACE_ERROR
EM_EJ_WRITE_ERROR
EM_EJ_NOT_INITIALISED
EM_EJ_NOT_PRESENT
EM_EJ_FILE_CHECKSUM_WRONG
EM_EJ_FULL
EM_EJ_NOT_THE_SAME
EM_EJ_UNSUPPORTED_MEDIUM
EM_EJ_WRITE_PROTECTED
EM_EJ_NOT_EMPTY
EM_EJ_NOT_INITIALISED
EM_EJ_SESSION_STILL_OPEN
EM_EJ_SESSION_NOT_OPEN
EM_EJ_TRACK_STILL_OPEN
EM_EJ_TRACK_NOT_OPEN
EM_EJ_FILE_NOT_FOUND
EM_EJ_WRITE_ERROR
EM_PARA_RANGE_ERROR

8.4 Supplementary checks error codes

The following list is a logical grouping of error codes that may be returned by some fiscal commands. They are grouped here to keep a clean overview of return codes listed in the commands.

EM_MFMEM_NOT_CONNECTED
EM_MFMEM_DEFECT
EM_RTC_BATTERY_EMPTY
EM_DATE_NOT_PRESENT
EM_MFC_CUSTOMER_DISP_NOT_PRESENT
EM_RAM_CHECKSUM_ERROR
EM_PRT_PAPER

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EM_EJ_NOT_PRESENT
EM_EJ_NOT_THE_SAME
EM_SNO_COMPARE_ERROR
EM_SNO_NOT_PRESENT

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9 INITIALIZATION / FISCALIZATION

To fiscalize a printer, the following step's has to be done.

1. Make sure the Tkd-Reset jumper is in its Default Position (see Chapter SERVICE RESET (TKD RESET))
2. Set the hardware data with the command **.**
3. **INIT**_HWDATA
4. Completely format the MFMEM with **INIT_FULL_FORMAT**
5. Set the current date and time with **SET_DATE_TIME**
6. Fiscalize the printer by omitting the command **INIT_WRITE_SNO**
7. Pair FFC and Tax Terminal with the command **TAX_TERMINAL_PAIR_REQUEST**
8. Insert a new WN EJ medium and initialize it with **EJ_INIT**
9. Set the receipt header with **SET_HEADER**
10. Set the VAT rates with **SET_VATRATES**

Now the printer is completely fiscalized. To start normal operation, send the command **DAY_BEGIN**.

10 SERVICE RESET (TKD RESET)

To perform a TKD reset procedure the following steps have to be done:

1. Carefully remove the MF interface from the printer unit to get access to the jumpers. Please refer to chapter 12.4.1 (Fiscal Family Controller (FFC)) for more details.
2. Change the state of the TKD Jumper (Jmp 1) by moving it to the opposite position ***The state of the TKD Jumper will be interpreted by the fiscal firmware. It is not necessary to move the jumper back to its original position!***
3. Power on printer.
4. If the date and time is valid, the TKD reset procedure starts automatically. If not please set date and time with the fiscal command SET_DATE_TIME and power printer off and on again to start TKD reset procedure

The TKD reset can take longer time if the content of MFMEM is big.

A TKD reset procedure can be forced without a blockage situation by changing the position of the TKD Jumper.

After TKD reset, all the fiscal data (including NV-RAM) will be recovered from EJ.

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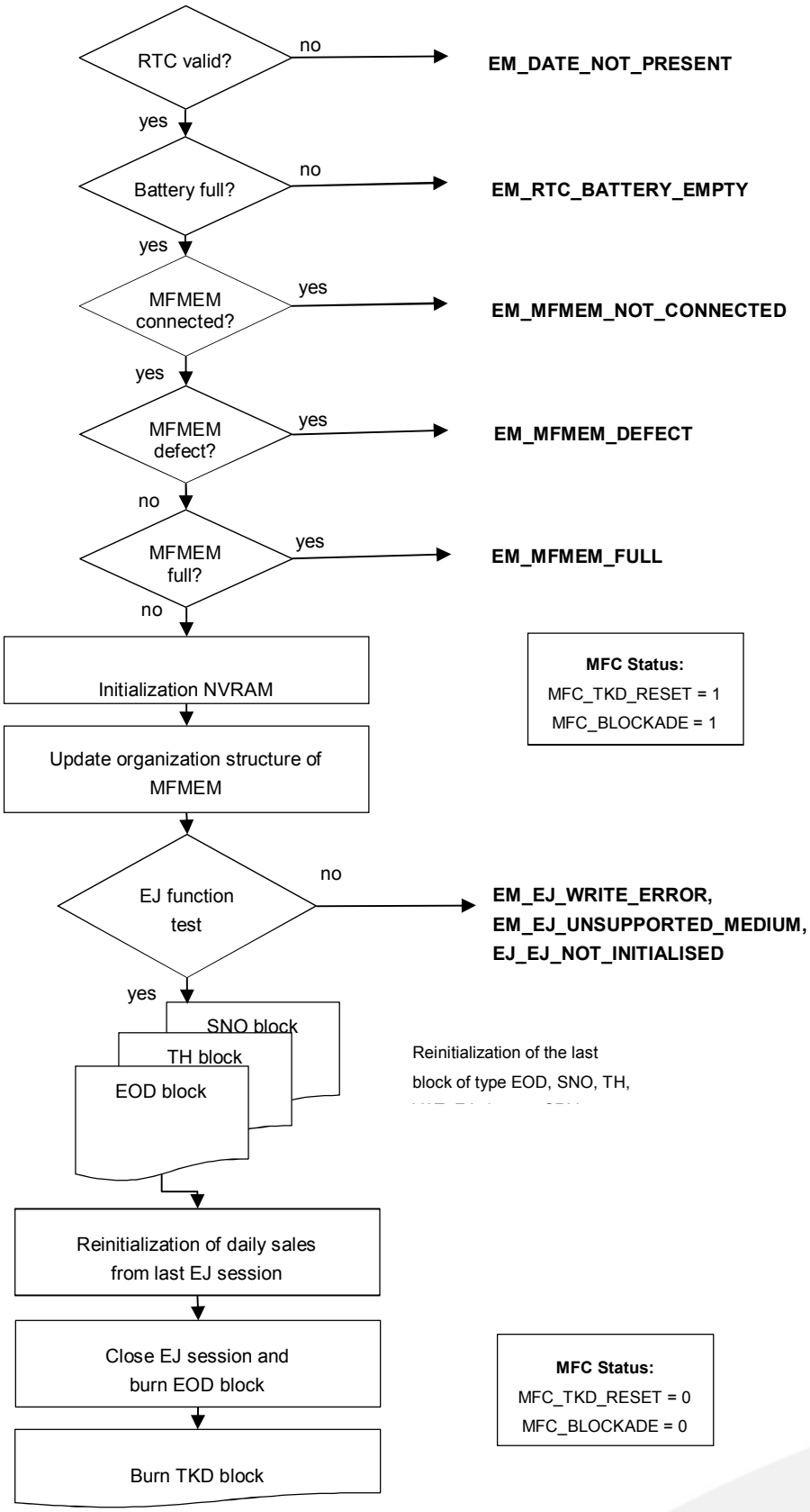

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11 SERVICE MENU

A service menu is integrated in the TH230-MF printer, which allows the user to easily to do some basic fiscal operations, configure the printer, run several diagnostic tests and gain some information about the printer state and the built-in functionality.

This chapter gives an overview about the handling and the content of the service menu.

11.1 Menu handling

11.1.1 Selecting the service menu

To start the service menu you just have to press and hold the linefeed (LF) button during power up or printer reset.

If there is no paper inserted in the printer there are only some major functions, which do not require paper, available via the LED driven menu.

With paper inserted, the print menu will be printed out and all of the functionality of the service menu is available.

The flowchart below shows the process to reach the service mode menu.

If a PAPER_END situation was detected during self-test print out, printer will be reset.

If an error occurred during the service menu, it will be aborted.

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11.1.2 Controlling the Service Menu

The linefeed button (LF) controls the service menu. A short click of the LF button increments the selection and a long click (approx. 1 sec) executes the selected function or rather prints out the selected sub-menu.

Action	Effect
Short click	Increment selection
Long click (approx. 1 sec.)	Execute selected function / Confirm

The current selection is indicated by the 3 LED's.

Function	LED code ● LED on / ● LED off ○ LED flashing
Please select	○○○
1	●●●
2	●●●
3	●●●
4	●●●
5	●●●
6	●●●
7	●●●
8	●●○
9	●●○
10	●●○

If the print menu is active, the current menu or sub-menu will also be printed.

For example if you wish to choose the function with the number 3, you have to click the LF button 3 times and then confirm the selection with a long click.

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11.2 LED Service menu

Please refer to the TH230+ Programmers Guide for a detailed description of the LED driven menu.

11.2.1 Display Service menu

The print menu allows you to change the most important printer settings and gives you access to all diagnostic and information functions. Unlike the LED menu, the print menu expands over several menus with sub-menus. One of these menu entries is the **MF Service Menu**.

Below you will find a detailed description of the overall structure and the functions of the MF Service Menu.

MF Service Menu	
1	Language Romanian
2	Print Selftest
3	XML files to USB
4	TT Test
5	Exit

11.2.2 Language

This command changes the menu language.

11.2.3 Print Selftest

This command prints the MF and the printer self-test.

11.2.4 XML files to USB

This command starts the XML file export from tax terminal to a medium connected to USB port. The export contains the last 3 month.

11.2.5 TT Test

This command starts a tax terminal test.

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11.2.6 Exit

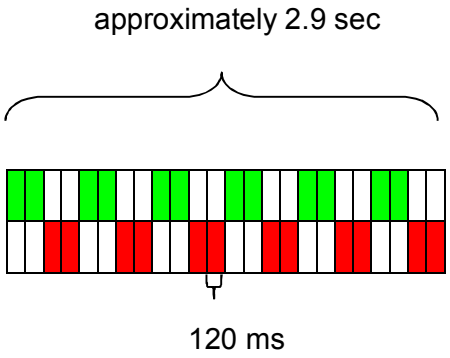
This command leaves the service menu and performs a warm reboot.

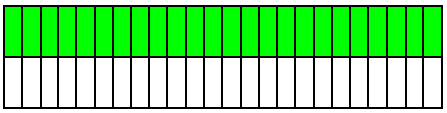
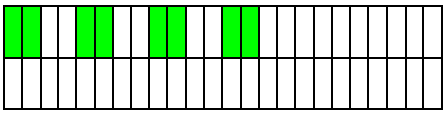
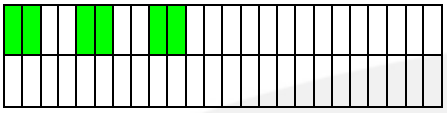
12 APPENDIX

12.1 Printer and MF LED's

There are 3 Printer LED's and 2 MF LED's. The printer LED's are located on the operator panel and described in the TH230+ Programmers Guide. The MF-LED's are located on the right side of printer.

The following table shows the possible different blink codes of the MF-LED's and there meaning.



State	Description	Blink code
Normal Operation	HWDATA present in the MFMEM. SNO Block already burned.	
MFMEM HWDATA not present	There is no HWDATA in the MFMEM. The MFMEM maybe empty	
MFMEM HWDATA present	Only HWDATA present in MFMEM. MFMEM is not formatted	

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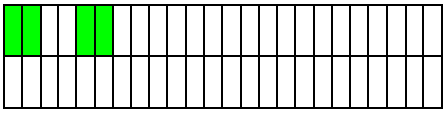
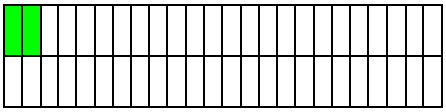
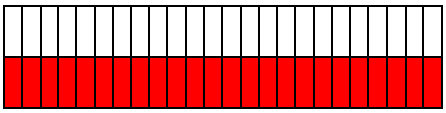
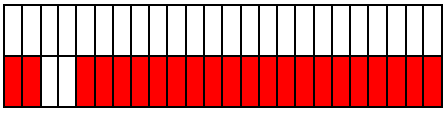
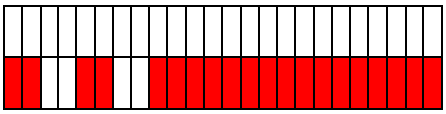
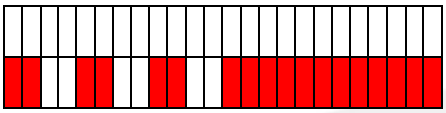
MFMEM partially formatted	HWDATA present in the MFMEM. MFMEM partially formatted.	
MFMEM fully formatted	HWDATA present in the MFMEM. MFMEM is fully formatted	
Blockade (General)	The MF Controller is in the fiscal blockage mode.	
Blockade (Hardware)	The MF Controller is in the fiscal blockage mode. Possible reasons: • *EM_CPU_DEFECT* • *EM_RAM_DEFECT* • *EM_EPROM_CHECKSUM_ERROR* • *EM_RAM_CHECKSUM_ERROR* • *EM_NVRAM_ERROR* • *EM_TKD_JUMPER_IN_RESET*	

Blockade (MF Memory)	The MF Controller is in the fiscal blockage mode. Possible reasons: • *EM_MFMEM_DEFECT* • *EM_MFMEM_BL_CHECKSUM_ERROR* • *EM_MFMEM_BL_READ_ERROR* • *EM_MFMEM_NOT_CONNECTED* • *EM_HWDATA_CHECKSUM_ERROR* • *EM_MFMEM_PTR_ERROR* • *EM_MFMEM_BL_NOT_BURNED* • *EM_MFMEM_BYTE_NOT_BURNED*	
Blockade (Fiscal)	The MF Controller is in the fiscal blockage mode. Possible reasons: • *EM_GRAND_TOT_OVERFLOW* • *EM_SNO_COMPARE_ERROR* • *EM_FW_CHANGE_WHILE_DAY_OPEN* • *EM_MFMEM_FULL* • *EM_TKD_RESET_LIMIT_OVERRUN* • *EM_EJ_DEFECT*	

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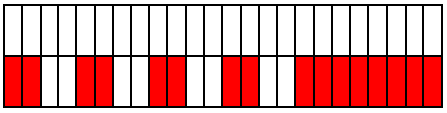
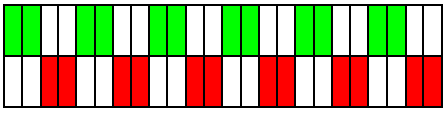
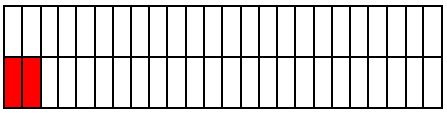
Blockade (RTC)	The MF Controller is in the fiscal blockage mode. Possible reasons: • *EM_RTC_BATTERY_EMPTY* • *EM_RTC_RAM_ERROR* • *EM_DATE_NOT_PRESENT*	
TKD Reset in progress	A TKD reset procedure is currently in progress.	
Parsing MFMEM	The MF controller is busy while parsing the complete MFMEM. This operation does take some time.	

12.2 Tax Terminal LED's

Detailed information of the Tax Terminal LED's can be found in [R8].

12.3 Updating the fiscal firmware

The firmware update protocol is described in [R7]. To update the fiscal firmware all of the following conditions must be fulfilled:

1. The service menu is not active
2. MFC_DAY_OPEN flag is not set (see 5.1 Fiscal Status)
3. Jumper JMP0 is open (see 12.4)

After the fiscal initialization the firmware change event is stored inside the fiscal memory in a separate block (see 5.7.9 FW block). If the maximum of block number (see Block restriction chapter 5.7.10) is reached, the firmware responds with an EM_MFMEM_BL_LIMIT_OVERRUN error. A blockade error is not generated.

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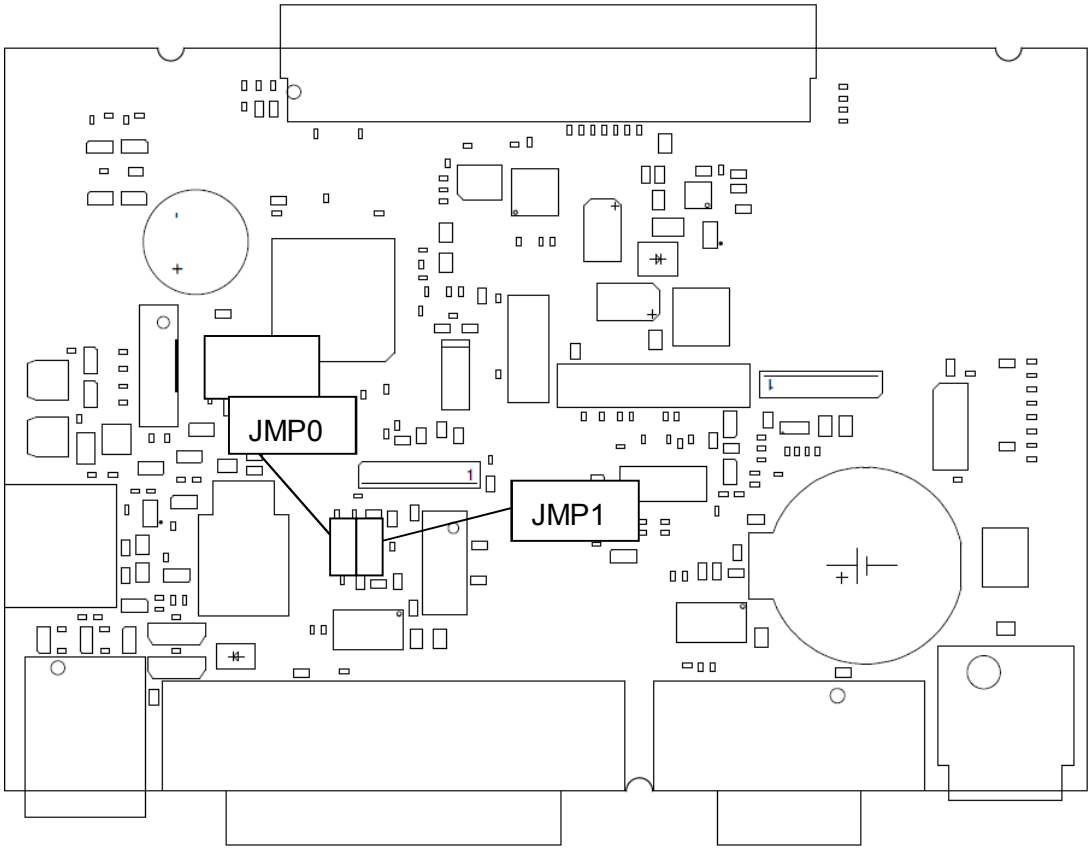
12.4 Fiscal hardware

12.4.1 Fiscal Family Controller (FFC)



Detailed information of the hardware can be found in the Hardware Specification for Fiscal Family Controller [R6].

12.4.2 Tax Terminal

Detailed information of the hardware can be found in [R8].

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12.5 RS232 interface

Data transmission:	Serial
Synchronization:	Asynchronous
Handshaking:	DTR/DSR control; XON/XOFF control not supported
Signal levels:	MARK = -3 to -15 V: Logic "1" / OFF
SPACE:	+3 to +15 V: Logic "0" / ON
Baud rate:	115200 bps (bits per second)
Data word length:	8 bits
Stop bits:	1 bit
Parity settings:	none

Receipt data buffer size: 4096/45 bytes (normal/line mode)
handshake off, if remaining space falls below 384/27 bytes

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12.6 Codepage 852 (Multilingual I + Euro)

HEX	8	9	A	B	C	D	E	F
0	Ç 128	É 144	á 160	░ 176	└ 192	đ 208	Ó 224	- 240
1	ü 129	Ĺ 145	í 161	▒ 177	┴ 193	Đ 209	ß 225	˝ 241
2	é 130	ĺ 146	ó 162	▓ 178	┬ 194	Ď 210	Ô 226	˛ 242
3	â 131	ô 147	ú 163	│ 179	├ 195	Ě 211	Ń 227	ˇ 243
4	ä 132	ö 148	Ą 164	┤ 180	─ 196	ď 212	ń 228	˘ 244
5	ů 133	Ľ 149	ą 165	Á 181	┼ 197	Ň 213	ň 229	§ 245
6	ć 134	ľ 150	Ž 166	Â 182	╞ 198	ť 214	Š 230	÷ 246
7	ç 135	Ś 151	ž 167	Ě 183	╟ 199	Ť 215	š 231	¸ 247
8	ł 136	ś 152	Ę 168	Ş 184	╚ 200	ę 216	Ř 232	° 248
9	ë 137	Ö 153	ę 169	╣ 185	╔ 201	┘ 217	Ú 233	¨ 249
A	Ő 138	Ü 154	¬ 170	║ 186	╩ 202	┌ 218	ř 234	· 250
B	ő 139	Ť 155	Ź 171	╗ 187	╦ 203	█ 219	Ů 235	ů 251
C	Î 140	ť 156	Č 172	╝ 188	╠ 204	▄ 220	ý 236	Ř 252
D	Ź 141	Ł 157	ş 173	Ż 189	═ 205	Ţ 221	Ý 237	ř 253
E	Ä 142	× 158	« 174	ż 190	╬ 206	Ů 222	ţ 238	■ 254
F	Ć 143	č 159	» 175	┐ 191	¤ 207	▀ 223	´ 239	255

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